Population-Accurate Adaptive Data Analysis and Adversarially Robust Second-Moment Estimation

A Hierarchical Learner–Corrector Approach

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Abstract

We develop a hierarchical learner–corrector approach to repeated adaptive access to a small randomized summary. A learner maintains a prediction, a protected comparison detects a discrepancy, and the resulting correction is charged to a persistent progress measure. The hierarchy coordinates the frequency of correction with the strength of statistical protection. We apply this approach to population-accurate adaptive data analysis and adversarially robust turnstile streaming.

For adaptive data analysis, we answer k bounded statistical queries with simultaneous population error at most α using $\widetilde{O}(\sqrt{k}/\alpha + \alpha^{-2})$ independent samples. The mechanism is distribution free, has evaluation access, and stores a constant number of scalar states per sample point. A hierarchy of weighted predictors combines small population comparators with coupled corrections: a certified disagreement decreases a global potential without identifying which predictor is correct.

For streaming, we give a $(1+\varepsilon)$-approximation to F_2 after every update of an adversarial turnstile stream using $\varepsilon^{-2}\,\mathrm{polylog}(n, m, U, 1/\varepsilon, 1/\beta)$ bits, for stream length m, frequency bound U, and failure probability β. A hierarchy of geometric learners preserves each descendant's comparison target in normalized coordinates after an ancestor corrects. Residual contraction then bounds the total corrective workload of each reused sketch. Its corrector contains the statistical hierarchy, whose validation cost depends on rejections and numerical answers, with only logarithmic dependence on the number of accepted tests.

The shared ingredients are protected correction certificates, persistent comparison objects, and potential-based accounting across levels. The statistical and geometric potentials serve different roles; their compatible correction budgets yield the streaming space bound. We include finite-bit implementations and the analysis of the information revealed by their actual adaptive executions.

1 Introduction and results

A small randomized summary is useful because it can answer many questions about data it no longer stores explicitly. Adaptivity makes this reuse delicate: an earlier answer can influence the next query and thereby correlate it with the summary's private randomness. We study this difficulty in two settings. In adaptive data analysis, the summary is an independent sample from an unknown population. In turnstile streaming, it is the memory retained while processing insertions and deletions chosen from earlier reported estimates.

A natural strategy is to learn from the occasions on which a proposed answer fails. An *estimator* forms the proposal from current state. A *corrector* checks it and supplies information when a change is necessary. A *learner* uses that information to improve the state used by future

proposals. This viewpoint underlies reusable holdouts in adaptive data analysis [3] and the estimator–corrector–learner framework for robust streaming of Gribelyuk et al. [9]. It raises two quantitative questions: how can the corrections themselves remain statistically valid, and how can progress survive interactions among several learners?

A hierarchical learner–corrector approach. We organize learners and correctors into a hierarchy, with persistent state at each level. The central requirement is that a correction have two consequences at once: its disclosed information is protected, and its effect can be charged to progress relative to a comparison object that remains meaningful throughout the learner's lifetime. The hierarchy must preserve this accounting when one level changes another. Merely placing independent correctors at several levels would not ensure either property.

We implement this principle in two forms. In the statistical hierarchy, levels are independent groups of sampled records with different privacy budgets. An intermediate level need not already give the correct population answer. Instead, privacy supplies a nearby comparison weighting for each level, and paired updates turn a disagreement between neighbors into a decrease of a common potential. The most strongly protected levels are required to correct least often. In the streaming hierarchy, levels correspond to nested time intervals. Each learner compares its iterate with a fixed prefix vector. Normalization preserves that target when an ancestor changes the represented vectors, so earlier geometric progress is retained.

Two results, with a shared quantitative mechanism. Our statistical result answers k arbitrary bounded adaptive queries to population error α using

$$\widetilde{O}\left(\frac{\sqrt{k}}{\alpha}+\frac{1}{\alpha^2}\right)$$

samples. Our streaming result maintains a multiplicative $(1+\varepsilon)$ estimate of F_2 throughout an adversarial turnstile stream in $\varepsilon^{-2}\,\mathrm{polylog}(n, m, U, 1/\varepsilon, 1/\beta)$ bits. Both are main results of the paper. The first applies to arbitrary finite domains and distributions; the second uses the geometry of squared Euclidean distance and allows the frequency vector to change after every answer.

The statistical construction also supplies the corrector used by the geometric hierarchy. It supports many tests of proposed answers while charging its polynomial sample cost to rejections and numerical releases. The geometric hierarchy proves that this corrective workload is small for each sketch, even over many updates. Their composition therefore needs both a statistical bound on the cost per corrective workload and a geometric bound on how much workload one sketch serves.

The common techniques. Three features organize the proofs. First, the comparison target is persistent: one hindsight population comparator serves an entire statistical transcript, and one normalized prefix serves an entire geometric learner lifetime. Second, a certified discrepancy supplies progress: opposing updates yield statistical potential decrease, while a signed cross term yields residual-energy contraction. Third, the whole interaction is protected: successful checks, correction signs, and stopping decisions belong to the analyzed transcript. The progress bound controls the cost of this protection; it does not replace it.

The two hierarchies consequently share a design and a composable interface, while retaining distinct correctness arguments. A completed unchanged scan transfers accuracy through the statistical chain. Recursive norm certificates and preserved geometric targets establish accuracy in the streaming tree. The overview makes this distinction explicit before developing the two constructions.

Notation and resource conventions. We write $[n] = \{1, \dots, n\}$, use e_i for the ith standard basis vector, and write $\|\cdot\|$ for the Euclidean norm. For $x \in \mathbb{R}^n$, $F_2(x) = \sum_i x_i^2 = \|x\|^2$. Logarithms written ln are natural, and $\log_2$ denotes base two. The notation $\widetilde{O}$ suppresses logarithmic factors in the displayed parameters, never a polynomial dependence on accuracy. Space is measured in bits and includes retained random seeds, counters, predictor states, and arithmetic workspace. No theorem asserts an update-time or query-time bound.

1.1 Population-accurate adaptive data analysis

We next define the statistical problem independently of streaming. For a distribution D on a finite set $\mathcal{U}$, let $Df = \mathbb{E}_{X\sim D} f(X)$.

Definition 1.1 (Adaptive statistical queries)**.** A mechanism receives $S = (X_1, \dots, X_N) \sim D^N$ for an unknown distribution D on a finite domain $\mathcal{U}$. In round $t \in [k]$, an analyst selects a function $f_t : \mathcal{U} \to [0,1]$ using its initial information, independent random tape, and the previous queries and answers. The analyst's initial information may depend on D but is independent of the mechanism's sample and private random tape. The mechanism returns $a_t \in [0,1]$. It is (α, ρ)-*population accurate* if, for every D and every such analyst,

$$\mathbb{P}[\,|a_t - Df_t| \le \alpha \text{ for all } t \in [k]\,] \ge 1 - \rho.$$

The mechanism has *evaluation access* if its only access to a query is through $f_t(X_1), \dots, f_t(X_N)$; it does not inspect a description of f_t or evaluate it at other domain points. The sequence of submitted queries and returned answers is the public transcript. All strategies and evaluations are assumed measurable. Query processing is atomic.

Sample accuracy is a different condition: it requires a_t to be close to $N^{-1}\sum_i f_t(X_i)$. Our statistical result requires population accuracy and places no such condition on the empirical mean of the entire sample. The mechanism is not given D and never evaluates Df_t.

Theorem 1.2 (Population-accurate adaptive query mechanism)**.** *There is an absolute constant C with the following property. For every finite domain $\mathcal{U}$, every integer $k \ge 1$, and every $0 < \alpha, \rho < 1/10$, a single mechanism independent of D satisfies Definition 1.1 for every distribution D on $\mathcal{U}$, using*

$$N \le C \log_2^5\!\left(\frac{Ck}{\alpha\rho}\right)\left(\frac{\sqrt{k}}{\alpha} + \frac{1}{\alpha^2}\right) \tag{1}$$

independent initial samples. It has evaluation access, stores a constant number of scalar states per sample point, and uses $O(\log(k/(\alpha\rho)))$ additional scalar states. The sample bound does not depend on $|\mathcal{U}|$.

Part I gives the complete construction and proof. The mechanism uses a hierarchy of private predictors, each represented by weights on an independent sample block, meaning a disjoint group of the initially sampled records. Its guarantee holds for arbitrary bounded queries, independently of any norm-estimation application. For finite computation, evaluations may be supplied on a common dyadic grid, or rounded to one before entering the mechanism. Corollary 7.4 gives the precise memory bound: for grid denominator D_0 and finite-sampling error allocation ζ, it is

$$O(N(L + \log_2 D_0) + (L + \log(1/\zeta))^2), \qquad L = \log_2 \frac{2^{30}k}{\alpha\rho},$$

in addition to sample representations and query-evaluation workspace. Its failure probability is at most $\rho + \zeta$. Rounding each query value to within $\alpha/4$, invoking the mechanism at accuracy $\alpha/2$, and allocating $\rho/2$ to each source of failure preserves the stated asymptotic sample bound and gives total error at most α. Thus the evaluation interface does not assume an unbounded-precision arithmetic operation in the streaming application.

1.2 Adversarially robust turnstile streaming

The second contribution concerns the frequency vector of an insertion-deletion stream. The algorithm must answer after every update, including updates chosen after observing its earlier answers. Here the resource is memory retained throughout the stream.

Definition 1.3 (Adversarial turnstile stream)**.** Fix public integers $n, m, U \geq 1$. The stream starts at $x^{(0)} = 0 \in \mathbb{Z}^n$. At time $t \in [m]$, an adversary supplies an update $(j_t, \Delta_t) \in [n] \times \mathbb{Z}$, and

$$x^{(t)} = x^{(t-1)} + \Delta_t e_{j_t}.$$

An admissible adversary maintains $\|x^{(t)}\|_\infty \leq U$ at every time and chooses its update as an arbitrary function of the public parameters, its own random tape, and the answers reported before time t. The algorithm processes the update and reports one answer $\widehat{F}_t$ before receiving the next update. Its private random tape is independent of the adversary's initial information and random tape. The adversary observes only the reported answers: it does not observe the algorithm's memory, measurements, internal decisions, or random bits. There is no computational restriction on the adversary.

Thus negative frequencies are allowed, the input stream itself is never rescaled, and the algorithm must succeed simultaneously at all m times. The magnitude bound implies $|\Delta_t| \leq 2U$, so an update has $O(\log(nU + 2))$ bits. The parameters may be upper bounds on the actual stream length and frequency magnitudes.

Theorem 1.4 (Adversarially robust second-moment estimation)**.** *For every $n, m, U \geq 1$ and $0 < \varepsilon, \beta < 1/10$, there is a randomized algorithm in Definition 1.3 such that, for every admissible adversary,*

$$\mathbb{P}\left[\frac{F_2(x^{(t)})}{1+\varepsilon} \leq \widehat{F}_t \leq (1+\varepsilon)F_2(x^{(t)}) \text{ for all } t \in [m]\right] \geq 1 - \beta.$$

For an absolute constant C, its space is at most

$$C\varepsilon^{-2}\Lambda^{20} \quad \textit{bits}, \qquad \Lambda = \log_2 \frac{2^{100}(n+2)(m+2)(U+2)}{\varepsilon\beta}.$$

The probability is over the algorithm's and adversary's random tapes. In particular, when $m, U \leq n^{O(1)}$, the space is $\varepsilon^{-2}\,\mathrm{polylog}(n, 1/\varepsilon, 1/\beta)$.

The logarithmic exponent is not optimized. The algorithm is stateful and initializes new independent sketches at predetermined block boundaries. Its internal vector rescalings are implemented by arithmetic on stored counters. This model permits fresh randomness during the stream; the separate fixed-measurement result below concerns a bounded number of queries.

1.3 Fixed measurements for a bounded number of queries

The statistical theorem also has a direct sketching consequence. Fix $n, K, U \geq 1$. Before an interaction begins, choose a private sign matrix A. At round t, an adversary supplies $x_t \in \mathbb{Z}^n$ with $\|x_t\|_\infty \leq U$, chosen from the previous reported answers and independent initial information. The estimator observes only Ax_t, retains private state between rounds, and reports an estimate of $F_2(x_t)$. All rows are fixed in advance.

Corollary 1.5 (Bounded adaptive second-moment queries)**.** *For every $0 < \varepsilon, \beta < 1/10$, this model admits simultaneous $(1+\varepsilon)$-multiplicative accuracy for all K rounds, with failure probability at most β, using*

$$\widetilde{O}\left(K + \frac{\sqrt{K}}{\varepsilon} + \frac{1}{\varepsilon^2}\right)$$

fixed sign measurements and the same number of bits up to logarithmic factors in $n, U, K, 1/\varepsilon, 1/\beta$. In particular, $K = \lceil \varepsilon^{-2} \rceil$ requires $\varepsilon^{-2}\,\mathrm{polylog}(n, U, 1/\varepsilon, 1/\beta)$ bits. Arbitrary changes to the vector are allowed between queries.

Proof. Apply Theorem 9.2, whose explicit row and memory bounds are proved in Section 11, and substitute the stated value of K. □

This corollary concerns K queries, whereas Theorem 1.4 reports after every stream update. The recursion reduces the latter problem to many sketch instances, each of which releases only a bounded number of corrections and numerical estimates. Only logarithmically many instances are retained at a time.

1.4 Comparison with previous approaches

The learner–corrector organization builds on established ideas. Reusable holdouts protect repeated validation, and the estimator–corrector–learner framework of [9] supplies the geometric starting point for streaming. Our contribution is the quantitative coordination of learning, protection, and persistence needed for the bounds above.

Adaptive data analysis. Dwork et al. [2] introduced the use of differential privacy to preserve statistical validity under adaptive query selection. Bassily et al. [4] give the general-purpose bound $\widetilde{O}(\sqrt{k}/\alpha^2)$ for bounded statistical queries. They also give a domain-dependent bound $\widetilde{O}(\sqrt{\log|\mathcal{U}|}\log k/\alpha^3)$, which can be preferable for small domains and many queries. Our comparison is with the domain-independent bound: Theorem 1.2 improves it to $\widetilde{O}(\sqrt{k}/\alpha + \alpha^{-2})$ while retaining evaluation access. For $k = \lceil \alpha^{-2} \rceil$, the two bounds are $\widetilde{O}(\alpha^{-3})$ and $\widetilde{O}(\alpha^{-2})$, respectively.

The accuracy requirement is essential to this comparison. Lyu and Talwar [7] distinguish population-only accuracy from simultaneous sample and population accuracy. In the large-domain, few-query regime, the known population-only lower bound has dependence $\widetilde{\Omega}(\sqrt{k}/\alpha)$, whereas the stronger accuracy requirement admits a $\Omega(\sqrt{k}/\alpha^2)$ lower bound. Theorem 1.2 does not assert that stronger requirement. The reusable-holdout work of Dwork et al. [3] already shows how to protect repeated validation. Our statistical contribution concerns the learning and privacy allocation across a hierarchy: a strongly protected level is needed less frequently, because neighboring learners make coupled progress. We compare unrestricted distributions and arbitrary bounded queries; we do not claim to dominate every bound that depends on a restricted domain or distribution class.

Robust turnstile streaming. The algorithm of Gribelyuk et al. [9] gives polylogarithmic space in n for constant accuracy, with polynomial dependence on $1/\varepsilon$. In its recursive construction, a learner makes $L = \widetilde{O}(\eta^{-2})$ corrections at local accuracy η. Corrections create new descendant blocks. The branching parameter is chosen at least as large as L, and each corrector is charged for $BL + B$ numerical interactions over its children. With $B = \widetilde{O}(\eta^{-2})$, this gives $\widetilde{O}(\eta^{-4})$ interactions per corrector. The analysis uses bounded computation paths and independent rows for numerical releases, giving $\widetilde{O}(\varepsilon^{-6} \log^{11} n)$ bits. Remark 2.6 of the same paper improves this to $\widetilde{O}(\varepsilon^{-4})\,\mathrm{polylog}(n)$ bits by using a differential-privacy-based robustification inside each block. The latter is the stronger prior bound for comparison, for polynomially bounded streams and the paper's high-probability guarantee.

Our improvement addresses both sources of accuracy dependence. We use a fixed binary time tree, so each corrector serves at most two learner lifetimes and handles only $\widetilde{O}(\eta^{-2})$ corrections and numerical requests. The adaptive statistical mechanism handles this workload in $\widetilde{O}(\eta^{-2})$ space. Merely replacing the statistical mechanism in the previous recursion would leave its larger interaction budget; merely changing the recursion would leave the cost of the previous validation methods, including the differential-privacy variant.

2 Technical overview: hierarchical learners and correctors

We begin with the common mechanism, then explain how each application supplies its statistical and geometric ingredients. The discussion keeps three quantities separate: the number of questions received, the number of corrections performed, and the information disclosed by the entire sequence of responses. Many questions can be answered with few corrections, but that fact is useful only when the responses remain valid under adaptive reuse.

Section 2.1 gives the organizing framework. Sections 2.2–2.4 develop its statistical realization in detail. Section 2.5 identifies the shared potential and persistence requirements and states the interface through which the two hierarchies compose. Sections 2.6–2.9 then explain the geometric realization and its finite implementation. Calculations in this overview suppress logarithmic factors; the formal statements retain them.

2.1 The common mechanism: protect, correct, and preserve

An *estimator* maps the current learner state and query to a candidate answer. A *learner* is the state-update procedure used when a discrepancy is detected. A *corrector* is the procedure that tests the candidate or compares it with another prediction and returns the information allowed to drive that update. These are roles: a corrector may itself contain learners, and an intermediate learner may help correct one neighbor while being corrected by the other.

We call a comparison *protected* when its information disclosure is covered by the statistical analysis of the complete interaction. Its returned sign or flag is a *certificate* if, on that analysis's simultaneous success event, it implies a specified inequality. The inequality need not say that one endpoint is correct. In ADA it can certify a signed gap between two current predictions. In streaming it certifies a signed discrepancy from the target norm, up to explicit error margins. This distinction permits the statistical hierarchy to learn even without access to a correct label.

The lifetime of a learner. A *comparison object* is the fixed object against which learning progress is measured; it need not be known or stored. A learner's *lifetime* is the interval of the execution over which this object and the accumulated learning state are retained. Our statistical

learners live for the whole interaction and compare with weights chosen in hindsight. A streaming learner lives on a predetermined time interval and compares with the prefix present at its birth, expressed in normalized coordinates. New questions may arrive throughout the lifetime without creating a new learning problem.

This persistence is essential to amortization. If the comparison object were chosen separately on every query, or the state were reset after each neighboring correction, a bound for one learning trajectory could be charged repeatedly. The hierarchy must specify which changes preserve an existing trajectory and which events start a genuinely new one.

Four requirements for the hierarchy. The constructions verify the following requirements separately.

1. *Validity of the disclosed comparisons.* Every update uses only a protected output, and the analysis covers all accepted checks and any terminal rejection.
2. *Progress on a certified discrepancy.* A correction decreases a nonnegative progress measure, or satisfies an amortized inequality that yields a bound on the total corrections. An individual prediction need not improve on every query.
3. *Persistence across neighboring updates.* The state and comparison object remain usable after other levels correct. Any additional perturbation cost is explicitly included in the progress bound.
4. *Certification of the answer actually returned.* The final answer is justified by comparisons and estimates for the states used to compute it. Certificates for states that have since changed cannot be reused without an argument.

These requirements describe the structure proved for our two constructions. They are not an assumption that an arbitrary hierarchy automatically has the desired statistical or space guarantees.

What a level means in each application.

	Statistical hierarchy	Geometric hierarchy
Levels	Sample groups with different privacy budgets	Learners on nested time intervals
Learner state	Scores defining capped sample weights	Linear measurements of a vector iterate
Persistent target	One population comparator per sample group	One prefix per learner in normalized coordinates
Correction certificate	A signed gap between adjacent predictions	A signed error in an energy decomposition
Progress measure	A weighted sum of regularized comparison gaps	Squared residual norm, with a perturbation allowance
Accuracy transfer	An unchanged scan ending at an accurate anchor	Protected norm tests and recursive error margins

The statistical hierarchy balances protection with correction frequency: levels with stronger privacy have smaller comparison potentials and geometrically smaller correction budgets. The geometric hierarchy balances lifetime with memory: a parent correction preserves its descendants, and a fixed binary time tree bounds how many lifetimes one sketch must serve. In the final algorithm the hierarchies are nested. Each geometric corrector contains a statistical hierarchy; the geometric progress bound determines that hierarchy's external workload.

Why checking and correcting must be analyzed together. A correction bound is deterministic once its certificates are valid. Statistical validity, however, depends on how much corrective information has already been revealed. We resolve this apparent circularity by fixing caps before the interaction. Privacy is proved for the capped algorithm, including executions that halt. On the resulting success event, progress rules out the first cap failure. Accepted comparisons remain part of the privacy analysis even when they do not change the learner. Later subsections show how this argument is implemented at both levels of the construction.

2.2 ADA: why one predictor and one holdout are not enough

We first separate the statistical task from the machinery used to solve it. The unknown distribution is D, an incoming query is a function $f : \mathcal{U} \to [0, 1]$, and the desired answer is the population mean $Df = \mathbb{E}_{X \sim D} f(X)$. The mechanism has sampled records from D, but it cannot ask D for this expectation. Its required error is α on each of k adaptively selected queries, simultaneously except with probability ρ.

Sample blocks, holdouts, and transcripts. A *sample block* is a group of records reserved from the initial independent sample. For example, a block of size r is $S = (X_1, \ldots, X_r) \sim D^r$. Different blocks use disjoint groups of these initially independent records. Here a block is a dataset, not a time interval; the streaming construction will separately use intervals of updates. The empirical mean of f on this block is $Sf = r^{-1} \sum_i f(X_i)$.

A *holdout* is a sample block reserved for checking predictions made using other information. Reserving it prevents the initial predictor from fitting those records directly. However, if the same holdout repeatedly reports whether predictions are correct, later queries can depend on those reports and thereby become dependent on the holdout. Initial independence alone does not justify indefinite reuse. Even a response saying that a prediction passed its check can reveal information.

A *transcript* is the record of responses in an interaction. The public transcript contains the queries and the answers seen by the analyst. Our privacy proof uses a larger, *augmented transcript*: it also records the outcomes of the mechanism's internal private comparisons. It does not include the samples, raw predictor values, or private noise values. This larger transcript is needed because internal comparison outcomes determine later updates, even when the analyst does not see those outcomes individually.

What privacy protects. A transcript-producing mechanism is (θ, δ)-differentially private if replacing one record in its protected block changes the probability of any transcript event by at most a multiplicative factor e^θ and an additive slack δ. More precisely, for neighboring datasets S, S' and every event E,

$$\mathbb{P}[T(S) \in E] \leq e^\theta \mathbb{P}[T(S') \in E] + \delta.$$

The same inequality must hold with S, S' interchanged. Smaller θ means stronger protection. The case $\delta = 0$ is called *pure privacy*; positive δ gives *approximate privacy*. The parameter δ is slack in this distributional inequality, not the final probability ρ that our accuracy theorem fails. Section 3 gives the formal interactive definition.

In this paper privacy is also a statistical tool: limiting what the transcript reveals about one record controls how conditioning on the transcript can distort that record's distribution. We will use this control to construct comparison weights, rather than require every empirical mean to be accurate.

Where the usual two accuracy factors arise. Suppose we simply answer using noisy empirical means on a block of size r. Replacing one record changes a bounded empirical mean by at most $1/r$. Under the usual private-release composition calculation, protecting k releases at total privacy level θ requires noise of order $\sqrt{k}/(r\theta)$, up to logarithmic factors. Requiring this noise to be at most α gives

$$r \gtrsim \frac{\sqrt{k}}{\theta\alpha}.$$

This controls the distance from the released answer to the empirical mean. It does not yet control the distance from that empirical mean to Df. A direct privacy-to-generalization argument takes $\theta = O(\alpha)$ for the latter purpose, leading to the familiar $\widetilde{O}(\sqrt{k}/\alpha^2)$ bound. One factor of $1/\alpha$ pays for numerical precision; the other pays for keeping adaptive empirical bias small. This calculation explains a limitation of this approach, rather than a lower bound for every ADA mechanism.

Why rare corrections alone do not remove the loss. An alternative is to learn a predictor on one block and consult a protected holdout to discover its mistakes. Suppose the predictor needs only $L = O(\alpha^{-2})$ corrections. A private threshold test can charge its principal composition cost to the number of corrections rather than to every successful check; we explain that test in Section 2.4. Nevertheless, if the holdout's empirical answers must themselves be accurate, the direct calculation still uses $\theta = O(\alpha)$. Its sample cost can therefore be

$$\widetilde{O}\left(\frac{\sqrt{L}}{\theta\alpha}\right) = \widetilde{O}(\alpha^{-3}).$$

Thus it is not enough to show that the learner changes infrequently. We need a relationship between a block's privacy requirement and the number of times it supplies corrective information.

The role of the hierarchy. We use several weighted predictors, arranged in a chain. A *weighted predictor* assigns nonnegative weights summing to one to its block and reports the corresponding weighted query average. The first predictor, called the *root*, ultimately answers the analyst. The last model, called the *empirical anchor*, uses an ordinary empirical mean on its own strongly protected block. Models between them are used to compare neighboring predictions.

Only the anchor is required to be directly population accurate before comparisons. For each intermediate predictor, privacy will instead imply the existence of a nearby set of weights that would answer all selected queries accurately. These weights are an object for the analysis, called a *comparator*; they are not available to the algorithm. We bound the number of disagreements by comparing the actual predictors with these weights in a learning inequality. The next two subsections explain these two steps separately.

2.3 ADA: privacy supplies a small population comparator

Fix a block $S = (X_1, \dots, X_r)$ and a transcript T that is private with respect to this block. Include enough independent auxiliary information in T that it determines the selected queries $f_1, \dots, f_k$. For example, this includes the analyst's initial random tape. Privacy is required for this enlarged transcript; the blockwise privacy proof checks that requirement explicitly. Let h be a small error allowance; later we choose $h = \alpha/(256p)$ for a hierarchy with p weighted predictors. A comparator is a single vector of weights u satisfying

$$u_i \geq 0, \qquad \sum_i u_i = 1, \qquad |uf_t - Df_t| \leq h \quad (t \in [k]), \qquad uf := \sum_i u_i f(X_i).$$

The word *single* matters: the same u must work for the whole realized sequence, rather than being chosen separately for each query. The word *nearby* means that u has small quadratic distance from uniform weights, measured by

$$R(u) = \frac{r}{2}\sum_i (u_i - 1/r)^2.$$

Uniform weights have $R(u) = 0$. We seek $R(u) = O(\theta^2)$, where θ is the block's privacy parameter, even when θ is much larger than the fitting error h.

Conditioning creates bias; inverse weighting removes it. Before observing the transcript, each X_i has distribution D. After observing it, some values can become more likely. If an output is positively correlated with the occurrence of a particular record, seeing that output increases the conditional probability of that record. This is a change in our conditional distribution of the data, not a change to the data themselves.

For illustration, suppose an initially uniform bit has conditional probabilities 0.6 for one and 0.4 for zero. Weighting a one by $0.5/0.6 = 5/6$ and a zero by $0.5/0.4 = 5/4$ restores both population masses to 0.5 in expectation. Consequently it restores the expectation of every function of this bit. Inverse weighting applies the same idea to every record, with a separate conditional ratio for each record and its already exposed history.

More precisely, for pure privacy, write

$$a_i(x) = \frac{\mathbb{P}[X_i = x \mid T, X_1, \ldots, X_{i-1}]}{D(x)} \quad \text{for } D(x) > 0, \qquad v_i = 1/a_i(X_i).$$

The ratio $a_i(x)$ is the factor by which conditioning has changed the probability of x. Pure privacy bounds it between $e^{-\theta}$ and e^{θ}, almost surely. Therefore the inverse weights also lie in $[e^{-\theta}, e^{\theta}]$. For a query f_t determined by T, cancellation of this density ratio gives

$$\mathbb{E}[v_i f_t(X_i) \mid T, X_1, \ldots, X_{i-1}] = \sum_x D(x) a_i(x) \frac{f_t(x)}{a_i(x)} = D f_t.$$

The displayed conditional probabilities are an explanatory notation. Lemma 4.1 formulates the argument through joint probability measures, so it does not require conditioning on an individual transcript event of positive probability.

Why expose the records sequentially? Conditioning on T generally makes the records dependent. We cannot apply an independent-sample concentration bound to their conditional law. Instead, for the proof, reveal X_1, then X_2, and so on. The preceding identity says that, at each reveal, the weighted evaluation has the correct conditional mean given everything already revealed. Its deviation from that conditional mean is a *martingale difference*: a random fluctuation with conditional expectation zero. A concentration inequality for sums of bounded martingale differences is exactly the tool needed here. This sequential reveal is a proof device; the algorithm does not expose its records to the analyst.

What "drift" means under approximate privacy. When $\delta > 0$, the density ratios need not satisfy pointwise bounds. We therefore clip, meaning truncate, the inverse weights to the interval $[e^{-\theta}, e^{\theta}]$:

$$v_i = \mathrm{clip}_{[e^{-\theta}, e^{\theta}]}(1/a_i(X_i)).$$

Clipping keeps the variables bounded but can spoil their exact conditional expectations. Define the resulting *conditional drift* by

$$e_{i,t} := \mathbb{E}[v_i f_t(X_i) \mid T, X_1, \ldots, X_{i-1}] - Df_t.$$

Thus drift here means a predictable error in the conditional mean. It is distinct from the zero-mean random fluctuation around that mean, and has nothing to do with updates to a streaming vector.

The two directions of the approximate-privacy inequality control both tails of the density ratio. The formal proof constructs nonnegative d_i, independent of the query index t, such that

$$|e_{i,t}| \le d_i \quad \text{for every selected query,} \qquad \mathbb{E}d_i \le 2\delta.$$

This is an expectation bound over the joint experiment, not a claim that the drift is small for every fixed transcript. Markov's inequality controls its average with high probability when δ is sufficiently small. The concentration argument controls the remaining fluctuation. Explicitly, the error splits as

$$\frac{1}{r}\sum_i v_i f_t(X_i) - Df_t = \frac{1}{r}\sum_i \left(v_i f_t(X_i) - \mathbb{E}[v_i f_t(X_i) \mid T, X_1, \ldots, X_{i-1}]\right) + \frac{1}{r}\sum_i e_{i,t}.$$

The first sum is the martingale fluctuation; the second is the drift. This decomposition explains why bounded weights alone would not finish the proof: we must also control what clipping does to the mean.

Normalization and the common comparator. The inverse weights do not necessarily sum to one in a realized sample. Apply the same argument also to the constant query $f_0 = 1$. It implies that $W = r^{-1}\sum_i v_i$ is close to one. Then set $u_i = v_i / \sum_a v_a$. Dividing the weighted averages by W only slightly enlarges their errors. A union bound over the selected queries and f_0 supplies one normalized u that fits all of them. The construction of v_i depends on the transcript and record history, but not on which of these queries is being checked.

For a per-block failure allowance ξ, the precise lemma needs

$$r \ge 32e^{2\theta}h^{-2}\ln\frac{4(k+1)}{\xi}, \qquad \delta \le \frac{h\xi}{32}.$$

Thus, for $\theta \le 1$, fitting to error h costs $\widetilde{O}(h^{-2})$ samples, while the distance of the comparator from uniform is governed separately by θ:

$$ru_i \in [e^{-2\theta}, e^{2\theta}], \qquad R(u) \le \tfrac{1}{2}\sinh^2\theta \le \theta^2.$$

To interpret this quadratic dependence, each weight differs from $1/r$ by at most $O(\theta/r)$ when $\theta \le 1$. Squaring these differences, summing over r records, and multiplying by $r/2$ gives $O(\theta^2)$.

What the lemma does and does not give the learner. The comparator can depend on the entire completed transcript and on the unknown distribution. It is not computed or stored. Instead, a deterministic learning inequality will compare the actual sequence of predictors against every fixed feasible weight vector, including this one chosen in hindsight. Separate comparators for separate queries would not support that inequality. Likewise, existence of a good comparator does not by itself prove that the learner's actual answers are accurate. We also do not require the actual weights to equal the comparator or remain close to it coordinate by coordinate. The next subsection supplies the learning and validation argument that connects this existence statement to the actual answers.

2.4 ADA: a disagreement can pay for itself without a true label

We now explain the executable mechanism and its accounting. There are p weighted sample blocks $S_0, \dots, S_{p-1}$, of sizes $n_0, \dots, n_{p-1}$, and one additional block H for the empirical anchor. All records and all independent private random tapes are drawn at initialization. The same blocks are reused across external queries; a correction does not draw a new dataset. On a current query f, model j predicts

$$\mu_j = \sum_i w_{j,i} f(X_{j,i}), \qquad \mu_p = Hf = \frac{1}{n_H} \sum_i f(H_i).$$

An *edge* is an adjacent pair of models: edge j compares μ_j to μ_{j+1}. The anchor is indexed by p for this notation, but it has no learned weights and is never updated.

How a predictor changes its weights. At level j, each record has an accumulated score $G_{j,i}$. The score initially equals zero. A signed query update adds a multiple of the current evaluation $f(X_{j,i})$ to it. Let $\eta > 0$ denote the learning rate. The predictor chooses

$$w_j = \arg\max_{w \in \mathcal{W}_j} \left\{ \eta \sum_i G_{j,i} w_i - R_j(w) \right\}, \quad \mathcal{W}_j = \left\{ w \geq 0 : \sum_i w_i = 1,\ w_i \leq 8/n_j \right\},$$

where $R_j(w) = (n_j/2) \sum_i (w_i - 1/n_j)^2$. The score term favors records that were useful on past updates; the regularizer penalizes moving too far from uniform. The *weight cap* $8/n_j$ prevents concentration on a single record. Initially the weights are uniform. The implementation stores the scores, not the list of old queries; new weights can be recomputed from the scores and one normalization constant.

The cap has a privacy role as well as a learning role. At a fixed augmented history, replacing one record changes the next prediction by at most $16/n_j$. This maximum change is called its *replacement sensitivity*. The bound accounts for both the changed query evaluation and the change in all the normalized weights. It would not follow merely by looking at the weight of the replaced record. Lemma 5.1 proves the bound using the common normalization shift in the weight formula.

The error scales. Use the same parameters as the formal construction:

$$\tau = \frac{\alpha}{4p}, \qquad h = \frac{\tau}{64}, \qquad \eta = \frac{\tau}{4}.$$

The threshold τ is the size of an adjacent disagreement worth correcting. We divide the target error by p because transferring accuracy across p edges adds their errors. The smaller allowance h covers comparator inaccuracies, private noise, and rounding. The learning rate η controls how much cumulative score affects the weights. In particular, η here is not a privacy parameter.

Give model j the accounting weight and privacy parameter

$$\lambda_j = 4^j, \qquad \theta_j = (p\lambda_j)^{-1/2}.$$

An accounting weight is a multiplier in the proof, not a sample weight and not an additional update. Moving one level toward the anchor multiplies λ_j by four and halves θ_j. The depth is chosen so that $4^{p-1}\tau^2$ lies between one and four; this gives $p = O(\log(1/\alpha))$. The anchor has the stronger privacy parameter $\theta_H = h/8$. Its near-uniform comparator is then so close to uniform that the anchor's ordinary empirical mean has error less than $2h$ on all selected queries (Lemma 6.3).

How a comparison is protected, and what a phase is. We cannot inspect the raw gap $\mu_{j+1} - \mu_j$ and expose its outcome for free. Each edge uses the private procedure *AboveThreshold*. In one *phase*, it draws a noisy threshold once and retains it. Each invocation compares a fresh noisy score against that same noisy threshold. The phase ends at its first positive outcome. If many checks are negative, they remain in the same phase; after a positive outcome, the next phase uses independent randomness.

The privacy guarantee covers the entire phase, including all its negative outcomes and an all-negative stopped prefix. Thus we pay for the number of phases in composition. This does not mean that negative outcomes reveal nothing: their joint disclosure is already included in the phase's privacy guarantee. A larger total number of checks still increases the logarithmic noise allowance needed for all checks to be accurate simultaneously. Lemma 3.4 gives the precise noise distributions and proof.

For a two-sided comparison, test the scores $\mu_{j+1} - \mu_j$ and $\mu_j - \mu_{j+1}$ against threshold τ. Two negative outcomes accept the edge. A positive outcome supplies a sign $q \in \{-1, 1\}$ indicating the direction of the detected disagreement. On the event that all individual noises have magnitude at most h, this sign certifies

$$q(\mu_{j+1} - \mu_j) \geq \tau - 2h.$$

The sign used for learning is the sign returned by this private comparison. Reading the unperturbed sign afterwards would be an additional disclosure not covered by this transcript.

The coupled update. On a positive outcome at an internal edge, add qf to model j's score function and add $-qf/4$ to model $j+1$'s score function. For example, if the right-hand prediction is larger, the left model is encouraged to put more weight on records where f is large, while the right model receives an opposing update of one quarter the magnitude. Neither model is declared to be correct. At the last edge only the weighted model changes; the anchor remains fixed.

Why can this work without knowing Df? Let u_j and u_{j+1} be the single hindsight comparators from the previous subsection. The gain of comparator u over a prediction w on an update g is $\langle u - w, g\rangle$. Multiplying the two comparator gains by their accounting weights gives

$$\begin{aligned}\lambda_j q(u_j f - \mu_j) - \frac{\lambda_{j+1}}{4} q(u_{j+1} f - \mu_{j+1}) \\ = \lambda_j q(\mu_{j+1} - \mu_j + u_j f - u_{j+1} f).\end{aligned}$$

The equality uses $\lambda_{j+1} = 4\lambda_j$. The unknown population answer has effectively canceled: both $u_j f$ and $u_{j+1} f$ are within h of the same number Df, so their difference has magnitude at most $2h$. The displayed gain is therefore at least

$$\lambda_j(\tau - 4h) \geq \lambda_j \tau/2.$$

The analogous last-edge gain is at least $\lambda_{p-1}(\tau - 5h)$, because the anchor has error less than $2h$. This calculation explains the opposing updates and the factor $1/4$: together they make a disagreement yield positive combined comparator gain without deciding which endpoint has the better answer.

Why there can be only a bounded amount of disagreement. The learner's *regret bound* limits the total gain of any fixed comparator over its changing predictions. For any chronological sequence of gain vectors g_t, the quadratic predictor satisfies

$$\sum_t \langle u - w_t, g_t\rangle \leq \frac{R(u)}{\eta} + \frac{\eta}{2} \sum_t \|g_t\|_\infty^2.$$

The first term pays for how far the comparator is from uniform. The second pays for the learner's finite response to its updates. This inequality holds for every realized sequence, including adaptive ones, so it is valid for a comparator selected in hindsight. Here the chronological updates of a model include updates from either neighbor, with their actual signs and magnitudes.

The comparator lemma and our choice of privacy parameters imply

$$P := \sum_{j=0}^{p-1} \lambda_j R_j(u_j) \le \sum_{j=0}^{p-1} \lambda_j \theta_j^2 = 1.$$

Let T_j be the number of corrections on edge j, and let $W = \sum_j \lambda_j T_j$ be their weighted count. One internal correction contributes at most $\lambda_j + \lambda_{j+1}/16 = 5\lambda_j/4$ to the weighted sum of squared gain magnitudes. Summing the regret inequalities and the lower bounds on comparator gain therefore yields

$$\frac{\tau}{2} W \le \frac{P}{\eta} + \frac{5\eta}{8} W \le \frac{P}{\eta} + \eta W.$$

Substituting $\eta = \tau/4$ and $P \le 1$ gives

$$W \le 16/\tau^2, \qquad T_j \le \frac{16}{4^j \tau^2}.$$

This is an aggregate progress argument. It does not assert that each individual predictor's population error decreases at every update. Corrections become geometrically rarer as we move toward the more strongly protected end of the chain.

Why the privacy and sample costs now balance. For a block of size r, a sensitivity-$O(1/r)$ mechanism that has C private phases or numerical releases needs, up to logarithms,

$$r \gtrsim \frac{\sqrt{C}}{\theta h}$$

to keep its noise below h at total privacy level θ. The comparator lemma separately requires $\widetilde{O}(h^{-2})$ records. An internal block participates only in phases on its two incident edges. Its number of such components is $\widetilde{O}(4^{-j}\tau^{-2})$. Consequently

$$\frac{\sqrt{4^{-j}\tau^{-2}}}{\theta_j h} = O\left(\frac{\sqrt{p}}{\tau h}\right).$$

The factors depending geometrically on j cancel. Put informally, halving the allowed privacy parameter would double the required sample size, but allowing one quarter as many corrections halves the square-root composition cost.

Only the root additionally pays for all k external numerical answers. Its cost includes $\sqrt{k}/(\theta_0 h)$, which is $\widetilde{O}(\sqrt{k}/\alpha)$. The last edge has only a constant number of phases because $4^{p-1}\tau^2 = \Theta(1)$; even the strongly protected anchor therefore costs only $\widetilde{O}(h^{-2})$. Summing the block sizes gives

$$N = \widetilde{O}\left(\frac{\sqrt{k}}{\alpha} + \frac{1}{\alpha^2}\right).$$

The suppressed factors include the logarithmic depth p. The formal sample-count calculation keeps these factors explicitly.

How a query is processed, and why its actual answer is accurate. For the current query, scan the edges in order from root to anchor. If an edge corrects, update its endpoint scores and restart the scan at the root. When an entire scan accepts without any update, release the current root prediction after adding the prescribed private noise and rounding. Then proceed to the next external query.

Restarting is essential. A successful comparison between models zero and one can become obsolete if a later comparison updates model one. The final unchanged scan certifies all adjacent gaps for the same current collection of predictors. On the simultaneous success event, each gap is at most $\tau + 2h$, so

$$|\mu_0 - Df| \le \sum_{j=0}^{p-1} |\mu_j - \mu_{j+1}| + |\mu_p - Df| \le p(\tau + 2h) + 2h.$$

The root release adds noise of magnitude at most h and rounding error at most $h/2$. Thus its actual answer a, rather than an unimplemented comparator or an earlier state, satisfies

$$|a - Df| \le p(\tau + 2h) + \tfrac{7}{2}h < \alpha.$$

Why the argument does not assume its own correction bound. Privacy parameters must be fixed before we know how many corrections occur. We therefore impose a deterministic *correction cap* $B_j = \lceil 32/(4^j\tau^2) \rceil$ on edge j. This count cap is different from the earlier cap on each sample weight. The specified algorithm stops doing data-dependent work if any count reaches its cap. Privacy is proved for this capped algorithm on every execution, including those that stop. Its complete private transcript then supplies the comparators. On their success event, the regret argument applies even to the prefix ending just after a first alleged cap. That prefix would have $W \ge 32/\tau^2$, contradicting $W \le 16/\tau^2$. Hence the cap is not reached on the success event.

There is also no claim that different blocks remain independent after conditioning on the transcript. To protect one block, the proof treats the other blocks and suitable independent random tapes as auxiliary information, and accounts for every private component that accesses the protected block. Components on different edges can be interleaved; Lemma 3.5 justifies composition in that setting. Raw predictions are not supplied to the process scheduling those components. These details are established before applying the comparator lemma.

For verification, the dependency order is therefore: deterministic caps and sensitivity (Lemmas 6.1 and 5.1); privacy of the entire capped execution (Lemma 6.2); simultaneous comparators, anchor accuracy, and bounded noise (Lemma 6.3); and finally nonexhaustion and accuracy of the released answers (Lemma 6.4). This order separates a statistical existence statement from the algorithmic claim it is used to prove.

2.5 Shared techniques: persistent potentials and protected corrections

We now make the connection between the two applications quantitative. There are two related uses of hierarchy. The statistical chain makes corrections reliable at a sample cost matched to their frequency. The geometric tree limits how often each corrector is needed while preserving the state learned by its descendants. The final algorithm contains both: every geometric corrector uses the statistical chain. Their common analysis has three ingredients: a persistent comparison object, progress certified by the disclosed outcome, and a budget that is valid for the complete adaptive execution.

One comparison object throughout a trajectory. For statistical block j, privacy supplies one feasible weighting u_j that approximates every selected population answer, with $R_j(u_j) = O(\theta_j^2)$. The regret inequality applies to every fixed feasible comparator, so this one can be chosen in hindsight. For a streaming learner, the comparison object is a prefix vector a. If u is the learned iterate and q the subsequent updates, its residual is $r = a - u$ and its outgoing vector is $y = u + q$. The current input is $x = r + y$. An ancestor correction can rescale all of these virtual vectors. Dividing by the product C of those ancestor multipliers preserves the comparison prefix a/C. Consequently one progress argument covers the entire learner lifetime, rather than restarting whenever an ancestor changes.

The statistical regret bound is a potential decrease. The preceding ADA analysis can be written in a form that exposes its geometric content. Retain its learning rate η, score vector G_j, regularizer R_j, and accounting weight λ_j. For a fixed comparator u_j, define

$$\Psi_j(G_j; u_j) := R_j(u_j) + \max_{w \in \mathcal{W}_j} \{\eta\langle G_j, w\rangle - R_j(w)\} - \eta\langle G_j, u_j\rangle.$$

The maximization is at least its value at u_j, so $\Psi_j \geq 0$. At initialization $G_j = 0$, it equals $R_j(u_j)$. Thus the weighted potential $\mathcal{P} = \sum_j \lambda_j \Psi_j$ starts at most one under the parameter choices above.

Lemma 5.3 proves that a score increment g satisfies

$$\Psi_j(G_j + g; u_j) - \Psi_j(G_j; u_j) \leq -\eta\langle u_j - w_j, g\rangle + \frac{\eta^2}{2}\|g\|_\infty^2.$$

For a certified edge-j disagreement, the paired gains have total weighted comparator gain at least $\lambda_j\tau/2$ and weighted squared magnitude at most $5\lambda_j/4$. With $\eta = \tau/4$,

$$\mathcal{P}^+ - \mathcal{P} \leq -\frac{11}{128}\lambda_j\tau^2 \leq -\frac{1}{16}\lambda_j\tau^2.$$

This proves the same bound $\sum_j \lambda_j T_j \leq 16/\tau^2$ as the regret calculation. The potential is analytical; computing the unknown u_j is unnecessary. An individual model's potential can increase when a neighbor corrects, but the weighted total decreases at every certified paired update. This is the precise sense in which the hierarchy coordinates progress across levels.

The geometric potential and the meaning of a correction. The streaming potential is the residual energy $E = \|a - u\|^2$. The ideal prediction omits one cross term:

$$\|x\|^2 - (\|r\|^2 + \|y\|^2) = 2\langle r, y\rangle.$$

With sufficient margins for the estimates used in the proposal, a rejection certifies a nonzero cross term of the indicated sign. Updating u by a suitable signed multiple γy gives

$$E^+ = E - 2\gamma\langle r, y\rangle + \gamma^2\|y\|^2.$$

The first-order improvement dominates the quadratic cost, producing a multiplicative decrease. Section 2.6 derives its size and the permitted interval of coefficients. In normalized coordinates the same calculation survives ancestor rescalings. The implemented proof adds the cost of virtual guard coordinates and uses an energy envelope to accommodate it.

The two progress calculations therefore share an improvement term minus a quadratic update cost. Their certificates and their targets are different. ADA obtains improvement by canceling the common population answer in two neighboring comparator gains; streaming obtains it from a squared-norm identity. Neither calculation says that all future queries will be easier. Each bounds the aggregate number of corrections against its persistent target.

A hierarchy must preserve what has already been learned. In ADA, score functions are retained through the entire interaction. Updating one endpoint invalidates earlier comparisons involving that endpoint, so the algorithm restarts its scan, but retains all learned scores. In streaming, correcting a parent changes its outgoing vector by a common scalar. The child retains its decomposition in normalized coordinates instead of beginning a new learner lifetime. These are different operations implementing the same requirement: rechecking validity must not unnecessarily erase accumulated progress.

The correction certificate is also an information boundary. Every sign used for learning comes from a protected comparison. Examining a raw empirical prediction after a protected test would create an extra dependence on the sample. Likewise, a streaming correction coefficient cannot be chosen by inspecting unreleased row measurements. The permitted transcript contains the signed test outcomes and numerical estimates used by the actual updates. Privacy accounts for the entire capped interaction, including accepted tests; progress subsequently proves that its caps are not exhausted.

How the two hierarchies compose. The standalone ADA mechanism releases a number on every query. For streaming we use the same internal learner but expose a richer interface: tests of proposed intervals and occasional numerical answers. Internal disagreements among statistical levels and external rejections of a geometric prediction have separate counters. The former are bounded by $\mathcal{P}$; the latter will be bounded by residual energy and the number of learner lifetimes per bank. Lemma 8.2 accounts for both within one protected transcript.

This interface is a statistical contribution for arbitrary bounded queries. Its norm-estimation realization uses compact linear measurements, as explained next. The subsequent geometric analysis proves that each realization receives a workload for which this statistical bound is affordable. Thus the shared framework connects the proofs through an implemented mechanism and through compatible progress inequalities.

The representation that makes composition space efficient. The ADA learner retains one accumulated score per sample point and updates it from the current query evaluations. It does not need the query's description or access to the unknown population mean. This evaluation interface is important for space: the streaming algorithm can supply the same evaluations from short row seeds and maintained linear counters, without reconstructing its current vector.

Concretely, a statistical sample point is a seed specifying a sign row $b \in \{-1, 1\}^n$, and D is uniform on the seeds. A query is specified by a vector v and a positive scale c. The stored row counter is $y_b = \langle b, v\rangle$; a sparse update $v_i \leftarrow v_i + \Delta$ changes it by Δb_i. Different sample rows are independently sampled. Within a single row, only a bounded number of signs need to be independent, allowing a short seed. Independence between rows supplies the ADA sample assumption; independence within a row supplies its moment bounds.

For pairwise independent unbiased signs,

$$\mathbb{E}_b\langle b, v\rangle^2 = \sum_i v_i^2 + \sum_{i\neq j} v_i v_j \mathbb{E}[b_i b_j] = \|v\|_2^2 =: F.$$

Thus a norm is a population statistic of the row distribution. This identity holds for every fixed vector. It does not imply that the empirical average of reused rows is accurate on adaptively chosen vectors. Population accuracy for the actual interaction is supplied by the statistical hierarchy.

Making the query bounded without a polynomial accuracy loss. The squared measurement need not lie in $[0,1]$. If c is within a constant factor of F, use

$$f_{v,c}(b) = \min\left\{\frac{\langle b,v\rangle^2}{Lc}, 1\right\}.$$

Division puts typical measurements on a controlled scale. Clipping makes the query bounded even when a measurement is unusually large. The parameter L balances the bias removed by a larger clipping threshold against the finer ADA accuracy needed when multiplying the answer back by Lc.

Let $Z = \langle b, v\rangle$. With $2s$-wise independent signs, $\mathbb{E}Z^{2s} \le (2s)^s F^s$. Choose $s = \Theta(\log(1/\varepsilon))$ and $L = 16s$. For $c \ge F/2$,

$$0 \le F - Lc\, Df_{v,c} = \mathbb{E}[(Z^2 - Lc)_+] \le \frac{\mathbb{E}Z^{2s}}{(Lc)^{s-1}} \le 2s\, 4^{-(s-1)} F \le \varepsilon F/8.$$

If also $c \le 2F$ and a estimates $Df_{v,c}$ to additive error α, then

$$|Lc\, a - F| \le 2LF\alpha + \varepsilon F/8.$$

Taking α to be a sufficiently small constant multiple of ε/L gives the desired relative accuracy. Rounding evaluations to a finite grid uses another small part of this error budget. These steps incur logarithmic, rather than polynomial, losses in accuracy.

Fresh scales for numerical queries. For a bounded sequence of K numerical norm queries, reserve a separate small group of rows for each query. A median of averages of squared measurements in that group provides a constant-factor scale c_t. The group is fresh in an informational sense: its measurements have influenced no earlier answer. Its rows can all be chosen at initialization, and its counters maintained from the start; neither replay nor a newly chosen measurement is required.

The coarse groups are independent of the shared ADA sample. In the ADA reduction their seeds are initial auxiliary information for the analyst. The current vector and coarse scale then define a valid bounded query, even if the scale happens to be inaccurate. We apply the ADA guarantee to that entire interaction and intersect its success event with the separate coarse-estimation success event. We do not condition the ADA analysis on coarse success or assume that current vectors are independent of the reused rows (Lemma 10.4).

The resulting row count is

$$\widetilde{O}\left(K + \frac{\sqrt{K}}{\varepsilon} + \varepsilon^{-2}\right).$$

Hence $K = \widetilde{O}(\varepsilon^{-2})$ adaptive numerical queries fit within $\widetilde{O}(\varepsilon^{-2})$ rows. An unrestricted stream can contain far more queries, however. We must use the additional information already present in the streaming learner's predictions.

The common validation interface. A client that has a candidate answer supplies a *band*, meaning an interval $[l,u]$ proposed to contain Df. With statistical tolerance $a > 0$, a band test returns one of the certificates

$$\begin{aligned} &\mathsf{ACCEPT}: && Df \in [l-a, u+a],\\ &\mathsf{HIGH}: && Df \ge u - a,\\ &\mathsf{LOW}: && Df \le l + a. \end{aligned}$$

The latter two are rejections. The overlapping margins allow uncertainty near a boundary. A numerical request instead asks for an additive-a estimate of Df itself.

Before either kind of request, the internal weighted-consensus hierarchy prepares an accurate root prediction as before. For a band test, only a private comparison with the two endpoints is exposed. One persistent outer AboveThreshold phase serves many accepted tests; a rejection ends that phase. These outer phases are additional to the internal edge phases. Thus external rejections, counted by J, and internal weighted-learner corrections, counted by the T_j, are different events covered by the same privacy analysis.

If M bounds all band tests and R bounds numerical requests, Lemma 8.2 uses

$$\widetilde{O}\left(\frac{\sqrt{J+R+1}}{a}+\frac{1}{a^2}\right)$$

samples, with M only in logarithmic factors. Negative outcomes do reveal information; the phase guarantee accounts for their joint disclosure. The certificate for a terminal rejection is valid even if that rejection exceeds the declared budget. A later progress argument can therefore rule out exhaustion without assuming it away.

Why both sets of techniques are needed for the streaming bound. A *norm bank* contains one shared sign-row sample, its statistical validation state, and coarse groups reserved for numerical requests. On a check, the proposed norm estimate supplies the scale, so successful checks do not consume fresh coarse groups. A poor proposal may have a poor scale; Lemma 12.2 uses global moment inequalities to validate or reject it without assuming that it is already accurate. The lemma also accommodates small regularization and rounded counters.

For local norm accuracy η, the bank's row count is bounded by

$$\widetilde{O}\left(J+R+\frac{\sqrt{J+R+1}}{\eta}+\eta^{-2}\right).$$

The streaming learner and its persistent recursion will establish $J+R=\widetilde{O}(\eta^{-2})$ for each bank. Substitution gives $\widetilde{O}(\eta^{-2})$ rows. This is where the statistical improvement affects the streaming exponent. Even with the same correction count, the direct private-empirical calculation of Section 2.2 would contribute $\widetilde{O}(\sqrt{\eta^{-2}}/\eta^2)=\widetilde{O}(\eta^{-3})$ to such a validation strategy. Conversely, without a geometric bound on $J+R$, the improved statistical mechanism alone does not give a small streaming bank.

The resulting algorithm therefore has two coupled learners: sample weights inside each bank, and a vector decomposition outside it. Their progress arguments control different corrections, and the protected transcript is their interface. The following subsections explain the outer learner, how it preserves descendants, and why its implemented requests satisfy that interface.

2.6 Streaming: a quadratic learner for the missing prefix

We follow the estimator–corrector–learner viewpoint of Gribelyuk et al. [9, Section 1.2.1]. We first explain why this framework can make progress without recovering an old stream prefix, and then explain how our statistical mechanism protects the information used to make that progress. For this discussion we work in exact arithmetic and temporarily set aside the auxiliary coordinates used to control precision.

Starting a new sketch in the middle of the stream. Suppose the current frequency vector can be written as $x = a + q$, where a consists of all updates up to some time t_0, and q consists of the updates after t_0. We call a the *prefix* and q the *suffix*. During this discussion, a is fixed and q evolves as more updates arrive.

Consider two sketch matrices. The older matrix A has seen the prefix, so we can retain Aa and maintain $Ax = Aa + Aq$. A fresh matrix B, initialized just after time t_0, can maintain Bq from zero. If we also knew Ba, then we could form $Bx = Ba + Bq$ and use the new sketch to estimate $\|x\|^2$. However, B was not present when the prefix arrived, and we cannot replay that part of the stream. Thus Ba is precisely the information that the new sketch is missing.

The basic idea is that either we can estimate $\|a + q\|^2$ without knowing Ba, or the current suffix reveals information that helps us learn the missing prefix. This is the starting point of the framework in [9].

An inaccurate estimate gives a useful direction. First try $\|a\|^2 + \|q\|^2$ as an estimate of $\|a + q\|^2$. The first term can be estimated using the old sketch and the second using the new one. If q is nearly orthogonal to a, then this estimate is accurate, even though q need not tell us much about a. On the other hand, the identity

$$\|a + q\|^2 = \|a\|^2 + 2\langle a, q\rangle + \|q\|^2$$

shows that a large error can occur only when the inner product is large. In that case q has a nontrivial alignment with a, and we can use a suitable multiple of q to approximate part of a. For example, orthogonal unit vectors give an exact estimate of two, whereas $a = q$ with $\|a\| = 1$ gives the estimate two for a true answer of four. In the latter case the query direction also points directly toward the unknown prefix.

The distinction is deterministic: an arbitrary adaptive suffix either has a small cross term, or gives a direction along which learning is possible. We do not assume that the adversary chooses random directions or that the prefix and suffix are independent.

The estimator, corrector, and learner. To isolate this idea, first suppose that we can accurately estimate the norm of every vector represented by the new sketch B. It is helpful to think temporarily of B as the identity matrix. This is a simplifying assumption for the explanation; the recursion below supplies these estimates without storing the vectors. Our immediate task is to answer the evolving queries $\|a + q\|^2$ while limiting the corrective information released by the old sketch.

The algorithm has three roles:

1. The *estimator* proposes an answer to the current norm query using what has already been learned about a.
2. The *corrector* checks whether this proposed answer is accurate. If it rejects the proposal, it supplies a reliable answer for the current query and the sign needed for a learning step.
3. The *learner* uses a rejected proposal to improve its approximation to a, making future proposals more accurate.

These are roles within one algorithm, not three independent copies of the stream. In our implementation, the corrector is the norm bank from Section 2.5; its older measurement matrix plays the role of A.

The learner maintains an *iterate* u, initially zero, representing the part of the prefix that it has learned. Define

$$r = a - u, \qquad y = u + q, \qquad x = r + y.$$

We call r the *residual*, because it is the part of the prefix not yet explained by u. We call y the *outgoing vector*, because its norm will be estimated by a recursive child. The estimator's ideal prediction is

$$\|a-u\|^2+\|u+q\|^2.$$

As sanity checks, when $u=0$ this is the original prediction $\|a\|^2+\|q\|^2$, and when $u=a$ it is exactly the desired quantity $\|a+q\|^2$. For a general iterate, the missing term is

$$\|x\|^2-(\|a-u\|^2+\|u+q\|^2)=2\langle a-u,u+q\rangle.$$

Thus the same reasoning applies after learning has begun: either the residual is nearly orthogonal to the outgoing vector, or the outgoing vector provides another useful direction for learning.

What is actually stored and returned. The algorithm does not store a, u, or q as dense vectors. It maintains linear measurements, such as Au and Bu, that can be updated when u changes. A correction of the form $u^+=u+\alpha(u+q)$ is implementable from measurements of u and q already available to the algorithm. Starting from $u=0$, the iterate is a linear combination of suffix vectors on which the estimator was corrected.

Write $E=\|a-u\|^2$ and $Q=\|u+q\|^2$. A protected numerical request initializes an estimate P of E. We keep this value as a *residual cache*: ordinary suffix updates change q but not $a-u$, so the estimate can be reused until the learner changes u. The new sketch, and eventually the child, supplies an estimate $\widehat{Q}$ of Q. The actual proposal is $S=P+\widehat{Q}$.

The corrector tests S against the current input norm. If it accepts, the algorithm returns S. Otherwise, the algorithm returns the corrector's protected numerical estimate of $\|x\|^2$ for this query, updates u, and obtains a new residual estimate for future queries. In particular, it does not keep asking the child to recompute the same query until the proposal is accepted. Each learner makes at most one correction in a physical round.

Why a correction improves the learner. Following [9], use the squared distance $\|a-u\|^2$ as the *progress measure*. If the proposal is inaccurate, then $a-u$ and $u+q$ must be far enough from orthogonal to permit a decrease in this distance. Geometrically, the learner moves u toward $-q$ or away from $-q$. The direction away from $-q$ is $u+q$: if $\langle a-u,u+q\rangle>0$, moving a short distance in that direction brings u closer to a; if the inner product is negative, the learner moves in the opposite direction.

There is one important issue in applying this intuition. The corrector sees the approximate proposal S, not the exact sum $E+Q$. Its rejection threshold must therefore leave room for the errors in both P and $\widehat{Q}$. Our thresholds ensure that every rejection certifies a genuine discrepancy in $E+Q$, with the sign indicated by the corrector (Lemma 16.4). A rejection caused only by the allowed estimation errors would not justify a learning step.

To see the decrease explicitly, let $0<\eta\le 1/100$ be the local accuracy parameter. Suppose that a rejection certifies

$$|\|x\|^2-(E+Q)|\ge(\eta/2)(E+Q)>0.$$

Then $E,Q>0$, and expanding the left-hand side gives

$$|\langle r,y\rangle|\ge\frac{\eta}{4}(E+Q)\ge\frac{\eta}{2}\sqrt{EQ}.$$

Let s be the sign of this inner product. Set $\alpha=st\sqrt{E/Q}$, where $\eta/32\le t\le\eta/8$, and update $u^+=u+\alpha y$. The new residual energy satisfies

$$E^+=\|a-u^+\|^2=\|r-\alpha y\|^2$$

$$= E - 2\alpha\langle r, y\rangle + \alpha^2 Q$$
$$\leq E(1 - \eta t + t^2) \leq (1 - \eta^2/64)E.$$

Thus every correction makes multiplicative progress. We use an interval of valid step sizes because the algorithm has approximate norm estimates and finite precision. It chooses α from P, $\widehat{Q}$, and the protected sign, and then rounds it; it does not require the inner product itself. Here α is a geometric correction coefficient, not the accuracy parameter of the ADA result. Lemma 13.1 also shows that $1/2 \leq 1 + \alpha \leq 3/2$, a fact used in the recursion.

Robustness and the number of corrections. The corrector still has to check every proposal. Each check can reveal information about its old sketch, even when the proposal is accepted, so a bound on corrections alone is not an adversarial robustness argument. The norm bank accounts for the full adaptive sequence of checks. Its statistical cost depends polynomially on the number of rejections and numerical requests, and only logarithmically on the total number of checks.

This is where we combine the geometric framework with our ADA techniques. The geometry limits how often the estimator needs a correction; the reweighting and validation hierarchy makes those corrections reliable using a small reused sample. In the simplified setting, a learner uses one numerical request to initialize P, and two more at each rejection: one for the current answer and one for the updated residual cache. If there are $\widetilde{O}(\eta^{-2})$ corrections, the norm-bank bound from Section 2.5 is therefore $\widetilde{O}(\eta^{-2})$ rows.

To understand the suggested correction count, after j corrections with a fixed prefix and no auxiliary perturbations, the preceding calculation gives

$$E_j \leq (1 - \eta^2/64)^j E_0 \leq \exp(-j\eta^2/64)E_0.$$

If every correction must start with energy at least $E_{\min} > 0$, then a run with at least one correction contains at most $1 + (64/\eta^2)\log(E_0/E_{\min})$ corrections. We still have to justify such a floor and preserve the same learning target in the recursive algorithm. The integrality of the top-level frequency vector does not by itself supply a floor for its real-valued recursive subproblems. Sections 2.8 and 2.9 explain how the auxiliary coordinates supply the needed bounds while preserving the progress argument.

Finally, we have so far assumed that the new sketch can accurately estimate $\|u+q\|^2$ throughout the interaction. This is again a streaming norm-estimation problem, and the natural approach is to solve it recursively. The next subsection explains how to do so without paying for a new child every time the learner updates u.

2.7 Streaming: a persistent hierarchy of learners

Recursive estimator. To extend the simplified setting to a full streaming algorithm, we must remove the assumption that the new sketch can answer every query accurately. Recall that the estimator uses $\|a - u\|^2 + \|u + q\|^2$. We have already explained how to maintain the first term. Estimating the second term is a problem of the same form on a shorter stream: learn a prefix of the outgoing vector while processing its subsequent updates. We therefore use another estimator, corrector, and learner as a *child* of the current one, and continue recursively until the relevant time interval contains a single physical update.

One caveat is that the iterate u may change while this child is processing its updates. The recursion in [9] handles such a change by starting a new child block, meaning a new interval of processing with a newly initialized learner. A parent corrector must then pay for the corrections

in all those child lifetimes. Consequently, bounding the corrections of each individual learner does not by itself give the same bound for one corrector. Our goal is to keep the existing child and the progress it has already made, even when its parent changes u.

A correction preserves the child's problem after normalization. Our update has a special algebraic form:

$$y^+ = u^+ + q = u + \alpha(u + q) + q = (1 + \alpha)y.$$

The child's entire current input is multiplied by the positive scalar $c = 1 + \alpha$. Suppose its decomposition before this change is $y = a_{\text{ch}} + q_{\text{ch}}$ with iterate u_{ch}. Multiply all three vectors $a_{\text{ch}}, q_{\text{ch}}, u_{\text{ch}}$ by c, and multiply its cached squared residual by c^2. This preserves both the decomposition and all relative geometric statements. The same operation propagates to deeper descendants.

For a descendant, let C be the product of the correction multipliers of its strict ancestors since its birth. Its normalized prefix a/C is unchanged by these operations. Therefore the child continues learning the same target in normalized coordinates; it does not start a new learning problem each time an ancestor corrects. Lemma 13.2 proves this invariance for every interleaving of updates, corrections, and ancestor rescalings in exact vector semantics. It is the streaming counterpart of retaining the same comparator in Lemma 5.3.

The physical stream is not rescaled. An arriving physical update is still added with its original value. Ancestor corrections act on virtual states and their stored counters, and normalization describes the target used in the proof. Linearity implements a scalar change using counter multiplication, without recovering the represented vectors. Rounding these operations requires the separate stability argument below.

A bank and a learner have different jobs. A learner owns a decomposition and decides whether to correct it. A bank owns a fixed random matrix, row counters, and its ADA state. At a nonleaf learner, their responsibilities are:

Object	Role at the current learner
Parent bank	Already tracks the incoming vector x; checks its candidate and estimates the residual $x - y$ from counters for both vectors.
Current learner	Stores the residual cache and scalar state; forms the candidate from that cache and its child's answer.
Own outgoing bank	Starts when $y = 0$, tracks y, and acts as the parent bank for the current learner's children.
Child learner	Treats y as its incoming vector and recursively returns an estimate of $\|y\|^2$.

The top learner has a permanent incoming-vector bank initialized at time zero. In particular, a learner's own outgoing bank is not a fresh sketch of its old prefix.

Births are determined by time, not by corrections. Fix a binary tree of time intervals before the stream. A learner lives on each interval; only the root-to-leaf path containing the current time is active. Its two children live on the two halves of its interval. Retire an expired child before allocating its successor. The tree depth is $H = O(\log(m + 1))$ for m physical updates. Corrections rescale the active descendants but never change these birth and retirement times.

At a child's birth, its incoming vector can contain an unknown past prefix. Its parent bank already tracks that vector and supplies the initial residual estimate. The child sets its own iterate and suffix to zero, making its outgoing vector zero. Its fresh outgoing bank therefore initializes all

counters at zero and sees every subsequent update to the vector it must track. No fresh matrix is asked to measure an unobserved old prefix. At a leaf, the lifetime contains only one physical update and the current auxiliary sparse updates, so its outgoing norm can be computed from a short explicit list.

Why persistence preserves the correction budget. Let J_* bound the corrections in one learner lifetime. A bank serves at most two successive learner clients, because its owner has two time-tree children. Each client needs one numerical request at birth and, per rejection, one for its current input and one for its refreshed residual. Consequently each bank has budgets

$$J \le 2J_*, \qquad R \le 4J_* + 2.$$

Ancestor corrections create neither new clients nor new numerical initializations. Once $J_* = \widetilde{O}(\eta^{-2})$ is proved, the bank cost from Section 2.5 is therefore $\widetilde{O}(\eta^{-2})$, and at most $H+1$ banks are live at any time. This is the connection between preserving a learning target and preserving the space exponent.

2.8 Streaming: preserving progress with finite precision

At this point the algorithm has a way to learn from inaccurate estimates and a way to preserve that learning across recursive corrections. It remains to implement these operations in small space. Linearity lets us rescale a represented vector by multiplying its stored counters, but those counters must have finite precision. A small rounding error introduced now could grow under later rescalings.

The decrease in residual energy does not immediately resolve this issue: large scaled components may cancel in the actual vector, whereas a stored error need not undergo the same cancellation. Moreover, we need to bound amplification starting at the time of *each* rounding operation. A bound on the product of multipliers from a learner's birth would not control an error introduced after a long contraction period and followed by an expansion. We address this by adding a small fresh coordinate that records each possible interval of subsequent rescalings.

The counter-error invariant. For one scalar counter, write its exact recurrence as $z_{t+1} = a_t z_t + d_t$ and its implemented recurrence as $\widehat{z}_{t+1} = a_t \widehat{z}_t + d_t + \zeta_t$, with $|\zeta_t| \le h_b$. The coefficients and increments are the ones actually selected by the algorithm. If every forward interval product of the a_t has magnitude at most G, then, from an exact initial counter and at most T operations,

$$|\widehat{z}_t - z_t| = \left| \sum_{s<t} \zeta_s \prod_{s<j<t} a_j \right| \le TGh_b.$$

Thus a fixed fine grid suffices once we bound *every* such product. The calculation is deterministic and remains valid for adaptively chosen multipliers (Lemma 13.4).

A fresh coordinate records each possible amplification interval. Reserve for every learner a family of coordinates on which its incoming vectors are always zero. Physical updates use the original coordinates; ancestors use their own disjoint reserved families. Before a query, add a small amount δ_g in a previously unused coordinate of the current learner's family to both u and y. The residual receives the opposite amount, so $r + y = x$ is unchanged. This auxiliary coordinate is a *guard*: it records scaling without changing the norm-estimation target of this learner.

Work in normalized coordinates and let U be an upper envelope for residual energy. Choose the guard amplitude between constant multiples of $\gamma\sqrt{U}$. Because it is orthogonal to the existing

residual, it adds exactly δ_g^2 to that energy. Taking $T\gamma^2 \le 1/64$ makes the total injected energy small. An *epoch* is a group of $\lceil 128/\eta^2 \rceil$ corrections; their contraction permits halving U at the end of each epoch, even after accounting for all guard insertions.

Now consider any interval of the learner's own corrections. The fresh guard just before its first correction is never touched by an incoming update or a later guard. If P is the product of the interval's correction multipliers, its residual coordinate becomes $-\delta_g P$. If U_s was the envelope when it was inserted, the nonincreasing envelopes imply

$$\delta_g^2 P^2 \le E \le U_s, \qquad \delta_g \ge (\gamma/16)\sqrt{U_s} \quad \Longrightarrow \quad P \le 16/\gamma =: B.$$

There is no cancellation in this coordinate. Applying the argument to the guard at the start of each interval gives all interval-product bounds, not just the product from birth.

Why guards cost bits rather than a large vector. The algorithm stores each guard's contribution through its sign-row counters. Its coordinate index is determined by time and depth, and no dense array of guard coordinates is retained. There are only polynomially many possible indices in the stream length and tree depth, so their contribution to seed length is logarithmic. These deterministic virtual coordinates are not new statistical samples and do not refresh a bank's randomness.

Along an active path, scalar multipliers commute. Grouping them by learner gives a global interval bound of the form $G = B^{O(H)}$. We choose γ inverse-polynomial in the operation bound, so $\log G = O(H \log(1/\gamma))$ is polylogarithmic in the stream parameters. The grid h_b can therefore be fine enough to make TGh_b negligible using only polylogarithmically many bits. The same calculation applies to normalization tags and, with squared multipliers, to residual caches. Lemmas 13.3 and 16.1 give the precise bounds.

2.9 Certifying the implemented hierarchy

The statistical and geometric analyses must apply to the states actually used to choose future queries. An argument for accurate unimplemented predictors would not suffice. We explain how the transcript interface, arithmetic bounds, and recursive error margins fit together.

The same information boundary in both applications. Within an ADA hierarchy, updates use queries and protected comparison outcomes, not raw signs obtained after a private test. Within the streaming algorithm, geometric corrections likewise use protected high/low flags and numerical answers, together with the candidate and child estimates already available. Raw row-counter values and raw weighted predictions are never exposed for choosing a correction. This is the same disclosure rule at two levels of the algorithm.

Fix one bank's permitted history and independent auxiliary randomness. Its rounded counter recurrence then defines a function of any hypothetical sign-row seed, not just of the sampled seeds. Squaring, normalizing, and clipping that function gives a legitimate bounded statistical query. The uniform counter-error bound compares its population mean with the intended norm statistic. Thus the actual rounded implementation is covered by the ADA theorem; we do not assert that its counters are exact measurements of some independently chosen rounded vector. The proof of Theorem 1.4 checks this information flow for each bank, including executions that halt at a declared cap.

Exact transcript coupling for finite ADA computations. ADA scores lie on a common rational grid, and the capped projection is computed exactly with rational arithmetic. The finite private-noise sampler is coupled to the ideal sampler until their first transcript disagreement. On the event of agreement, the adaptive analyst sees identical answers and chooses identical subsequent queries. Merely showing that two answers are numerically close would not justify that conclusion: an analyst can react discontinuously to a small change. Section 7 accounts for the probability of the first disagreement.

Shadow vectors follow the implemented choices. For streaming, define *shadow vectors* by exact arithmetic using the coefficients and guard amplitudes actually chosen by the rounded algorithm. They describe what its counters are intended to measure. They are not vectors in another execution with different decisions. The counter-error invariant compares stored counters with these vectors, while the geometric proof analyzes their residuals and corrections. This choice makes the arithmetic and adaptive analyses refer to the same interaction.

Why small additive errors fit a relative guarantee. For a nonzero physical integer vector, $\|x\| \geq 1$. A recursive virtual vector can nevertheless be much smaller because of cancellation. The root's fresh guard, inserted before the round's recursive evaluations, supplies a nonzero coordinate in every descendant input. It cannot be canceled by that round's other updates, and at most H factors, each at least $1/2$, can shrink it. The resulting norm floor $\nu > 0$ has a logarithmic-size description (Lemma 16.3). It allows the absolute counter and cache errors to fit within relative-error margins.

A separate, tiny regularizing coordinate in a norm-bank query keeps its statistical normalization positive. Its known squared magnitude is explicitly included in the certificates. This regularizer and the learner's guards have distinct purposes. At a physical zero vector, the algorithm instead returns exact zero using a short independent fingerprint, a linear checksum analyzed by coupling to the true zero decisions (Lemma 15.1). This avoids relying on a relative-error margin where none exists.

Guards smaller than their storage grid are omitted. The proof does not assume that every attempted guard is representable. A genuine correction implies $\|r\| \geq \eta\nu/16$; the energy envelope then forces its prescribed guard to exceed the chosen grid. Consequently every correction has the fresh nonzero coordinate needed for the amplification argument (Lemma 16.2).

The recursive margin and the order of the induction. At height ℓ, the child and residual cache make the candidate accurate to roughly $8\ell\eta$ relative to $E+Q$. The bank tests it with threshold $(8\ell + 4)\eta$. The gap between these values pays for statistical and arithmetic errors. Acceptance implies an accurate returned candidate; rejection has the sign and magnitude needed for geometric contraction. We take $\eta = \Theta(\varepsilon/(H+1)^2)$, with the small constant specified in the algorithm, so errors over the full depth fit within ε.

The proof proceeds chronologically and, inside one round, from leaves to root. Previously proved product bounds control the counters before the next test. Those counter bounds justify the current statistical certificate. The child's established accuracy and the threshold margin justify the current correction. That correction's contraction and fresh guard establish the next energy and product bounds, which then justify the residual refresh. This order avoids assuming the stability of a new correction before proving its geometric premise. It is the streaming analogue of proving privacy for the capped ADA execution before using comparators to exclude cap exhaustion.

Closing the correction and space bounds. The normalized residual envelope halves after $O(\eta^{-2})$ corrections. A correction requires a residual at least a fixed fraction of the target norm,

and the target floor and ancestor-product bounds give an explicit lower bound on that normalized residual. Only polylogarithmically many envelope halvings are possible before a further correction is excluded. Therefore $J_* = \widetilde{O}(\eta^{-2})$ throughout a learner's lifetime, including the prefix that would end in a first cap violation. Lemma 16.5 combines this count with the preceding invariants and rules out such a violation.

There are at most $H + 1$ live banks, each has $\widetilde{O}(\eta^{-2})$ rows by the shared statistical mechanism, and each row and scalar uses polylogarithmically many bits by the amplification bounds. With $H = O(\log(m + 1))$ and the stated choice of η, their product is $\widetilde{O}(\varepsilon^{-2})$ space. Thus statistical learning controls the cost of reliable validation, geometric learning controls how often that validation releases corrective information, and persistence and precision control how those costs compose in an actual stream.

Organization of the proofs. The two hierarchies are analyzed separately so that their invariants remain explicit. The statistical part proves the protected correction interface, and the streaming part proves that a persistent geometric hierarchy meets its workload requirements. The link is both a common potential-based design and an actual nesting of the statistical learner inside every geometric corrector.

Part I proves the statistical result and the more general band-validation interface: private population comparators, capped quadratic learning, weighted consensus, and finite computation. Part II instantiates that same interface with sign-row evaluations, obtaining the bounded-query theorem and norm banks. Part III supplies the geometric correction count, persistent recursion, and arithmetic invariants needed to run those banks throughout an unrestricted adaptive stream. The two main progress arguments are in Sections 6 and 16; their connection is the common validation budget made explicit above.

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Part I
Population-accurate adaptive data analysis

The first instance of the hierarchical learner–corrector approach uses a chain of sample-weight learners. Its fixed comparison objects are supplied by privacy. Lemma 5.3 makes the quadratic progress measure explicit; the subsequent hierarchy uses it to coordinate the correction budgets of adjacent levels.

This part proves Theorem 1.2 for arbitrary distributions and bounded statistical queries. Its construction does not assume that queries arise from vectors or streams. We develop the mechanism in three stages: privacy supplies population comparison weights; quadratic online learning turns these weights into a correction bound; and the hierarchy uses that bound to allocate privacy across levels. The final validation interface will also be used in the streaming construction.

3 Probability and privacy preliminaries

The relevant object is the entire transcript, including comparisons that return a negative result. A future query may depend on any permitted earlier response, so protecting only the numerical answers would not justify a population guarantee for the subsequent queries. We use replacement differential privacy separately for each independent sample block.

Three facts will be used in different roles. Martingale concentration controls reweighted samples after fixing an adaptive transcript. AboveThreshold protects a sequence of comparisons at a cost that is charged when a phase ends in a positive answer. Concurrent composition allows these phases to be interleaved across the hierarchy. The last point matters because an update at one level can change the next query issued to a still-active phase at another level.

We prove the specific tools needed below. The Laplace and AboveThreshold constructions are standard [5]. The concurrent composition argument is the finite-transcript randomized-response simulation of Vadhan and Wang [8], included here to make the interleaving step explicit.

Lemma 3.1 (Bounded martingale differences)**.** *Let integrable Y_i be adapted to a filtration $\mathcal{F}_i$. Suppose that, conditionally on $\mathcal{F}_{i-1}$, Y_i lies in an interval of length at most c_i, where c_i is deterministic. Then, for $u > 0$,*

$$\mathbb{P}\left[\left|\sum_{i=1}^{r}(Y_i - \mathbb{E}[Y_i \mid \mathcal{F}_{i-1}])\right| \geq u\right] \leq 2\exp\left(-\frac{2u^2}{\sum_i c_i^2}\right).$$

The corresponding one-sided bound omits the factor two.

Proof. For any random variable Y in an interval of length c, put $\psi(t) = \ln \mathbb{E}e^{tY}$. Under the exponentially tilted distribution, $\psi''(t)$ is the variance of Y, at most $c^2/4$: the squared distance from the interval midpoint is at most $c^2/4$, and variance is the minimum mean squared distance from a constant. Integrating twice gives $\ln \mathbb{E}e^{t(Y-\mathbb{E}Y)} \leq t^2c^2/8$. This proof also applies to conditional distributions. Iterating conditional expectation therefore bounds the moment generating function of the sum by $\exp(t^2 \sum_i c_i^2/8)$. Exponential Markov, optimized at $t = 4u/\sum_i c_i^2$, proves the upper-tail bound. Apply the same argument to the negative sum and take a union bound. □

Definition 3.2 (Replacement differential privacy)**.** Two ordered datasets of the same size are neighbors if they differ in one coordinate. A randomized channel M is (θ, δ)-DP if for all neighboring S, S' and every measurable event E,

$$\mathbb{P}[M(S) \in E] \leq e^{\theta}\mathbb{P}[M(S') \in E] + \delta.$$

For an interactive mechanism this requirement applies to the complete view of every adaptive analyst whose auxiliary input is independent of the protected dataset. Pure privacy means $\delta = 0$.

Lemma 3.3 (Postprocessing and independent auxiliary input)**.** *Data-independent randomized postprocessing preserves (θ, δ)-DP. If Z is independent of S and, for every fixed z, the channel $M_z(S)$ is (θ, δ)-DP with the same parameters, then $(Z, M_Z(S))$ is (θ, δ)-DP.*

Proof. An event after postprocessing has conditional probability $g(y) \in [0, 1]$ given the original output y. Integrate the DP inequality for the events $\{g > v\}$ over $v \in [0, 1]$. The resulting additive term is δ, proving the first assertion. For the second, apply privacy to the section E_z of an output event and integrate over the common, data-independent law of Z. □

Write $\mathrm{Lap}(b)$ for the distribution with density $(2b)^{-1}e^{-|z|/b}$. Its tail is

$$\mathbb{P}[|Z| > u] = e^{-u/b} \quad (u \geq 0). \tag{2}$$

Lemma 3.4 (Laplace and AboveThreshold)**.** *Suppose a real-valued query has replacement sensitivity at most Δ. Adding independent $\mathrm{Lap}(b)$ noise is pure Δ/b-DP. The following interactive mechanism has the same privacy parameter for any fixed finite horizon: draw $Z \sim \mathrm{Lap}(2b)$, and, on each adaptively supplied sensitivity-Δ query g, output whether*

$$g(S) + V \geq T + Z, \qquad V \sim \mathrm{Lap}(4b)$$

with a fresh independent V. Stop after the first positive outcome. Here T is fixed independently of the protected dataset. Any prefix, including an all-negative stopped prefix, obeys the same guarantee.

Proof. For the numerical release, the density ratio at any output is at most $\exp(|g(S) - g(S')|/b) \leq e^{\Delta/b}$ by the triangle inequality.

For AboveThreshold fix the analyst's coins. Before the first positive outcome, its successive queries are fixed by the all-negative history. Consider a transcript whose first positive outcome is at position r. Fix the first $r - 1$ check noises. Map a successful pair (Z, V_r) for dataset S to $(Z + \Delta, V_r + 2\Delta)$ for S'. Every earlier negative test remains negative because its score increases by at most Δ. The final test remains positive because its score decreases by at most Δ. This translation is injective, and the ratio of the original density to the translated density is at most

$$\exp\left(\frac{\Delta}{2b} + \frac{2\Delta}{4b}\right) = e^{\Delta/b}.$$

Integrating the fixed earlier noises proves the bound for that transcript. For an all-negative transcript translate only Z by Δ; the density ratio is at most $e^{\Delta/(2b)}$. Summing over transcript events and integrating the analyst's coins proves privacy. Stopping based only on exposed outputs and independent coins is postprocessing of the interaction, so it is covered as well. □

Lemma 3.5 (Concurrent composition, finite transcripts)**.** *Let at most C interactive components have mutually independent random tapes. Each component is pure e_0-DP, where $e_0 > 0$, for arbitrary adaptive inputs in its allowed query set. An orchestrator with no access to the protected dataset may adaptively interleave them and postprocess their outputs. Suppose the interaction has a finite horizon and finite message sets. For every $0 < \delta < 1$, its complete transcript is (θ, δ)-DP for*

$$\theta = e_0\sqrt{2C\ln(1/\delta)} + \frac{Ce_0^2}{2}. \tag{3}$$

Components that are never invoked may be included in C.

Proof. Fix neighboring datasets S_0, S_1. We first construct a simulator for one component that works for every input strategy. Put $c = e^{e_0}$. For a local history v containing its received messages and its emitted outputs, let $P_b(v)$ be the probability of those outputs on S_b when the indicated inputs are supplied along that history. Pure privacy gives

$$c^{-1}P_1(v) \le P_0(v) \le cP_1(v).$$

Indeed, an analyst can supply that fixed input sequence and inspect the event of seeing the indicated output prefix. Define

$$R_0(v) = \frac{cP_0(v) - P_1(v)}{c-1}, \qquad R_1(v) = \frac{cP_1(v) - P_0(v)}{c-1}.$$

Both are nonnegative and equal one at the empty history. For any possible next input q, the values on the children (v, q, a) sum, over outputs a, to $R_z(v)$. Hence they specify an interactive probability kernel: in state v, output a with probability $R_z(v, q, a)/R_z(v)$ when the denominator is positive. If it is zero, choose any kernel; that state is unreachable under this simulator.

First sample a bit Z that equals b with probability $c/(1+c)$ and equals $1-b$ otherwise. Then run the simulator with prefix probabilities R_Z. The identities

$$\frac{cR_0(v) + R_1(v)}{1+c} = P_0(v), \qquad \frac{R_0(v) + cR_1(v)}{1+c} = P_1(v)$$

show that the interaction has exactly the original distribution, for every input strategy. The simulator depends on the fixed pair S_0, S_1, but not on which member is the actual dataset or on the analyst's strategy.

Use independent such bits and simulator coins for all C components. The orchestrator's entire concurrent view is now a randomized postprocessing of these C bits. Under S_0, each bit's log likelihood ratio against S_1 is $+e_0$ with probability $c/(1+c)$ and $-e_0$ otherwise. Its mean is

$$e_0 \frac{c-1}{c+1} = e_0 \tanh(e_0/2) \le e_0^2/2.$$

The last inequality follows from $\tanh u \le u$ for $u \ge 0$, which in turn follows by integrating its derivative $1/\cosh^2 u \le 1$. The C privacy losses are independent and have interval length $2e_0$. Lemma 3.1 shows that their sum exceeds (3) with probability at most δ. On the complementary event the likelihood ratio is at most e^θ. Integrating any event therefore proves (θ, δ)-DP for the bit vector. The same argument with S_0, S_1 exchanged gives the reverse inequality. Apply Lemma 3.3 to the simulation. Randomized orchestrators are handled by conditioning on, and then integrating, their independent coins. □

4 Simultaneous population comparators from privacy

The starting point for learning is not that a private transcript makes the empirical mean accurate enough. Instead, it implies that a nearby choice of sample weights is accurate enough. We will compare the learner against these weights in a deterministic regret inequality.

Why should such weights exist? Conditioning on a transcript changes the distribution of the sample points. Inverse conditional density ratios correct that change of measure. We expose the sample in order because the points need not remain independent after conditioning on the transcript. At each step, inverse weighting gives the desired conditional mean, which is the condition needed for martingale concentration. Including the constant query controls the normalizing denominator.

For approximate privacy, inverse density ratios may be large on exceptional events. Clipping them to $[e^{-\theta}, e^{\theta}]$ keeps the variables bounded. The two directed privacy inequalities control the total conditional drift caused by this clipping. The proof then separates this drift from the martingale fluctuations.

The conclusion has two quantitatively different parameters. The fitting error γ costs $O(\gamma^{-2})$ samples up to logarithms when $\theta \le 1$, while the quadratic radius is $O(\theta^2)$. We do not have to set $\theta = O(\gamma)$. This separation permits the hierarchy's upper levels to answer often and its lower levels to have smaller comparator cost.

The weights form a single hindsight comparator for all selected queries. The algorithm never computes them or uses D. The transcript may include independent auxiliary variables of any measurable type, even though the sample domain itself is finite.

Lemma 4.1 (Two-sided posterior reweighting)**.** *Let $X_1, \dots, X_r$ be independent with law D on a finite domain, and let T be a (θ, δ)-DP channel of this ordered sample. Assume T determines measurable queries $f_t(T, \cdot) \in [0,1]$, $t \in [k]$. If $0 < \gamma \le 1$, $0 < \xi < 1$, and*

$$\delta \le \frac{\gamma\xi}{32}, \qquad r \ge 32 e^{2\theta} \gamma^{-2} \ln \frac{4(k+1)}{\xi}, \tag{4}$$

then with probability at least $1 - \xi$ there are weights $v_i \in [e^{-\theta}, e^{\theta}]$, whose normalization $u_i = v_i / \sum_{a=1}^{r} v_a$ satisfies

$$\left| \sum_i u_i f_t(T, X_i) - D f_t(T, \cdot) \right| \le \gamma \quad \text{for every } t \in [k]. \tag{5}$$

On every outcome the weights can be chosen so that

$$r u_i \in [e^{-2\theta}, e^{2\theta}], \qquad R(u) := \frac{r}{2} \sum_i u_i^2 - \frac{1}{2} \le \frac{1}{2} \sinh^2 \theta. \tag{6}$$

In particular $R(u) \le \theta^2$ when $0 \le \theta \le 1$.

Proof. Discard points of zero D-probability and put $C = e^{\theta}$. Let $V_i = (T, X_1, \dots, X_{i-1})$, $J_i = \mathcal{L}(V_i, X_i)$, and $P_i = \mathcal{L}(V_i) \otimes D$. These two joint measures satisfy both directed (θ, δ) inequalities. To verify this, draw an independent prefix p, old row x, replacement row x', and suffix y from their product-D laws. Compare

$$(M(p, x, y), p, x) \quad \text{and} \quad (M(p, x', y), p, x).$$

For fixed p, x, x', y, apply privacy to the transcript section of the desired joint event, keeping the displayed x fixed. Integrating gives the asserted inequalities. The second joint law is P_i because its displayed x is independent of its other coordinates.

Since the sample domain is finite and D is positive on its support, $J_i \ll P_i$. Write $a_i = dJ_i / dP_i$. The two directed inequalities imply

$$\int (a_i - C)_+ \, dP_i \le \delta, \qquad \int (1 - C a_i)_+ \, dP_i \le \delta. \tag{7}$$

For example, the first integral is the difference of the two measures on the measurable set where $a_i > C$; the second is analogous. Choose a measurable version of a_i, and define

$$v_i = \mathrm{clip}_{[C^{-1}, C]}(1 / a_i(V_i, X_i)),$$

taking $v_i = C$ when $a_i = 0$. The latter event has zero J_i-measure. For each value v of V_i, the identity

$$a_i(v,x)\,\mathrm{clip}_{[C^{-1},C]}(1/a_i(v,x)) - 1 = C^{-1}(a_i(v,x) - C)_+ - (1 - Ca_i(v,x))_+ \tag{8}$$

holds, also with the stated convention when $a_i = 0$. Define the nonnegative, V_i-measurable variable

$$d_i = \sum_x D(x)\left[C^{-1}(a_i(V_i,x) - C)_+ + (1 - Ca_i(V_i,x))_+\right].$$

Equation (7) gives $\mathbb{E}d_i \le (1 + C^{-1})\delta \le 2\delta$. Multiplying (8) by any selected query and summing over x gives

$$\mathbb{E}[v_i f_t(T, X_i) \mid V_i] = Df_t(T,\cdot) + e_{i,t}, \qquad |e_{i,t}| \le d_i. \tag{9}$$

This uses only conditional probabilities for a finite-valued X_i; no conditioning on a probability-zero transcript is required. The same equation holds for the constant query $f_0 = 1$.

Let $\overline{d} = r^{-1}\sum_i d_i$. Markov's inequality yields

$$\mathbb{P}[\overline{d} > \gamma/8] \le 16\delta/\gamma \le \xi/2.$$

For each fixed $t \in \{0,\dots,k\}$, use the filtration $\mathcal{F}_i = \sigma(T, X_1, \dots, X_i)$. The random variable $v_i f_t(T, X_i)$ lies in $[0, C]$. By Lemma 3.1, its centered average differs from zero by more than $\gamma/8$ with probability at most $2\exp(-r\gamma^2/(32C^2))$. A union bound over $k+1$ queries and (4) bounds this failure by $\xi/2$.

On the intersection, put $W = r^{-1}\sum_i v_i$ and $V_t' = r^{-1}\sum_i v_i f_t(T, X_i)$. Equation (9) gives $|W - 1| \le \gamma/4$ and $|V_t' - Df_t| \le \gamma/4$. Since $0 \le Df_t \le 1$,

$$\left|\frac{V_t'}{W} - Df_t\right| \le \frac{\gamma/2}{1 - \gamma/4} \le \frac{2\gamma}{3} < \gamma.$$

This proves simultaneous accuracy.

For the radius calculation, take uniform expectation over the indices of any deterministic vector $v \in [a,d]^r$, with $a = e^{-\theta}$, $d = e^{\theta}$, and $m = \mathbb{E}v$. The inequality $(v - a)(v - d) \le 0$ gives

$$\frac{\mathbb{E}v^2}{m^2} \le \frac{a+d}{m} - \frac{ad}{m^2} \le \frac{(a+d)^2}{4ad}.$$

The last inequality follows by completing the square in $1/m$. As $R(u) = \frac{1}{2}(\mathbb{E}v^2/m^2 - 1)$, this is $R(u) \le \frac{1}{2}\sinh^2\theta$. Also $a \le m \le d$, proving the coordinate bounds. Finally, convexity of sinh on $[0,1]$ implies $\sinh\theta \le \theta\sinh 1$, and $\sinh 1 < \sqrt{2}$ (for example $e < 3$ and $e^{-1} > 1/3$ give $\sinh 1 < 4/3 < \sqrt{2}$). Thus $R(u) \le \theta^2$. □

5 Capped quadratic predictors

We now need an executable learner that can compete with the comparison weights supplied by the preceding section. There are two requirements: small regret against a comparator close to uniform, and small sensitivity when a single sample point changes. The capped quadratic predictor provides both through the same regularizer.

The regularizer measures squared distance from uniform. Thus the $O(\theta^2)$ radius from Lemma 4.1 becomes a small initial cost in the regret inequality. The coordinate cap limits the influence of a changed record even when the cumulative scores are highly adaptive. It is important that the

sensitivity calculation allow all the other weights to renormalize; a bound on one weight alone would not suffice.

The predictor retains only one accumulated score for each sample point. It does not store the preceding query functions. Its weight vector can be recomputed from these scores whenever needed. This is also where the evaluation-access property becomes useful: every update can be performed from the current query's sample evaluations.

For a block with r samples and a fixed cap $A \geq 1$, let

$$\mathcal{W} = \{w \in \mathbb{R}^r_{\geq 0} : \textstyle\sum_i w_i = 1,\ w_i \leq A/r\}, \qquad R(w) = \tfrac{r}{2}\|w\|_2^2 - \tfrac{1}{2}.$$

The regularizer is nonnegative, and its value at the uniform vector is zero. Given a cumulative score vector G and a learning rate $\eta > 0$, the predictor uses

$$w(G) = \arg\max_{w \in \mathcal{W}} \{\eta \langle G, w\rangle - R(w)\}. \tag{10}$$

Strict concavity gives uniqueness.

Lemma 5.1 (Projection formula and sensitivity)**.** *The weights in* (10) *have the form*

$$w_i(G) = \frac{1}{r}\operatorname{clip}_{[0,A]}(\eta G_i + c), \qquad \sum_i \operatorname{clip}_{[0,A]}(\eta G_i + c) = r.$$

Suppose $G_i = G(X_i)$ for a fixed function G and the prediction is $\mu(S) = \sum_i w_i(G(S))f(X_i)$ for a fixed $f : \mathcal{U} \to [0,1]$. Then μ has replacement sensitivity at most $2A/r$.

Proof. The normalization equation has a solution by continuity and monotonicity, since its left side ranges from 0 to rA. To check optimality, put $z_i = \eta G_i + c$. The objective's gradient at the proposed w is $z_i - \operatorname{clip}_{[0,A]}(z_i) - c$ in coordinate i. For any feasible u, the term $-c$ cancels in its inner product with $u - w$. A coordinate with $z_i \leq 0$ contributes $z_i u_i \leq 0$; one with $z_i \geq A$ contributes $(z_i - A)(u_i - A/r) \leq 0$; and every other coordinate contributes zero. The inner product is therefore nonpositive, proving optimality by concavity. If shifts are not unique, the resulting weight vector is nevertheless unique.

Replace row i. Every unchanged-coordinate weight moves in the same direction, determined by the change of the common normalizing shift. Their signed mass change sums to the opposite of the change at row i. Consequently the total variation distance between the two weight vectors equals $|w_i - w_i'| \leq A/r$. For a fixed evaluation vector in $[0,1]^r$, changing weights therefore changes the prediction by at most A/r. Changing the evaluation at row i costs at most another $w_i' \leq A/r$. Adding the two bounds proves the claim. □

Lemma 5.2 (Quadratic regret)**.** *Let $G_0 = 0$, $w_t = w(G_{t-1})$, and $G_t = G_{t-1} + g_t$ for any finite sequence of gain vectors, including adaptively chosen ones. For every $u \in \mathcal{W}$,*

$$\sum_t \langle u - w_t, g_t\rangle \leq \frac{R(u)}{\eta} + \frac{\eta}{2}\sum_t \|g_t\|_\infty^2. \tag{11}$$

This is a pathwise statement, so u may be chosen after the sequence.

Proof. Let $\Phi(G) = \max_{w \in \mathcal{W}}\{\langle G, w\rangle - R(w)/\eta\}$. If w, w' maximize at $G, G+g$, first-order optimality at w and the quadratic expansion of R imply

$$\Phi(G+g) - \Phi(G) - \langle w, g\rangle \leq \langle w' - w, g\rangle - \frac{r}{2\eta}\|w' - w\|_2^2$$

$$\leq \frac{\eta}{2r}\|g\|_2^2 \leq \frac{\eta}{2}\|g\|_\infty^2.$$

The middle inequality follows by completing the square. Telescope, use $\Phi(0) = 0$, and lower-bound the final potential by $\langle \sum_t g_t, u\rangle - R(u)/\eta$. Rearranging proves (11). □

Lemma 5.3 (Potential for a persistent comparator)**.** *Use $\mathcal{W}, R, w(G)$ from this section and fix any $u \in \mathcal{W}$. Define*

$$\Psi(G; u) = R(u) + \max_{w\in\mathcal{W}}\{\eta\langle G, w\rangle - R(w)\} - \eta\langle G, u\rangle.$$

Then $\Psi(G; u) \geq 0$, $\Psi(0; u) = R(u)$, and every increment $g \in \mathbb{R}^r$ satisfies

$$\Psi(G + g; u) - \Psi(G; u) \leq -\eta\langle u - w(G), g\rangle + \frac{\eta^2}{2}\|g\|_\infty^2. \tag{12}$$

The inequality holds for all u and all increments on each realized trajectory, so u may be selected after that trajectory is fixed.

Proof. Feasibility of u makes its value a lower bound for the maximization, which proves nonnegativity. The uniform weighting belongs to $\mathcal{W}$ and minimizes R with value zero, giving $\Psi(0; u) = R(u)$. Let $w = w(G)$ and $w' = w(G + g)$. Optimality at w implies $\langle \eta G - \nabla R(w), w' - w\rangle \leq 0$. The exact quadratic expansion

$$R(w') = R(w) + \langle \nabla R(w), w' - w\rangle + \frac{r}{2}\|w' - w\|_2^2$$

therefore shows that the change in the maximized objective is at most

$$\eta\langle w, g\rangle + \eta\langle w' - w, g\rangle - \frac{r}{2}\|w' - w\|_2^2 \leq \eta\langle w, g\rangle + \frac{\eta^2}{2r}\|g\|_2^2.$$

The last inequality follows by completing the square in $w' - w$. Subtracting $\eta\langle g, u\rangle$ and using $\|g\|_2^2 \leq r\|g\|_\infty^2$ proves the step bound. Every step is a deterministic inequality for every feasible u, which proves the final assertion. □

The potential is a gap between a regularized optimum and the value of one fixed comparator. The cap can make it piecewise quadratic as a function of G; the proof uses the quadratic regularizer and does not assume that this gap is a squared Euclidean distance. It is not stored by the mechanism. Its role is to make the progress of coupled learners explicit, in the same way that residual energy will expose the progress of a geometric learner.

6 The weighted-consensus hierarchy

The preceding sections give a stable learner and a small comparator, but neither supplies the unknown population answer to use as a label. The hierarchy replaces that label by comparisons between learners. Only its final predictor is required to be directly population accurate.

There are two directions of influence. Accuracy is transferred from the empirical anchor toward the root through successful comparisons. Corrections update both endpoints of an edge, so learning can also move a more private predictor in response to a less private one. The weighted regret argument accounts for both directions. We do not assume that an intermediate level is correct every time it is queried.

The analysis is organized so that privacy does not depend on an unproved progress bound. We first specify a mechanism that stops at fixed caps. We prove privacy for all its executions, construct

population comparators for the resulting transcript, and then show that those comparators exclude the first cap failure. A final unchanged sweep proves accuracy of the answer actually returned.

We now specify every parameter and every branch of the mechanism in Theorem 1.2. All sample blocks below are independent and drawn at initialization. Random tapes of different private components are also independent. The use of cap 8, instead of a different exponential cap at each level, makes exact rational projection possible.

6.1 Parameters

The parameter choices express a balance between how private a level is and how often it is allowed to disclose information. The depth p makes the last correction budget constant. The comparison tolerance τ leaves room to sum p adjacent errors. The smaller scale h pays for comparator inaccuracies, private noises, and rounding. The potential weights λ_j increase geometrically, the privacy parameters θ_j decrease, and the correction caps B_j decrease at the matching rate.

The notation in this part is local: η is the quadratic learner's learning rate, H denotes the empirical anchor, and δ is a privacy failure parameter. In the streaming part these letters have separately defined roles. All shared interfaces explicitly specify their own accuracy and failure parameters.

Let p be the least positive integer satisfying

$$4^{p-1}\left(\frac{\alpha}{4p}\right)^2 \geq 1, \qquad \tau = \frac{\alpha}{4p}, \quad h = \frac{\tau}{64}, \quad \eta = \frac{\tau}{4}. \tag{13}$$

Then $p \geq 2$ and

$$1 \leq 4^{p-1}\tau^2 < 4. \tag{14}$$

The strict upper bound follows from minimality at $p-1$: $4^{p-1}\tau^2 < 4(p-1)^2/p^2$. Also $p = O(\log_2(2/\alpha))$. For a direct bound, put $L = \log_2(16/\alpha) \geq 7$. The integer $p' = \lceil 4L \rceil$ satisfies $2^{p'-1} \geq 2^{4L-1} \geq 4p'/\alpha$; the last inequality follows from $2^{3L+1} \geq 4L+1$, which holds at $L = 7$ and is preserved as L increases. Thus $p \leq p'$.

There are p weighted blocks S_j of sizes n_j, indexed by $j = 0, \ldots, p-1$, and an empirical anchor H of size n_H. Define

$$\lambda_j = 4^j, \qquad \theta_j = (p\lambda_j)^{-1/2}, \qquad B_j = \left\lceil \frac{32}{\lambda_j \tau^2} \right\rceil, \tag{15}$$

$$\theta_H = h/8, \qquad \xi = \rho/(p+2), \qquad \delta = h\xi/64. \tag{16}$$

Edge j joins model j to model $j+1$, with model p meaning the anchor. Let

$$Q = 10(p+1)\left(k + \sum_{j=0}^{p-1} B_j + 1\right), \qquad \ell = \lceil \log_2(Q/\xi) \rceil, \qquad b = \frac{h}{4\ell}, \tag{17}$$

$$L_\delta = \lceil \log_2(1/\delta) \rceil, \qquad L_g = \lceil \log_2(8(k+1)/\xi) \rceil. \tag{18}$$

The integer logarithms dominate the corresponding natural logarithms. The maximum numbers of protected components are

$$C_0 = k + B_0, \qquad C_j = B_{j-1} + B_j \ (1 \leq j < p), \qquad C_H = B_{p-1}. \tag{19}$$

Take the least integers satisfying

$$n_j \geq \max\left\{ \frac{128}{b}\sqrt{p\lambda_j C_j L_\delta}, \frac{1024 L_g}{h^2} \right\}, \quad 0 \leq j < p, \tag{20}$$

$$n_H \geq \max\left\{\frac{128}{hb}\sqrt{C_H L_\delta}, \frac{1024 L_g}{h^2}\right\}. \tag{21}$$

All thresholds, caps, and sizes are selected before seeing data or answers. These formulas are also explicit integer prescriptions: a ceiling involving a square root can be found by comparing integer squares after clearing rational denominators.

6.2 State, private tests, and scheduler

A query is processed against the current states of all models. An accepted edge provides a certificate only for those states. For example, an edge between models 0 and 1 may accept, after which the edge between 1 and 2 updates model 1. The first certificate is then stale. The scheduler below restarts at edge zero after every update to ensure that the final answer is supported by a whole scan with no state changes.

The two signed comparisons serve another purpose: the update sign comes from a private output. Using the sign of the unperturbed difference, even after privately deciding to correct, would reveal additional information not covered by the comparison transcript.

Weighted model j uses the capped simplex and regularizer of Section 5 with $r = n_j$ and $A = 8$. Its cumulative function G_j starts at zero. In an implementation only the evaluations $G_j(X_{j,i})$ are stored. On the current query f, write

$$w_j = w(G_j(S_j)), \quad \mu_j = \sum_i w_{j,i} f(X_{j,i}), \qquad \mu_p = Hf = \frac{1}{n_H}\sum_{i=1}^{n_H} f(H_i).$$

Each edge has a counter starting at zero and a current AboveThreshold phase with public threshold τ. A phase draws one threshold noise from $\mathrm{Lap}(2b)$ and retains it. When edge j is called, set $d = \mu_{j+1} - \mu_j$, recomputing both predictions from the current states. Test d and then $-d$, with independent fresh $\mathrm{Lap}(4b)$ check noise for every tested score. If both tests are negative, the edge accepts. If the first test is positive, set $q = +1$; if the second is positive, set $q = -1$. A positive outcome closes the phase and causes the updates

$$G_j \leftarrow G_j + qf, \qquad G_{j+1} \leftarrow G_{j+1} - qf/4 \quad \text{if } j+1 < p. \tag{22}$$

The anchor is never updated. Increment the edge counter. If it now equals B_j, abort immediately after these updates, without starting a new phase or making another comparison. Otherwise start a fresh independent phase on that edge. Both updates use the crossing direction determined by the private test; no raw sign is used.

For each external query, sweep edges $0, 1, \ldots, p-1$. Whenever a correction occurs, perform the updates and restart the sweep at edge zero. When a whole sweep accepts, return

$$a = \mathcal{R}_h(\mathrm{clip}_{[0,1]}(\mu_0 + Z)), \qquad Z \sim \mathrm{Lap}(b), \tag{23}$$

where $\mathcal{R}_h$ rounds to the nearest point of

$$\mathcal{G}_h = \{0, h, 2h, \ldots, \lfloor 1/h \rfloor h, 1\},$$

with a fixed tie-breaking rule. The noise Z is fresh and independent. If any edge has aborted, return zero for the current and all subsequent external queries and perform no further data-dependent work.

Lemma 6.1 (Deterministic execution bounds)**.** *Every execution terminates on every query. Across all k queries, there are at most $p(k + \sum_j B_j)$ edge calls, at most $\sum_j B_j$ threshold draws, at most twice as many check draws as edge calls, and at most k numerical-release draws. Their total is less than Q. Edge j uses at most B_j AboveThreshold phases.*

Proof. Every sweep that does not produce an answer has a correction, and there are at most $\sum_j B_j$ corrections before abort. A sweep has at most p edge calls, each with at most two checks. On edge j, before a cap is reached, the number of phases is one plus the number of its corrections. If no cap is reached this number is at most B_j; if a cap is reached, no final new phase is drawn and the number is exactly B_j. The numerical releases occur at most once per external query. These bounds sum to less than the expression for Q in (17). □

6.3 Privacy of the complete capped execution

For a fixed protected block, the other blocks and their independent random tapes can be treated as auxiliary information. Once the augmented history is fixed, the cumulative score functions are also fixed, because their updates use only queries and protected flags. The sensitivity lemma therefore applies to the next prediction.

This does not assert that the blocks remain independent after an adaptive transcript is observed. Rather, the proof describes, for each protected block separately, an interactive orchestrator whose initial auxiliary randomness is independent of that block. The orchestrator can interleave its incident private phases, but it never reads the block's raw predictions.

Define the augmented transcript T to contain every individual negative or positive test flag, the sign associated with each positive flag, and all rounded public answers. Query indices and edge identities are determined by this transcript and the scheduler. The transcript does not contain raw predictions, threshold noises, or check noises. Fixing the analyst's independent coins makes every external query a function of the preceding public answers, and hence of T.

Lemma 6.2 (Blockwise transcript privacy)**.** *The following joint channels are private. For weighted block S_j, include in the channel all other sample blocks, the analyst's coins, and all component tapes except those of its incident edges and, when $j = 0$, the root numerical releases. Together with T, this is a (θ_j, δ)-DP channel of S_j. For the anchor, include all weighted blocks, analyst coins, and nonincident tapes; together with T, the result is (θ_H, δ)-DP in H.*

Proof. Fix the stated auxiliary information for the protected block. These are complete preassigned independent tapes, including unused entries; we do not condition on a data-dependent list of consumed draws. At any fixed augmented history, all G_i are fixed functions because their updates use only the queries and the private flags in that history. Lemma 5.1 gives sensitivity $\Delta_j = 16/n_j$ for μ_j.

Only the two incident edges (one for the root) directly access μ_j. Every other prediction, test, and update can be computed from the fixed auxiliary information and the history. Thus an orchestrator independent of S_j can simulate this work. At an incident test it supplies the component with a *function* of S_j of sensitivity at most Δ_j; it does not request or receive that function's raw value. The neighbor's prediction is computable from the fixed auxiliary information and history. An incident component need retain only its own threshold noise; its current query function includes the currently applicable cumulative score function. This is precisely an allowed interactive input. Only for $j = 0$ must numerical root releases also be protected.

By Lemmas 3.4 and 6.1, there are at most C_j independently randomized pure-DP components,

each with privacy parameter at most

$$e_0 = \frac{16}{n_j b} \le \frac{\theta_j}{8\sqrt{C_j L_\delta}}.$$

Unused component slots are retained as dormant components. Since the augmented output alphabet and the horizon are finite, there are only finitely many possible augmented histories under the fixed auxiliary information. Consequently only finitely many input query functions are needed by this orchestrator. Lemma 3.5 therefore applies, even though the sensitivity queries themselves are real-valued. Its privacy parameter is at most

$$\frac{\sqrt{2}}{8}\theta_j + \frac{\theta_j^2}{128 L_\delta} < \theta_j,$$

using $\theta_j \le 1$ and $L_\delta \ge 1$.

For the anchor, the empirical prediction has sensitivity $1/n_H$; we use the upper bound $2/n_H$. Only the final-edge phases directly access it. Equation (21) gives

$$\frac{2}{n_H b} \le \frac{h}{64\sqrt{C_H L_\delta}} = \frac{\theta_H}{8\sqrt{C_H L_\delta}},$$

so the same composition proof applies.

Each fixed auxiliary variable is independent of the protected sample block. Lemma 3.3 therefore proves privacy of its joint release with the transcript. We use a different auxiliary channel for each block; we do not assert privacy after simultaneously revealing all these auxiliary channels. □

6.4 Simultaneous comparators, anchor accuracy, and noise

We collect the probabilistic statements before making the deterministic progress argument. The useful event provides a comparator for every weighted block, accuracy of the empirical anchor on all actual queries, and bounded values of every consumed noise. The queries here are those selected by the adaptive execution, including one that might stop at a cap. The event is not an oblivious concentration statement applied to a reused sample.

Lemma 6.3 (A common success event)**.** *With probability at least $1-\rho$, all of the following hold:*

(i) For each weighted block there is $u_j \in \mathcal{W}_j$ such that $|u_j f_t - D f_t| \le h$ for every actual external query, and

$$P := \sum_{j=0}^{p-1} \lambda_j R_j(u_j) \le 1.$$

(ii) For every actual external query, $|H f_t - D f_t| < 2h$.

(iii) Every threshold, check, and numerical-release noise drawn in the execution has magnitude at most h.

Proof. Apply Lemma 4.1 separately to the joint channel in Lemma 6.2 for each block, with $\gamma = h$ and failure probability ξ. Its size hypothesis follows from (20)–(21), since $\theta_j, \theta_H \le 1$, $32e^2 < 1024$, and $L_g \ge \ln(4(k+1)/\xi)$. Its approximate privacy hypothesis follows from $\delta = h\xi/64$. For weighted blocks, $n_j u_{j,i} \le e^{2\theta_j} \le e^2 < 8$, so the comparator is feasible. Moreover

$$P \le \sum_j \lambda_j \theta_j^2 = \sum_j 1/p = 1.$$

For the anchor, its comparator coordinates obey $n_H u_i \in [e^{-2\theta_H}, e^{2\theta_H}]$. Thus its distance in total variation from the uniform weights is at most $\frac{1}{2}(e^{2\theta_H} - 1) \le 2\theta_H = h/4$. For the inequality, $0 \le 2\theta_H < 1$ and convexity gives $e^z - 1 \le (e-1)z < 2z$ on $[0,1]$. Since $f_t \in [0,1]$, $|Hf_t - Df_t| \le h + h/4 < 2h$.

The largest noise scale is $4b = h/\ell$. Each newly consumed draw, conditionally on the previous execution, has its prescribed fresh law. By (2) its probability of exceeding h is at most $e^{-\ell}$. Lemma 6.1 and a union bound over at most Q draws give failure at most $Qe^{-\ell} \le \xi$. There are p weighted comparator events, one anchor event, and one noise event. Each has an unconditional failure bound before we take their union bound, which is $(p+2)\xi = \rho$. □

6.5 A decreasing potential prevents exhaustion

A correction has no known population label, and an individual model can be moved by either adjacent edge. For this reason, we do not try to prove that each model's error decreases at every update. Instead, combine the persistent comparison potentials of all models with weights λ_j. The factors 4 and 1/4 make the unknown common population answer cancel across an internal edge.

The proof has two separate conclusions. First, each correction contributes enough weighted comparator gain to bound the total number of corrections and rule out exhaustion. Second, an unchanged scan transfers the anchor's accuracy to the current root state. The correction count alone would not establish the latter.

Lemma 6.4 (Correction budget and actual prediction accuracy)**.** *On the event in Lemma 6.3, no correction budget is reached, and every public answer has error less than α. If T_j is the total number of edge-j corrections, then*

$$\sum_j \lambda_j T_j \le 16/\tau^2. \tag{24}$$

Proof. At a correction the signed test score satisfies

$$q(\mu_{j+1} - \mu_j) \ge \tau - 2h,$$

since both the threshold noise and the check noise have magnitude at most h. At an internal edge, the weighted comparator gain from the two updates is exactly

$$\begin{aligned}\lambda_j q(u_j f - \mu_j) - \frac{\lambda_{j+1}}{4} q(u_{j+1} f - \mu_{j+1}) \\ = \lambda_j q(\mu_{j+1} - \mu_j + u_j f - u_{j+1} f) \ge \lambda_j(\tau - 4h) \ge \lambda_j \tau/2.\end{aligned}$$

Here $\lambda_{j+1} = 4\lambda_j$, and the comparators both approximate the same population mean to error h. At the anchor edge, only the weighted parent moves. Its comparator error is at most h, and the anchor error is less than $2h$, so its gain is at least $\lambda_{p-1}(\tau - 5h) \ge \lambda_{p-1}\tau/2$.

Fix the simultaneous comparators and put

$$\mathcal{P} = \sum_{j=0}^{p-1} \lambda_j \Psi_j(G_j; u_j),$$

using Lemma 5.3 separately on every block. Initially $\mathcal{P} = P := \sum_j \lambda_j R_j(u_j) \le 1$, and it is always nonnegative. An internal edge correction has weighted squared gain magnitude at most

$$\lambda_j \|qf\|_\infty^2 + \lambda_{j+1} \| - qf/4\|_\infty^2 \le \lambda_j + \lambda_{j+1}/16 = 5\lambda_j/4.$$

At the last edge this quantity is at most λ_{p-1}, so the same upper bound applies. Combining the comparator gain proved above with (12) gives, at either kind of edge,

$$\mathcal{P}^+ - \mathcal{P} \le -\eta\lambda_j\tau/2 + 5\eta^2\lambda_j/8 = -11\lambda_j\tau^2/128 \le -\lambda_j\tau^2/16,$$

where $\eta = \tau/4$. A test or release that makes no score update does not change this potential. Summing over any executed prefix, with $W = \sum_j \lambda_j T_j$ on that prefix, yields

$$0 \le \mathcal{P} \le P - \tau^2 W/16, \qquad W \le 16P/\tau^2 \le 16/\tau^2.$$

The argument is pathwise and applies also to the prefix ending immediately after a first alleged cap. That cap would give $W \ge \lambda_j B_j \ge 32/\tau^2$, a contradiction. This proves nonexhaustion without assuming it when deriving privacy.

Every query therefore ends in a sweep containing no updates. For each edge, its two negative tests imply $|\mu_{j+1} - \mu_j| \le \tau + 2h$. Since all predictions refer to the same unchanged vector of model states during this sweep, telescoping and anchor accuracy give

$$|\mu_0 - Df| \le p(\tau + 2h) + 2h.$$

The numerical noise costs at most h. Clipping to $[0, 1]$ cannot increase error from $Df \in [0, 1]$, and the largest gap in $\mathcal{G}_h$ is h, so rounding costs at most $h/2$. Hence

$$|a - Df| \le p(\tau + 2h) + \frac{7}{2}h = \tau\left(p + \frac{p}{32} + \frac{7}{128}\right) < 4p\tau = \alpha.$$

This is a bound on the mechanism's actual adaptive predictions. □

6.6 Sample count

Only the root pays for all k external numerical releases. Every other block pays for phases on its two incident edges. The bound $\lambda_j T_j = O(\tau^{-2})$, together with $\theta_j^{-2} = p\lambda_j$, cancels the depth-dependent factor in its privacy cost. The anchor has an even smaller privacy parameter, but only a constant number of phases. Thus stronger accuracy at the end of the hierarchy does not introduce a second factor of $1/\alpha$ into the root's $\sqrt{k}$ term.

Proof of Theorem 1.2. Accuracy follows from Lemmas 6.3 and 6.4. We verify the sample bound. Since $\lambda_j\tau^2 < 4$, the expression inside each ceiling in (15) exceeds eight, whence

$$B_j \le \frac{36}{\lambda_j\tau^2}, \quad C_j \le \frac{180}{\lambda_j\tau^2}\ (1 \le j < p), \quad C_0 \le k + 36/\tau^2, \quad C_H \le 36.$$

The last inequality also uses $\lambda_{p-1}\tau^2 \ge 1$. Moreover $\sum_j B_j \le 48/\tau^2$, by summing the geometric series.

Let $\Lambda = \log_2(2^{30}k/(\alpha\rho))$. The depth bound gives $p = O(\Lambda)$, and the displayed formulas show $\ell, L_\delta, L_g = O(\Lambda)$. More explicitly, $h^{-1} = 256p/\alpha$, $\xi^{-1} = (p+2)/\rho$, $\delta^{-1} = 64h^{-1}\xi^{-1}$, and $Q \le 10(p+1)(k + 48/\tau^2 + 1)$; taking logarithms establishes all these bounds. Since $b^{-1} = 256\ell/\tau$, the root size is

$$O\left(\Lambda^{3/2}\sqrt{p}\left[\frac{\sqrt{k}}{\tau} + \frac{1}{\tau^2}\right] + \frac{\Lambda}{\tau^2} + 1\right).$$

Each of the other $p-1$ weighted blocks has size $O(\Lambda^{3/2}\sqrt{p}/\tau^2+\Lambda/\tau^2+1)$, and the anchor has size $O(\Lambda^{3/2}/\tau^2+1)$. Summing and substituting $\tau=\alpha/(4p)$ gives

$$\begin{aligned} N &= O\left(\Lambda^{3/2}\left[\frac{\sqrt{pk}}{\tau}+\frac{p^{3/2}}{\tau^2}\right]\right) \\ &= O\left(\Lambda^{3/2}\left[\frac{p^{3/2}\sqrt{k}}{\alpha}+\frac{p^{7/2}}{\alpha^2}\right]\right) = O\left(\Lambda^5\left[\frac{\sqrt{k}}{\alpha}+\frac{1}{\alpha^2}\right]\right). \end{aligned}$$

This proves (1). A block stores its cumulative scores and computes its weights from them; corrections use the current query evaluations. Besides these arrays, the scheduler needs one counter and one current threshold noise per edge, the current query index, and temporary scalar arithmetic. Thus only a constant number of scalar values per sample and $O(p)$ other scalar states are necessary. □

7 A finite-bit implementation of the hierarchy

The preceding analysis concerns an ideal randomized mechanism. There are two distinct implementation questions. The score and projection computations must have bounded bit length, and the continuous private noises must be replaced by finite random-bit procedures. We handle the first by exact rational arithmetic and the second by coupling.

Approximation of every answer by a small numerical error is not enough for this purpose: an adaptive analyst could choose a very different next query after an arbitrarily small answer change. Instead, public answers already lie on a fixed output grid. We couple the finite and ideal mechanisms so that their actual flags and rounded answers agree, except on a small failure event. Until the first disagreement, the analyst submits the same queries to both executions. This is the invariant that transfers the population guarantee.

In this section α and ρ are dyadic rationals. This entails no asymptotic restriction: replace any requested parameter by the largest power of two not exceeding it. We assume query evaluations are supplied on a common grid $\{0,1/D,\dots,1\}$, where D is a power of two. Section 10 constructs these evaluations exactly. There is no need to round an accumulated score after an update.

Lemma 7.1 (Exact finite-size state)**.** *Put $\Lambda=\log_2(2^{30}k/(\alpha\rho))$. Apart from random noises, the hierarchy can be implemented using $O(N(\Lambda+\log_2 D))$ bits for scores, weights, query evaluations, and scheduler state, in addition to sample representations and the workspace needed to evaluate the queries. Every projection and prediction can be computed exactly by rational arithmetic within this space bound.*

Proof. Let $B_*=\sum_j B_j=O(\tau^{-2})$. Each accumulated score is a multiple of $1/(4D)$, with absolute value at most B_*, because at most B_* corrections occur and each gain has magnitude at most one. Its numerator therefore requires $O(\log D+\log B_*)$ bits. All sizes, counts, and rational parameters other than square roots in the definition of sample sizes have $O(\Lambda)$-bit representations. Square roots are needed only for selecting integer sample sizes; integer-square comparisons compute those sizes with $O(\Lambda)$ bits. In particular $\log N$, $\log B_*$, and the bit lengths of η,τ,h,b are $O(\Lambda)$.

In one projection, express every ηG_i as r_i/T with a common positive denominator T having $O(\Lambda+\log D)$ bits. Sort the $2n_j$ breakpoints $-r_i/T$ and $8-r_i/T$ for the normalizing shift. Between consecutive breakpoints, let I be the active coordinates and J those at the upper cap. If I is nonempty, the candidate normalizing shift is

$$c=\frac{n_j-8|J|-\sum_{i\in I}r_i/T}{|I|}.$$

It is a solution exactly when it belongs to that interval, endpoints included. If I is empty, the interval is a solution precisely when $8|J| = n_j$; an endpoint suffices. The normalization function is continuous and ranges from zero to $8n_j$, so some interval or endpoint succeeds. This algorithm uses arrays of length $O(n_j)$.

All nonconstant coordinates of the resulting weight vector share a denominator dividing $n_j T|I|$; zero and capped coordinates can be expressed with the same denominator. If I is empty, denominator n_j suffices. Each numerator and denominator has $O(\Lambda + \log D)$ bits. Taking the inner product with the grid query values adds a factor D to a common denominator and increases numerator size by only $O(\log n_j)$ bits. The prediction therefore has the same asymptotic bit length. Differences between two freshly computed predictions require only the product of two such denominators, again of this bit length. These predictions are not accumulated in future states; only grid gains are accumulated. Consequently no denominator growth occurs across corrections or queries. Storing a constant number of these arrays and $O(p)$ counters proves the bound. □

We next supply a finite random-bit sampler and quantify its effect on the whole adaptive execution. This avoids a claim that arbitrary floating-point perturbations preserve differential privacy.

Lemma 7.2 (Finite approximation of a Laplace draw)**.** *Let b, ζ, a be positive rationals with $b < 1$, $\zeta < 1$, and $a < 1/4$, and let $\sigma \in \{b, 2b, 4b\}$. Choose an integer R with*

$$2^{-R} \le \min\{a/2,\ a\zeta/(32b)\}. \tag{25}$$

There is a finite algorithm using R independent unbiased random bits that outputs a dyadic $\widetilde{Z}$ and admits a coupling to $Z \sim \mathrm{Lap}(\sigma)$ with

$$\mathbb{P}[|Z - \widetilde{Z}| > \zeta] \le 2a.$$

If the parameters and R have total bit length at most B and $R + \log_2(1/\zeta) + \log_2(1/a) + \log_2(1/b) = O(B)$, the output uses $O(B)$ bits and the sampler uses $O(B^2)$ workspace.

Proof. Let J be uniform on $\{0, \dots, 2^R - 1\}$, and put $U_* = (J + 1/2)2^{-R}$. Couple this with a continuous uniform $U \in [0, 1]$ by making U uniform within the selected dyadic interval. Thus $|U - U_*| \le 2^{-R-1}$. The inverse Laplace distribution function is

$$F_\sigma^{-1}(u) = \begin{cases} \sigma \ln(2u), & u \le 1/2, \\ -\sigma \ln(2(1-u)), & u \ge 1/2. \end{cases}$$

On $U \in [a, 1-a]$, the segment joining U to U_* lies in $[a/2, 1 - a/2]$. On this segment the derivative of the inverse is at most $2\sigma/a \le 8b/a$. Equation (25) therefore gives

$$|F_\sigma^{-1}(U) - F_\sigma^{-1}(U_*)| \le (8b/a)2^{-R-1} \le \zeta/8.$$

Compute $F_\sigma^{-1}(U_*)$ to absolute error at most $\zeta/2$ and round the result to a dyadic grid of spacing at most $\zeta/4$. The resulting error from $Z = F_\sigma^{-1}(U)$ is less than ζ on that event. Its complement has probability $2a$.

For completeness, the logarithm computation can be implemented with finite rational arithmetic as follows. For a positive rational x appearing inside the logarithm, write $x = 2^j v$ with $1 \le v < 2$. Here $|j| \le R + 1$. Put $y = (v - 1)/(v + 1) \in [0, 1/3)$. The identity

$$\ln v = 2 \sum_{r=0}^{M} \frac{y^{2r+1}}{2r+1} + E_M, \qquad |E_M| \le \frac{2|y|^{2M+3}}{1 - y^2} \le 3^{-2M-2}$$

follows by integrating the geometric series for $2/(1-y^2)$. Use the same series at $y = 1/3$ to approximate $\ln 2$, and combine $j \ln 2 + \ln v$. Choose $M = O(R + \log(1/\zeta) + \log(1/b) + 1)$ so that the two tails, multiplied by $\sigma(|j| + 1)$, sum to at most $\zeta/4$. Such M exists because the tails decrease geometrically; it can be found by rational comparisons with powers of 3. If $y = A_0/B_0$, all terms through M have a common denominator dividing $B_0^{2M+1}(2M+1)!$. Its bit length is $O(MB + M\log(M+1)) = O(B^2)$. Exact summation with this denominator, followed by dyadic rounding, thus uses $O(B^2)$ workspace and attains the required error. Finally $|F_\sigma^{-1}(U_*)| \le 4b(R+1)\ln 2$, so the rounded output itself needs only $O(B)$ bits. □

Lemma 7.3 (Coupling the entire adaptive transcript)**.** *Fix the hierarchy's parameters, a dataset, and an adaptive analyst. Suppose its query evaluations and its deterministic arithmetic are implemented exactly, as in Lemma 7.1. Let $0 < \nu < 1/10$. Replace every Laplace draw with the sampler in Lemma 7.2, taking*

$$a = \frac{\nu}{16Q}, \qquad 0 < \zeta \le \frac{\nu bh}{128Q}, \tag{26}$$

where ζ is the largest dyadic number satisfying its bound and R is the least integer satisfying (25). There is a coupling in which all internal flags and public answers of the finite and ideal executions agree with probability at least $1-\nu$. This bound is uniform over datasets and analysts.

Proof. Fix the analyst's independent coins. At each next noise operation, use the uniform coupling in Lemma 7.2; independent uniforms for different operations give the required independent component tapes. There are at most Q operations, so the probability that any draw has approximation error exceeding ζ is at most $2Qa = \nu/8$.

Until the first differing flag or public answer, the two executions receive exactly the same queries and have exactly the same cumulative scores and raw predictions. Their current threshold noises differ by at most ζ on the complementary noise event. If a check gives different flags, its ideal noisy score must lie within 2ζ of its ideal threshold. Conditional on all preceding coupled randomness, the current ideal check noise is a fresh $\mathrm{Lap}(4b)$ variable, whose density is at most $1/(8b)$. This event has conditional probability at most $\zeta/(2b)$. This argument conditions on the actual previous threshold noises, so it remains valid after arbitrarily many uses of a persistent threshold.

For a numerical release, the map $z \mapsto \mathcal{R}_h(\mathrm{clip}_{[0,1]}(z))$ has fewer than $2/h$ decision boundaries, all lying in $(0,1)$. If the rounded releases differ when their noises differ by at most ζ, the ideal value is within ζ of one of these boundaries. Its fresh $\mathrm{Lap}(b)$ density is at most $1/(2b)$, giving conditional probability at most $2\zeta/(bh)$. This also upper-bounds the check estimate because $h < 1$. A union bound over the first possible disagreement at at most Q operations gives probability at most

$$\frac{\nu}{8} + \frac{2Q\zeta}{bh} \le \frac{\nu}{8} + \frac{\nu}{64} < \nu.$$

The density bounds are applied before conditioning on the current sampler's approximation event; intersection with that event can only decrease the probability. Thus the union bound is valid for these adaptive operations. Integrating the analyst's coins completes the proof. □

Corollary 7.4 (Finite-bit population mechanism)**.** *For grid queries and dyadic α, ρ, the hierarchy has a finite-bit implementation whose accuracy failure is at most $\rho + \nu$. In addition to sample representations and evaluation workspace, its memory is*

$$O\left(N(\Lambda + \log D) + (\Lambda + \log(1/\nu))^2\right).$$

Proof. The ideal hierarchy has failure at most ρ by Theorem 1.2. On the coupling event in Lemma 7.3, all answers and all selected queries coincide, so its accuracy guarantee transfers without any continuity assumption on the analyst. A union bound gives $\rho + \nu$. The formulas for Q, b, h imply that their logarithmic magnitudes are $O(\Lambda)$. Equations (26) and (25) therefore have $B = O(\Lambda + \log(1/\nu))$. Combine Lemmas 7.1 and 7.2. Current threshold approximations need $O(B)$ bits per edge, and $p = O(\Lambda)$, so they fit the stated quadratic additional term. □

What the statistical construction establishes. Theorem 1.2 and Corollary 7.4 together provide an implementable population-accurate mechanism in the evaluation-access model. The proof has accounted for the actual predictions, all comparison outcomes, the stopping rule, and finite random-bit sampling. No property of quadratic norm queries has been used. The next section derives a useful validation interface from the same construction.

8 Adaptive validation with a bounded number of corrections

The numerical query theorem charges the root for k releases. Some applications instead supply a candidate interval on every round and need a numerical answer only when a prediction changes. The distinction is substantive: charging every successful check as a numerical release would lose the benefit of a small correction budget.

The same hierarchy still prepares a population-accurate root prediction. We change only how that prediction is exposed. A persistent AboveThreshold phase checks the upper and lower endpoints of proposed bands. Negative outcomes do reveal information, but the privacy bound accounts for the entire phase at once, including all-negative prefixes. A rejection ends the phase and is charged to the rejection budget.

A corrector must validate many predictions while releasing relatively few corrections. The next lemma extends Theorem 1.2 to this interface: its sample complexity depends polynomially on the number of corrections and numerical answers, and only logarithmically on the number of accepted predictions.

Definition 8.1 (Statistical validation transcript)**.** A client submits either a band test (f, l, u), with $f : \mathcal{U} \to [0, 1]$ and rational $l \le u$, or a numerical request for Df. A band test returns one of ACCEPT, HIGH, and LOW; either of the last two outcomes is a rejection. A numerical request returns a number in $[0, 1]$. The permitted client transcript consists of these requests and responses, together with an explicit halt symbol if a declared budget is exceeded. The client has no access to sample points, raw empirical predictions, or the mechanism's private state. Each next request is a function of this transcript and initial auxiliary information independent of the protected sample and its private random tape. In the privacy proof, the transcript is augmented by the mechanism's protected internal comparison outcomes; raw predictor values and weights are not included.

Lemma 8.2 (Adaptive band tests)**.** *Let D be an unknown distribution on a finite domain $\mathcal{U}$, let M, J, R be nonnegative integers with $M + R \ge 1$, and let $0 < a, \delta < 1/10$. There is a mechanism with evaluation access that, for every adaptive client in Definition 8.1, handles at most M band tests and R numerical requests, with rejection budget J, using*

$$O\left(\log^5 \frac{C(M + R + J + 2)}{a\delta} \left[\frac{\sqrt{J + R + 1}}{a} + \frac{1}{a^2}\right]\right)$$

independent samples from D. With probability at least $1-\delta$, simultaneously over all operations it completes,

$$\begin{aligned} \mathsf{ACCEPT} &\implies Df \in [l-a, u+a], \\ \mathsf{HIGH} &\implies Df \geq u-a, \\ \mathsf{LOW} &\implies Df \leq l+a, \end{aligned}$$

and a numerical answer differs from Df by at most a. The $(J+1)$st rejection may halt the mechanism; the certificates include this rejection. There are no other cap failures on this event. For query values on a common rational grid, the state has polylogarithmically many bits per sample in the operation bounds, $1/a$, $1/\delta$, and the grid denominator, in addition to sample representations, the current request's encoding, and evaluation workspace. The probability guarantee is uniform over D and the client.

Proof. Use the hierarchy of Section 6 with $k = M+R$, ideal failure parameter $\rho = \delta/2$, and accuracy $\alpha = a/32$, first rounding evaluations to a grid with spacing at most $a/64$. Use exactly the parameters in (13)–(16), except replace the component counts and operation horizon by

$$C_0 = B_0 + J + 1 + R, \qquad C_j = B_{j-1} + B_j \ (1 \leq j < p), \quad C_H = B_{p-1}.$$

The operation horizon is

$$Q = 20(p+1)(k + \textstyle\sum_j B_j + J + R + 2).$$

Use this Q in (17) and these component counts in (20)–(21). Before every operation, run the internal correction sweeps to a sweep with no update. Numerical requests use (23). For band tests maintain an additional root AboveThreshold phase with threshold zero. Test $\mu_0 - u$, followed, if negative, by $l - \mu_0$. Two negatives accept. A positive returns its direction, closes the phase, and increments the external rejection count. Open a fresh phase unless that count exceeds J.

At a fixed augmented transcript, l, u and every cumulative score function are fixed. Both outer scores therefore have sensitivity $16/n_0$ by Lemma 5.1. Each entire phase is pure $16/(n_0 b)$-DP by Lemma 3.4; all-negative prefixes are included. There are $B_0 + J + 1 + R$ root components and the displayed numbers of components elsewhere. Their tapes are independent. Applying Lemma 3.5 gives exactly the privacy calculation in Lemma 6.2, with these counts. The horizon Q dominates all internal and outer check noises, threshold noises, and numerical noises. Consequently the proof of Lemma 6.3 applies to all k selected queries, including accepted band tests. The weighted regret inequalities in Lemma 6.4 are unchanged, because outer tests do not update a predictor. They prevent every internal cap and give $|\mu_0 - Df| < a/16$ after allowing for query rounding. Every consumed noise is at most $h < a/1024$ on the same event. A negative upper test gives $\mu_0 \leq u + 2h$, and a positive one gives $\mu_0 \geq u - 2h$; the lower test is analogous. These inequalities imply the stated certificates. Numerical errors are less than a by the same final-sweep calculation.

For the sample count, the summation in the proof of Theorem 1.2 has $J + R + 1$ in place of the root release count k; M enters only its logarithms. The other levels and their caps are unchanged. This gives the displayed bound. Lemmas 7.1 and 7.3 implement the mechanism finitely. The new tests are comparisons with fixed rational endpoints, so their first-disagreement probabilities have the same fresh-Laplace density bound. Assign the coupling failure parameter $\delta/2$. This proves all assertions without an assumption that the external cap is avoided. □

Part II
From statistical queries to norm certificates

This part connects the two main results. We first prove that fixed linear measurements support a bounded number of arbitrary adaptive norm queries. We then construct the more flexible interface needed by a recursive learner: many tests of proposed norm estimates, a bounded number of rejections, and accurate numerical estimates when requested. The latter interface also accounts for rounded measurements. These reductions use the statistical guarantee proved in Part I; they do not yet use a streaming correction bound.

9 The bounded-query model and theorem

The evaluation-access restriction in the ADA theorem is exactly what makes a small linear sketch usable as a statistical sample. One sample point is a row seed; one query evaluation is a bounded function of the corresponding linear measurement. The decoder keeps the ADA scores across rounds but need not see the full query vector. We first isolate this fixed-matrix statement before allowing sketches to be initialized during a stream.

All logarithms written ln are natural; $\log_2$ denotes base two. For an integer $a \geq 1$, write $[a] = \{1, \dots, a\}$. All randomized channels, analyst selection rules, and query evaluations are assumed measurable. Throughout, a failure probability is taken over all private randomness of the estimator and any independently randomized adversary. An adversary may know the algorithm and all its public parameters.

Definition 9.1 (Adaptive fixed-measurement model)**.** An estimator first chooses a private matrix $A \in \{-1, 1\}^{m \times n}$. In round $t \in [K]$, an adversary chooses $x_t \in \mathbb{Z}^n$ using its private randomness and the previously reported estimates. The estimator receives Ax_t and reports an estimate of $F_2(x_t)$. Its decoder may keep private state between rounds. The adversary sees the reported estimates, but not A, the measurements, the decoder state, or internal computation. The adversary's initial randomness and information are independent of the estimator's private randomness.

The corresponding turnstile implementation starts at $x = 0$ and processes updates (i, Δ) with $\Delta \in \mathbb{Z}$ by setting $x_i \leftarrow x_i + \Delta$. At most K queries are made. Between queries the updates are unrestricted and may depend on previous answers. Query processing is atomic. We assume $|x_i| \leq U$ at all times, with $U \geq 1$ an integer; this assumption accounts only for counter bit lengths. In the fixed-measurement model it is enough to impose this bound on the supplied vectors. There is no restriction on stream length.

Theorem 9.2 (Fixed measurements for adaptive second-moment queries)**.** *Let $n, K, U \geq 1$ be integers and $0 < \varepsilon, \beta < 1/10$. There is an estimator in the adaptive fixed-measurement model such that, simultaneously for every $t \in [K]$, with probability at least $1 - \beta$,*

$$\frac{F_2(x_t)}{1+\varepsilon} \leq \widehat{F}_t \leq (1+\varepsilon) F_2(x_t).$$

Put $\mathcal{H} = \log_2(2^{20} K/(\varepsilon\beta))$. For an absolute constant C, its matrix has at most

$$m \leq C\mathcal{H}^7 \left(K + \frac{\sqrt{K}}{\varepsilon} + \frac{1}{\varepsilon^2} \right) \tag{27}$$

rows, all selected before the interaction. It has a finite-bit implementation using at most

$$Cm\left[\log_2(nU+2)+\log_2(n+1)\log_2(2/\varepsilon)+\mathcal{H}^2\right] \tag{28}$$

bits of working memory. The same space bound holds for the turnstile implementation. No restriction on the adversary's computation is needed.

The theorem concerns a fixed linear measurement matrix with a stateful decoder. Its additional scores are counted in the space bound. The theorem imposes no fast update-time requirement. For arbitrary real vectors, the ideal arithmetic construction below still gives the measurement and accuracy guarantees; the bit statement uses the integer and magnitude assumptions above.

The reduction uses Theorem 1.2 exactly in the evaluation-access model of Definition 1.1. We now show how to supply its evaluations from fixed sign measurements, without exposing the queried vector to the decoder.

10 Fixed sign measurements for adaptive F_2 queries

Three issues separate the identity $\mathbb{E}\langle r,x\rangle^2=\|x\|^2$ from a space-efficient adaptive norm estimator. The squared measurement is unbounded; the scale $\|x\|^2$ is unknown; and a fully independent row would require n random bits. Clipping at a coarse scale addresses the first two issues, and limited independence addresses the third.

We use independent rows as the ADA sample. Within one row, only enough independence to reproduce the required moments is needed. These are different requirements: compact seeds do not replace the independence between the sample points on which the statistical theorem relies.

10.1 Compact sign rows and their moments

We describe the finite distribution used for the statistical samples. Let $d=\max\{1,\lceil\log_2 n\rceil\}$ and fix n distinct elements $z_1,\dots,z_n$ of $\mathbb{F}_{2^d}$. Fix a nonzero $\mathbb{F}_2$-linear functional $\ell_0:\mathbb{F}_{2^d}\to\mathbb{F}_2$, for example one coordinate in a polynomial basis. For an integer $q\ge 1$, choose independent uniform coefficients $a_0,\dots,a_{q-1}\in\mathbb{F}_{2^d}$ and set

$$r_i=(-1)^{\ell_0(P(z_i))},\qquad P(z)=\sum_{j=0}^{q-1}a_jz^j. \tag{29}$$

This uses qd seed bits. Finite fields of this size and polynomial representations are standard [6]; the independence property needed here is verified next.

Lemma 10.1 (Independence and moments)**.** *The signs in (29) are q-wise independent, in the sense that every subset of at most $\min\{q,n\}$ coordinates is uniform on its sign cube. For $q=2s$, $s\ge 2$, every fixed $x\in\mathbb{R}^n$, and $F=\|x\|_2^2$, the variable $Z=\langle r,x\rangle$ satisfies*

$$\mathbb{E}Z^2=F,\qquad \mathbb{E}Z^{2s}\le\frac{(2s)!}{2^s s!}F^s\le(2s)^sF^s. \tag{30}$$

For a four-wise independent row, $\mathbb{E}Z^4=3F^2-2\sum_i x_i^4\le 3F^2$, and $\mathrm{Var}(Z^2)\le 2F^2$.

Proof. For $t\le\min\{q,n\}$ distinct evaluation points, the map from coefficients to the t evaluations is a surjective linear map over the field: its first t columns form an invertible Vandermonde matrix.

Thus the evaluations are independent uniform field elements. A nonzero linear functional takes the values zero and one equally often, so the corresponding signs are independent uniform signs.

Every monomial in Z^{2s} involves at most $2s$ coordinates. Its expectation therefore equals that under fully independent signs. For independent signs the coefficient of t^{2s} in $\prod_i \cosh(x_i t)$ equals $\mathbb{E}Z^{2s}/(2s)!$. Coefficient by coefficient, the even-power series obeys

$$\cosh(x_i t) = \sum_{a\ge 0} \frac{x_i^{2a} t^{2a}}{(2a)!} \preceq \sum_{a\ge 0} \frac{x_i^{2a} t^{2a}}{2^a a!} = e^{x_i^2 t^2/2},$$

because $(2a)! \ge 2^a a!$. All coefficients are nonnegative, so this inequality is preserved by multiplication. Comparing with $e^{Ft^2/2}$ proves the first upper bound in (30). The coefficient $(2s)!/(2^s s!)$ is the product $1 \cdot 3 \cdots (2s-1)$ and is at most $(2s)^s$. Expansion of the second and fourth moments gives respectively $\sum_i x_i^2$ and $\sum_i x_i^4 + 6\sum_{i<j} x_i^2 x_j^2$. Subtracting F^2 proves the variance claim. □

10.2 A precommitted fresh coarse block per round

We use the second-moment estimator of Alon, Matias, and Szegedy [1], with its elementary analysis given below.

For every round t, reserve a block of $64r$ independent four-wise independent sign rows, where r is the least odd integer with

$$r \ge 8 \log_2(4K/\beta). \tag{31}$$

Split it into r groups of 64 rows. For the current vector, compute in each group the mean of the 64 squared measurements, and let c_t be the median of the r group means. All rows of every round's block are fixed at initialization and, in a stream, maintained from the beginning. Only the designated block is read for the coarse estimate in a round.

Lemma 10.2 (Adaptive coarse scales)**.** *The blocks use $64Kr = O(K \log(K/\beta))$ rows. With probability at least $1 - \beta/4$, simultaneously for all rounds,*

$$F_t/2 \le c_t \le 2F_t, \qquad F_t = F_2(x_t). \tag{32}$$

On this event, $c_t = 0$ if and only if $x_t = 0$.

Proof. For a fixed vector with $F > 0$, a group mean has expectation F and variance at most $2F^2/64$. Chebyshev gives probability at most $1/8$ of leaving $[F/2, 3F/2]$. The groups are independent. If their median leaves this interval, at least half of the r group indicators are bad. Applying the one-sided bound in Lemma 3.1, with mean at most $r/8$, bounds this probability by

$$e^{-2r(1/2-1/8)^2} = e^{-9r/32} \le e^{-r/8} \le \beta/(4K).$$

For $F = 0$, every measurement and the median are exactly zero.

For an adaptive x_t, the reserved round-t block has influenced no previous output. Conditional on the analyst's coins, the other randomness, and the previous interaction, x_t is independent of this block. The same failure estimate therefore holds for the current vector. Integrating and taking a union bound over rounds proves (32). If $F_t > 0$, its lower bound forces $c_t > 0$, proving the last assertion. □

10.3 The bounded query and its bias

For a finite parameter description, first replace the requested ε and β by the largest dyadic numbers not exceeding them, and retain the same notation. Each decreases by a factor less than two; achieving these stronger parameters implies the original guarantee and changes the bounds only by absolute constants. Set

$$s = \lceil \log_2(32/\varepsilon) \rceil, \qquad L = 16s, \tag{33}$$

and let α be the largest dyadic number at most $\varepsilon/(64L)$. Thus

$$\varepsilon/(128L) < \alpha \le \varepsilon/(64L). \tag{34}$$

Let ρ and ν both be the largest dyadic number at most $\beta/4$. Run the finite mechanism of Corollary 7.4 with $k = K$ and these parameters. Its sample domain is the set of polynomial seeds $(\mathbb{F}_{2^d})^{2s}$ in (29), and its population distribution is uniform on these seeds. All its weighted samples and its anchor samples are independently drawn seeds. We identify a seed with its induced row when writing a query evaluation.

Set $D = 4/\alpha$, a power of two. For $c_t > 0$ define

$$f_t(r) = \min\left\{ \frac{\langle r, x_t\rangle^2}{Lc_t}, 1 \right\}, \qquad \bar{f}_t(r) = \frac{\lfloor D f_t(r) \rfloor}{D}. \tag{35}$$

For $c_t = 0$ define both functions to be zero. Submit $\bar{f}_t$ to the hierarchy in every round, including zero-scale rounds, and let $a_t \in [0,1]$ be its answer. Report

$$\widehat{F}_t = Lc_t a_t. \tag{36}$$

In particular a zero scale gives the exact output zero.

Lemma 10.3 (Clipping and rounding bias)**.** *For every fixed vector with $F = \|x\|_2^2 > 0$ and every scale $F/2 \le c \le 2F$, the query in (35) satisfies*

$$0 \le F - Lc\,\mathbb{E} f(r) \le \varepsilon F/8, \qquad |\mathbb{E}\bar{f}(r) - \mathbb{E} f(r)| \le 1/D = \alpha/4. \tag{37}$$

Proof. Write $Z = \langle r, x\rangle$. Lemma 10.1 and $c \ge F/2$ give

$$\begin{aligned} F - Lc\,\mathbb{E} f(r) &= \mathbb{E}[(Z^2 - Lc)_+] \\ &\le \frac{\mathbb{E} Z^{2s}}{(Lc)^{s-1}} \le \frac{(2s)^s F^s}{(LF/2)^{s-1}} \\ &= 2s(4s/L)^{s-1} F = 2s\, 4^{-(s-1)} F. \end{aligned}$$

For $s \ge 2$, $2s \le 2^s$ by induction. Consequently the last coefficient is at most $2^{2-s} \le \varepsilon/8$, by (33). Nonnegativity follows from clipping. Pointwise, $0 \le f - \bar{f} < 1/D$, proving the rounding bound. □

Lemma 10.4 (Legitimacy under adaptive vector selection)**.** *The queries $\bar{f}_t$ are valid adaptive statistical queries for the hierarchy, and with probability at least $1 - \rho - \nu$,*

$$|a_t - \mathbb{E}_r \bar{f}_t(r)| \le \alpha \qquad (t \in [K]).$$

This assertion does not condition on success of the coarse estimates.

Proof. Regard all coarse seeds and all adversary coins as independent auxiliary randomness for an analyst interacting with the hierarchy. Given this auxiliary randomness and the hierarchy's past answers, this analyst simulates the original adversary. It computes each external report from (36), hence the next vector, and then the current coarse scale and the function $\bar{f}_t$. This is a well-defined bounded function for every scale, including zero and inaccurate scales. The auxiliary randomness is independent of every hierarchy sample. Thus it is an admissible analyst in Theorem 1.2 and Corollary 7.4.

The finite and ideal mechanisms can be coupled for this same analyst. On the event of identical hierarchical transcripts, their reported external estimates coincide, and hence their future input vectors and query functions coincide exactly. Therefore Corollary 7.4 gives the assertion. No independence between the adaptively selected vector and the reused hierarchy rows is asserted or needed. □

11 Correctness and space for the bounded-query construction

Proof of Theorem 9.2. Use the matrix consisting of all coarse rows from Lemma 10.2 and all weighted-block and anchor rows of the hierarchy in Section 10. These independent rows are all selected at initialization. Their entries are exactly ± 1.

First consider accuracy. On the simultaneous events of Lemmas 10.2 and 10.4, if $F_t = 0$ then $c_t = 0$ and (36) is exact. Otherwise $F_t/2 \le c_t \le 2F_t$, so Lemma 10.3 gives

$$\begin{aligned} |\widehat{F}_t - F_t| &\le Lc_t\,|a_t - \mathbb{E}\bar{f}_t| + Lc_t\,|\mathbb{E}\bar{f}_t - \mathbb{E}f_t| + |Lc_t\,\mathbb{E}f_t - F_t| \\ &\le 2LF_t(\alpha + \alpha/4) + \varepsilon F_t/8 \\ &\le \left(\frac{5}{128} + \frac{1}{8}\right)\varepsilon F_t = \frac{21}{128}\varepsilon F_t \le \varepsilon F_t/4. \end{aligned}$$

Here we used (34). For $0 < \varepsilon < 1/10$, $1 - \varepsilon/4 \ge 1/(1+\varepsilon)$, since $(1-\varepsilon/4)(1+\varepsilon) - 1 = \varepsilon(3-\varepsilon)/4 \ge 0$. The upper factor is also at most $1 + \varepsilon$. This proves the requested multiplicative estimate on every round on these events. The total failure is at most

$$\beta/4 + \rho + \nu \le 3\beta/4 < \beta.$$

This union bound uses unconditional guarantees for the coarse blocks and for the finite population mechanism, not a conditional accuracy claim about the reused measurements.

For the row count, $s = O(\log(2/\varepsilon)) = O(\mathcal{H})$ and $\alpha^{-1} = O(s/\varepsilon)$. Also $\rho^{-1} = O(1/\beta)$. Thus $\log_2(CK/(\alpha\rho)) = O(\mathcal{H})$, and the number of hierarchy rows obeys

$$\begin{aligned} N &\le C\mathcal{H}^5\left(\frac{s\sqrt{K}}{\varepsilon} + \frac{s^2}{\varepsilon^2}\right) \\ &\le C\mathcal{H}^7\left(\frac{\sqrt{K}}{\varepsilon} + \frac{1}{\varepsilon^2}\right). \end{aligned}$$

The $64Kr = O(K\log(K/\beta))$ coarse rows fit within the remaining term of (27). The dyadic replacements of the original parameters affect only constants in this estimate.

We next verify each component of working memory. Every hierarchy row has $2sd = O(\log(n+1)\log(2/\varepsilon))$ seed bits, and a coarse row has only $4d$ seed bits. One common irreducible polynomial of degree d specifies the field representation and costs $O(d)$ bits. If desired, it can be found by enumerating monic degree-d polynomials and testing divisibility by all monic polynomials of degrees

at most $d/2$. Polynomial division over $\mathbb{F}_2$, the trial polynomials, and the enumeration counters use $O(d)$ bits. Existence follows from the existence of $\mathbb{F}_{2^d}$. The points z_i can be the first n binary coordinate strings.

For integer vectors bounded by U, each linear measurement is an integer of magnitude at most nU. Its signed counter therefore requires $O(\log(nU+2))$ bits. Squares and sums of 64 squares have the same asymptotic bit length. The coarse scale has the form $C_t/64$ with an integer numerator of $O(\log(nU+2))$ bits. Its median can be selected or sorted using a temporary array of the r group means, within the stated total memory.

For a hierarchy row with counter z, if $c_t = C_t/64 > 0$, its grid query value can be computed exactly as

$$\bar{f}_t(r) = \frac{1}{D} \min\left\{ D, \left\lfloor \frac{64Dz^2}{LC_t} \right\rfloor \right\}.$$

All integer arithmetic in this expression uses $O(\log(nU+2)+\mathcal{H})$ bits, since $\log D = O(\mathcal{H})$. If $C_t = 0$, use zero. Thus the algorithm never needs the coordinates of x_t to evaluate a statistical query; only its linear counters and the scalar c_t are used.

The parameter Λ in Corollary 7.4 is $O(\mathcal{H})$, as are $\log D$ and $\log(1/\nu)$. That corollary provides $O(N\mathcal{H}+\mathcal{H}^2)$ bits for the entire finite learner and its noise-sampling workspace. It stores one accumulated grid score per weighted row, optionally the current query values and projected weights, one threshold approximation per edge, and the correction and round counters. No past vector, query description, or sequence of correction functions is retained. The final multiplication in (36) has a rational output with $O(\log(nU+2)+\mathcal{H})$ bits.

All memory just listed is bounded by (28), increasing the absolute constant if needed. In the fixed-measurement model the current input Ax_t can be stored in these counters. For a turnstile update (i,Δ), evaluate the sign of every row at coordinate i from its seed and set its counter to $z+\Delta r_i$. This maintains every counter exactly. Evaluating the polynomial by Horner's rule needs only $O(d)$ extra field workspace; the seed is already stored. All learner scores stay unchanged between queries. The current counter bounds follow from the frequency bound, independently of stream length. Arbitrary adaptive changes of the frequency vector between queries are therefore supported within the same memory. □

Corollary 11.1 (The exponent-two regime)**.** *For $K = \lceil \varepsilon^{-2} \rceil$, the estimator uses $\widetilde{O}(\varepsilon^{-2})$ fixed measurements and $\widetilde{O}(\varepsilon^{-2})$ bits, where the suppressed factors are polynomial in $\log(n+1)$, $\log(U+1)$, $\log(1/\varepsilon)$, and $\log(1/\beta)$. The conclusion also holds for $K = \widetilde{O}(\varepsilon^{-2})$ with corresponding polylogarithmic factors.*

Proof. For $K = \lceil \varepsilon^{-2} \rceil$, both K and $\sqrt{K}/\varepsilon$ are $O(\varepsilon^{-2})$. Substitute into Theorem 9.2. Multiplying K by polylogarithmic factors changes these terms only by polylogarithmic factors as well. □

The reduction uses precisely the population guarantee in Definition 1.1. Its relation to sample accuracy and the previous adaptive-data-analysis bounds is discussed in Section 1.4.

12 Norm estimation and validation from rounded measurements

A recursive learner supplies both a vector to estimate and a proposed answer. To obtain the final space bound, the corrector must do more than estimate a norm accurately on demand. It must cheaply accept good proposals, give a useful sign on rejection, and tolerate the small arithmetic discrepancy in its maintained measurements.

The candidate itself supplies the clipping scale for a test. At first this scale need not be near the true norm. The proof therefore uses two moment bounds in order: a global bound rules out a grossly incorrect scale on acceptance, and a sharper tail bound controls clipping bias once the scale is known to be within a constant factor. Numerical requests instead obtain a fresh coarse scale. A small regularization coordinate keeps both operations meaningful for zero or very small virtual vectors.

For rounded counters, the relevant object is a function of the row seed. The counter update rule defines that function on every possible row once the permitted history is fixed. The uniform perturbation lemma below compares it with an ideal measurement; it does not require the rounded counters to represent a single rounded vector.

We now construct the norm estimator and corrector used at each level of the recursion. A *norm bank* is one independently initialized sketch instance: it contains a shared statistical sample of sign rows, the state of its adaptive validation mechanism, and separate coarse estimation blocks reserved for numerical requests. All of its rows are chosen at initialization. The bank supports both a numerical norm estimate and a test of a proposed estimate.

Fix an integer $D_* \geq 2$, a local accuracy $0 < \eta \leq 1/100$, and a regularization parameter $\sigma > 0$. The queried vectors belong to $\mathbb{R}^{D_*-1}$. Their final, additional coordinate is reserved for regularization: write $v^+ = (v, \sigma) \in \mathbb{R}^{D_*}$ and $\omega = \sigma^2$. Thus $F^+(v) = \|v\|^2 + \omega \geq \omega$. Put

$$s = \lceil \log_2(2^{32}/\eta) \rceil, \qquad L_0 = 16s, \qquad \frac{\eta}{2^{17}L_0} < a \leq \frac{\eta}{2^{16}L_0},$$

where a is dyadic. Failure parameters are rounded downward to dyadic values within a factor two. A bank consists of the samples of Lemma 8.2, drawn from the $2s$-wise independent sign family of Lemma 10.1, and separate initially unused coarse blocks. All rows in a bank are selected when that bank is born. If a bank's failure allocation is δ, assign $\delta/2$ to its statistical mechanism and use δ in its coarse-block formula below.

Lemma 12.1 (Uniform evaluation perturbations)**.** *Suppose for every possible sign row b the available value $z(b)$ obeys $|z(b) - \langle b, v^+\rangle| \leq \xi$, with $\|v^+\| \leq V$, and*

$$2\sqrt{D_*}V\xi + \xi^2 \leq \eta\omega/2^{30}.$$

For every $c \geq \omega$, replacing the exact squared measurement by $z(b)^2$ in $\min\{z(b)^2/(L_0c), 1\}$ changes the function uniformly by at most $\eta/(2^{30}L_0)$. For $c \geq \omega/2$, twice this bound suffices. If $z(b)$ is determined rowwise by permitted transcript information, the perturbed function is an admissible statistical query.

Proof. Cauchy–Schwarz gives $|\langle b, v^+\rangle| \leq \sqrt{D_*}V$. The identity $z^2 - w^2 = (z-w)(z+w)$ bounds the squared error by $2\sqrt{D_*}V\xi + \xi^2$. Clipping is 1-Lipschitz, proving the uniform bounds. Fixing the transcript defines the same function of each possible row; it is a map into $[0, 1]$. This is precisely the sample-evaluation interface. No assertion that rounded measurements are exact measurements of one rounded vector is used. □

Lemma 12.2 (Norm-bank certificates)**.** *Fix D_*, η, σ as above, nonnegative integer budgets M, J, R with $M + R \geq 1$, and failure probability $0 < \delta < 1/10$. There is a norm bank with*

$$\widetilde{O}\left(J + R + \frac{\sqrt{J+R+1}}{\eta} + \eta^{-2}\right)$$

rows; the suppressed factors are logarithmic in $M + J + R + 2$, $1/\eta$, and $1/\delta$. It supports at most M checks, J rejected checks, and R numerical requests. Assume that every queried vector, candidate,

and rowwise counter recurrence is selected from the permitted transcript and independent auxiliary information. Assume also that the uniform measurement-error hypothesis of Lemma 12.1 holds at each use, and that the vector for each reserved coarse block is independent of that block before its first use. With probability at least $1-\delta$, all completed requests satisfy the following certificates simultaneously. A numerical request for $v \in \mathbb{R}^{D_*-1}$ returns $A \ge 0$ with

$$|A - \|v\|^2| \le (\eta/2^{12})\|v\|^2 + \omega. \tag{N}$$

A check with candidate $S \ge 0$ and $0 < \theta \le 1/64$ tests the regularized vector at scale $S+\omega$, with endpoints $(1-\theta)/L_0$ and $(1+\theta)/L_0$. Writing $r = (\|v\|^2+\omega)/(S+\omega)$, its certificates are

$$\begin{aligned} \mathsf{ACCEPT} &\Longrightarrow |r-1| \le \theta + \eta/1000, \\ \mathsf{HIGH} &\Longrightarrow r - 1 \ge \theta - \eta/1000, \\ \mathsf{LOW} &\Longrightarrow 1 - r \ge \theta - \eta/1000. \end{aligned}$$

The guarantee holds up to the first violation of the stated measurement-error hypothesis, and includes the certificate for a terminal $(J+1)$st rejection if that rejection occurs.

Proof. For a sign measurement Z of a vector of squared norm F', the moment proof in Lemma 10.1 gives $\mathbb{E}Z^2 = F'$, $\mathbb{E}Z^4 \le 3F'^2$, and $\mathbb{E}Z^{2s} \le (2s)^s F'^s$. For $u = \mathbb{E}\min\{Z^2/(L_0 c), 1\}$, Cauchy–Schwarz yields

$$\frac{F'}{L_0 c + 3F'} \le u \le \frac{F'}{L_0 c}. \tag{B}$$

Indeed, $\min\{w/t, 1\} \ge w/(t+w)$ for $w \ge 0$, and $(\mathbb{E}w)^2 \le \mathbb{E}[w/(t+w)]\mathbb{E}[w(t+w)]$. When $c \ge F'/2$, the moment bound further gives

$$0 \le F' - L_0 c u \le \frac{\mathbb{E}Z^{2s}}{(L_0 c)^{s-1}} \le 2s4^{-(s-1)}F' \le \eta F'/2^{20}. \tag{C}$$

For the last inequality, $2s4^{-(s-1)} \le 2^{-s+4}$ for $s \ge 4$, and $2^{-s} \le \eta/2^{32}$.

On a check, put $c = S + \omega \ge \omega$ and $F' = \|v\|^2 + \omega$. Lemma 12.1 and the statistical certificates change each endpoint by at most $2a$. If accepted, (B) implies $r \ge 1 - \theta - 2L_0 a > 1/2$ and $r < 2$: otherwise $L_0 u \ge 2/(1 + 6/L_0) > 3/2$, contradicting $L_0 u \le 1 + \theta + 2L_0 a < 3/2$. Now (C) implies $|r-1| \le \theta + \eta/1000$. For a high rejection, the upper bound in (B) directly gives $r \ge 1 + \theta - 2L_0 a$. For a low rejection, (B) again excludes $r \ge 2$, so (C) gives $r \le (1 - \theta + 2L_0 a)/(1 - \eta/2^{20}) \le 1 - \theta + \eta/1000$. These calculations also cover zero vectors.

On a numerical request, use the next fresh coarse block. Take the median of r_0 group means, each using 512 four-wise independent sign rows applied to v^+. An ideal group mean differs from F' by more than $F'/4$ with probability at most $1/16$, by Chebyshev. The probability that at least half the groups fail is at most $2^{r_0}(1/16)^{r_0/2} = 2^{-r_0}$. Choose an odd $r_0 \ge \log_2(8(R+1)/\delta)$. The uniform squared-measurement error in the hypothesis is less than $F'/100$, so the computed median c lies in $[F'/2, 2F']$ on success. Unused coarse rows have affected no earlier decisions. Conditional on the rest of the randomness, their current shadow vector is independent of them; hence this argument applies adaptively at first use. The union bound covers all numerical requests.

On arbitrary executions replace a computed median below $\omega/2$ by $\omega/2$; on the success event this replacement is inactive. Ask the statistical mechanism for the mean of the clipped query at scale c. If its answer is b, return $A = \max\{0, L_0 c b - \omega\}$. Equations (C), Lemma 12.1, and the numerical statistical error a give $|L_0 c b - F'| \le (\eta/2^{13})F'$. Clipping at zero cannot increase error from $\|v\|^2$; widening the constants gives (N). All coarse tapes are independent auxiliary randomness for the

shared statistical sample, so releasing this answer is covered by the statistical theorem. Add $512r_0R$ coarse rows to the bound in Lemma 8.2. This proves the row count and the joint failure guarantee. The same proof remains valid up to a stopping time at which the stated evaluation-error condition first fails. □

Part III
Adversarially robust turnstile streaming

The second instance uses a time tree of geometric learners. Its correctors contain the statistical hierarchy from Part I. The new issue is persistence: ancestor corrections must preserve the comparison target of each live descendant. Residual energy supplies the progress measure, and normalized coordinates preserve its meaning throughout a learner lifetime.

We now prove Theorem 1.4. The norm certificates from Part II provide reliable estimates and protected tests, but their space depends on the number of informative uses. The main task is to design and analyze a recursive learner that keeps this number small throughout an unrestricted adaptive stream. Its descendants persist across corrections, and its virtual guard coordinates control the finite-precision arithmetic.

13 Geometric progress and bounds on amplification

This part uses Euclidean geometry in addition to the statistical mechanism. A corrector now compares a proposed energy decomposition with the incoming vector's norm. A rejection certifies correlation between its residual and outgoing vectors. Moving the iterate along the outgoing vector reduces residual energy.

The first lemma deliberately allows a range of step sizes. The algorithm chooses its step from approximate norm estimates and then rounds it; it cannot implement an exact ideal minimizer. Lemma 13.2 makes the preservation of the comparison target explicit. Lemma 13.3 adds fresh coordinates to turn an energy bound into a bound on multiplier products. Lemma 13.4 converts that product bound into a counter-error bound. These are separate uses of geometry: progress controls how often the learner changes, whereas amplification control limits the bits needed to represent its state.

The following lemmas are deterministic. They explain why an inaccurate prediction permits a useful learner update, why fresh coordinates control products of these updates, and how that control bounds rounding error. They apply in any finite-dimensional Euclidean space.

Lemma 13.1 (Robust correction)**.** *Let $a, u, q \in \mathbb{R}^d$ and $0 < \eta \le 1/100$. Define the residual $r = a - u$, the outgoing vector $y = u + q$, the input $x = a + q = r + y$, and energies $E = \|r\|^2$, $Q = \|y\|^2$. Suppose*

$$|\|x\|^2 - (E + Q)| \ge (\eta/2)(E + Q) > 0.$$

Let s be the sign of $\langle r, y\rangle$ and choose

$$\alpha = st\sqrt{E/Q}, \qquad \eta/32 \le t \le \eta/8.$$

Then $1/2 \le 1 + \alpha \le 3/2$ and the update $u^+ = u + \alpha y$ satisfies

$$E^+ := \|a - u^+\|^2 \le (1 - \eta^2/64)E.$$

Moreover $E, Q \ge \eta^2\|x\|^2/256$ and $\eta/4 \le \sqrt{E/Q} \le 4/\eta$.

Proof. Put $c = |\langle r, y\rangle|$. The assumption says $c \ge \eta(E + Q)/4 \ge (\eta/2)\sqrt{EQ}$. Combining this with $c \le \sqrt{EQ}$ gives both ratio bounds and $|\alpha| \le (\eta/8)(4/\eta) = 1/2$. Expansion gives

$$E^+/E = 1 - 2t\frac{c}{\sqrt{EQ}} + t^2 \le 1 - \eta t + t^2 \le 1 - \eta^2/64.$$

For the last inequality, $\eta t - t^2$ is increasing on the stated interval and its value at $t = \eta/32$ is $31\eta^2/1024$. Also $2c \geq \eta(E+Q)/2 \geq \eta\|x\|^2/4$. If $\sqrt{E} < \eta\|x\|/16$, then $2c \leq 2\sqrt{E}(\|x\| + \sqrt{E}) < \eta\|x\|^2/4$, a contradiction. Interchanging r and y gives the same bound for Q. □

Lemma 13.2 (A persistent target under ancestor rescaling)**.** *Consider exact vectors $a, u, q \in \mathbb{R}^d$ and a scalar $C > 0$. Let $x = a + q$, $r = a - u$, and $y = u + q$. Starting from any initial state, interleave any of the following operations:*

1. *a suffix update $q \leftarrow q + d$, for any vector d;*
2. *an ancestor rescaling $(a, u, q, C) \leftarrow (ca, cu, cq, cC)$, for any $c > 0$;*
3. *a local correction $u \leftarrow u + \gamma y$, for any real coefficient γ.*

Then the normalized target $\bar{a} = a/C$ is constant throughout. Writing $\bar{u} = u/C$, $\bar{q} = q/C$, $\bar{r} = r/C$, and $\bar{y} = y/C$, the normalized residual energy $\bar{E} = \|\bar{a} - \bar{u}\|^2$ is unchanged by the first two operations. A local correction obeys

$$\bar{E}^+ = \bar{E} - 2\gamma\langle \bar{r}, \bar{y}\rangle + \gamma^2\|\bar{y}\|^2.$$

Whenever the premise and coefficient requirements of Lemma 13.1 hold, the same contraction holds for $\bar{E}$. Moreover $y^+ = (1+\gamma)y$ at a local correction. Thus, when $1 + \gamma > 0$, scaling a child's entire decomposition by $1 + \gamma$ represents the new outgoing vector while preserving that child's normalized target and normalized residual energy.

Proof. Suffix updates and local corrections do not change a or C. An ancestor rescaling multiplies both by c, so their ratio is unchanged. This proves constancy of $\bar{a}$ by induction over the operations. A suffix update leaves $a - u$ unchanged. An ancestor rescaling multiplies $a - u$ and C by the same scalar, leaving $(a - u)/C$ unchanged. These observations prove the two invariance claims.

At a local correction C is unchanged and $\bar{r}^+ = \bar{r} - \gamma\bar{y}$. Expanding its squared norm gives the displayed identity. Dividing all vectors in the robust correction lemma by C divides all squared norms and inner products by C^2, preserves the energy ratio and the inner-product sign, and hence preserves the premise, coefficient interval, and contraction factor. Finally $y^+ = u + \gamma y + q = (1+\gamma)y$. Scaling the child's prefix, iterate, suffix, and ancestor product by this same positive factor is exactly an operation of the second kind for that child, for which invariance was already proved. □

This lemma concerns exact vector semantics, including coefficients chosen by the implemented algorithm. It does not assert exact invariance of rounded counters. Virtual guard insertions also change the iterate and can increase residual energy. The next two lemmas account for these perturbations and for arithmetic error separately.

Lemma 13.3 (Guarded envelopes)**.** *Fix $0 < \eta \leq 1/100$, an integer $T \geq 1$, and $\gamma > 0$ with $T\gamma^2 \leq 1/64$. Consider a learner with fixed prefix a, iterate u, incoming suffix q, and energy $E = \|a - u\|^2$, all expressed in its normalized coordinates. Initially let $U_0 \geq 2E_0$. Between corrections the suffix may change; in addition, at most T guards are inserted by adding a scalar multiple of a standard basis vector to u. A nonzero guard uses a previously unused coordinate on which the prefix and every incoming query are zero. In epoch k its normalized amplitude lies in $[(\gamma/16)\sqrt{U_k}, \gamma\sqrt{U_k}]$, where $U_k = 2^{-k}U_0$. An epoch ends after $L_1 = \lceil 128/\eta^2 \rceil$ corrections obeying Lemma 13.1. Zero guards are allowed, provided every correction has a nonzero fresh guard immediately preceding it among that learner's own corrections. Then $E \leq U_k$ throughout epoch k, and $E \leq U_k/2$ at its start. Every forward interval product of the learner's correction multipliers is at most $B = 16/\gamma$.*

Proof. Fresh orthogonality increases energy by exactly the squared amplitude. Within an epoch the total increase is at most $T\gamma^2 U_k \le U_k/64$. Thus its energy is at most $U_k/2 + U_k/64 < U_k$. At its end the initial energy contributes at most $U_k/8$, because $(1-\eta^2/64)^{L_1} \le e^{-2} < 1/4$. All injected energies together contribute at most $U_k/64$, even if their subsequent contractions are ignored. The final energy is less than $U_k/4 = U_{k+1}/2$, proving the induction.

For an interval containing a correction, use the fresh guard before its first correction, with amplitude δ and envelope U_s. No incoming vector or later guard touches its coordinate. Its iterate coordinate at the end is exactly δG, where G is the product over that interval. The residual has the negative of this coordinate, so $\delta^2 G^2 \le E \le U_s$. Since $\delta \ge (\gamma/16)\sqrt{U_s}$, $G \le 16/\gamma$. An interval without a correction has product one. □

Lemma 13.4 (Counter propagation)**.** *Let $z_{t+1} = a_t z_t + d_t$ be a real scalar recurrence, and let $\widehat{z}_{t+1} = a_t \widehat{z}_t + d_t + \zeta_t$ use the same coefficients and increments, with $|\zeta_t| \le h$ and initial error $e_0 = \widehat{z}_0 - z_0$. Suppose $G \ge 1$ and $|\prod_{j=s}^{t-1} a_j| \le G$ for every forward interval $[s,t)$. Then*

$$|\widehat{z}_t - z_t| \le G(|e_0| + th).$$

The assertion holds for adaptive coefficients and increments.

Proof. Writing the local arithmetic error as ζ_t, subtraction gives $e_{t+1} = a_t e_t + \zeta_t$. Iteration gives $e_t = (\prod_{j<t} a_j)e_0 + \sum_{s<t}(\prod_{s<j<t} a_j)\zeta_s$. Take absolute values. This is a pathwise identity for the coefficients actually chosen; it compares no two executions with different answers. □

14 The recursive streaming algorithm

It is useful to read the algorithm as three operations at each physical update. First initialize the learners whose fixed time intervals begin, and propagate the known sparse stream update. Next insert the current fresh coordinates from the root downward. Finally evaluate from the leaves upward, so each learner receives its child's estimate before testing its own prediction.

On acceptance the learner returns its proposed estimate. On rejection it obtains a protected numerical answer, updates its iterate, refreshes its residual estimate, and returns that numerical answer. The update changes the future child state by a scalar multiplication. It does not ask the child to answer the current query again, and it does not allocate a new child.

The definitions below separate vector semantics from stored data. The vectors explain what is being estimated; the bank counters, cached residuals, and scalar tags are the actual implementation.

We specify the algorithm for the parameters in Theorem 1.4. A learner stores a decomposition of its current input; a norm bank acts as its corrector; and a child learner supplies the outgoing-vector estimate. The tree schedule is fixed before the stream, while the corrections are adaptive.

14.1 Parameters and precision

The choices are made in dependency order. We first bound the recursion depth and the number of operations. These determine a sufficiently small guard amplitude and hence a bound on multiplier products. That bound gives ranges for all virtual vectors and a norm floor for recursive queries. We can then choose the rounding grids so that amplified counter errors are smaller than the required statistical and geometric margins. None of these choices depends on an unknown future vector.

The main parameters have the following roles. Their values below are deliberately conservative to simplify the error estimates.

Parameter	Role
H	Number of active learner levels
η	Local relative accuracy and correction step scale
T	Upper bound on operations affecting a retained scalar
γ	Relative amplitude of a fresh guard coordinate
B	Bound on a learner's forward multiplier products
V_d, U_d	Initial norm and normalized energy bounds at depth d
G, V	Global amplification and queried-vector norm bounds
ν	Lower bound on nonzero recursive query norms
J_*	Maximum corrections in one learner lifetime

Put

$$H = 1 + \lceil \log_2 \max\{1, m\} \rceil, \quad T = 2^{12}(m+1)(H+1)^3, \quad D_* = n + T + 1, \quad W = nU + 1.$$

Choose dyadic parameters satisfying

$$\frac{\varepsilon}{2^{21}(H+1)^2} < \eta \le \frac{\varepsilon}{2^{20}(H+1)^2}, \qquad \frac{1}{16\sqrt{T}} < \gamma \le \frac{1}{8\sqrt{T}}, \qquad B = 16/\gamma.$$

Set $V_0 = W$, $V_{d+1} = (1 + 2B^d)V_d$ for $0 \le d < H$. Let U_d be the least power of two at least $2V_d^2$, and put

$$U_* = \max_{d<H} U_d, \quad V = 8B^{H+1}V_H + 2, \quad G = B^{H+2}, \quad \nu = \gamma\eta 2^{-H-20}.$$

All large magnitude parameters may be rounded upward to integers. Define

$$K = \left\lceil \log_2 \frac{2^{20} U_* B^{2H}}{\eta^2 \nu^2} \right\rceil, \quad L_1 = \lceil 128/\eta^2 \rceil, \quad J_* = L_1(K+1).$$

Choose positive dyadic grid sizes, within a factor two of the upper bounds below:

$$h_g \le \gamma\eta\nu/2^{20}, \qquad e_C = h_g/(2^{10}\gamma V), \qquad h_C \le e_C/(TG), \tag{38}$$

$$\sigma \le \frac{\eta^4\nu^2}{2^{40}TG^2(V+1)^2 D_*}, \qquad \omega = \sigma^2, \qquad h_P \le \frac{\eta^2\nu^2}{2^{40}TG^2}, \tag{39}$$

$$h_b \le \frac{\eta\omega}{2^{40}(T+1)GD_*(V+1)}, \qquad h_\alpha \le \eta^2/2^{20}. \tag{40}$$

Use a dyadic h_b dividing h_g and σ; decreasing it by at most a factor two suffices since all are powers of two.

14.2 The time tree and initialization

A bank and a learner have different lifetimes and responsibilities. A bank stores measurements with one fixed matrix. A learner uses its parent bank to check an incoming vector and uses a child to estimate its outgoing vector. A parent's bank can therefore serve two successive children without giving a fresh child matrix access to an old unknown prefix.

This distinction explains the constant number of numerical initialization requests per bank. It also explains why the tree is indexed by physical time rather than by correction events.

Partition $[1, 2^{H-1}]$ by its complete binary interval tree. At time t, the active learner nodes are its root-to-leaf path. Retire ended nodes before allocating their replacements. A learner is born

at the left endpoint of its interval and dies at the right endpoint. Births occur before that time's physical update, in increasing order of depth. Corrections never change this schedule.

There is a permanent top norm bank A_{top}, initialized at time zero. Every nonleaf learner v owns a fresh norm bank A_v, initialized when v is born. A learner's parent bank is its parent's bank, or A_{top} at the root. Every bank has at most two successive learner clients; the top bank has one. Give it rejection cap $2J_*$, numerical cap $4J_* + 2$, and check horizon $2m + 2$. Its failure allocation is the largest dyadic number at most $\beta/[16(m + 1)]$.

14.3 Vector semantics and stored state

Ideal vectors associated with the stored counters. For analysis, define vectors by exact arithmetic using the coefficients and guard amplitudes actually chosen by the finite-precision algorithm. We call these its *shadow vectors*; they do not refer to an alternative execution that makes different decisions. The original vector is padded by zeros to dimension $D_* - 1$. At depth d the learner's incoming vector is X_v and its outgoing vector is Y_v. At birth set $a_v = X_v$, $u_v = q_v = Y_v = 0$, and $C_v = 1$. Subsequently $X_v = a_v + q_v$, $Y_v = u_v + q_v$. The root's incoming vector is the physical x; a child's incoming vector is its parent's outgoing vector. When a strict ancestor corrects by multiplier c, multiply $a_v, u_v, q_v, X_v, Y_v, C_v$ by c. Thus a_v/C_v stays fixed. These are analytic vectors, not stored arrays.

Stored counters and scalars. Each bank stores row counters for its own tracked vector Z and for its active client's outgoing vector Y_v. Here $Z = x$ for the top bank and $Z = Y_w$ for bank A_w. All newly born bank counters and new client-outgoing counters start at zero. A bank already tracks Z when a new client is born. It can therefore evaluate that client's residual using the difference $Z - Y_v$ without reconstructing the old prefix for a fresh matrix. Counters are updated by additions of known sparse increments and multiplications by actual correction coefficients, rounded to grid h_b after each operation. They are clipped to $[-2D_*V, 2D_*V]$; this clipping affects no control flow. Each learner also stores: its correction count; $\widehat{C}_v$, on grid h_C, initialized to one and multiplied by ancestor multipliers; and a nonnegative residual estimate P_v, on grid h_P.

At birth request a numerical estimate of $\|Z\|^2$ from the parent bank and store its rounding as P_v. After an own correction refresh P_v by a numerical request for $Z - Y_v$. Ancestor scaling multiplies P_v by c^2. An own guard of amplitude δ adds δ^2 to P_v. Round these scalar operations to grid h_P and clip at zero. If a bank reports failure, or an explicitly declared scalar magnitude or correction cap is exceeded, halt and report zero henceforth. Counter clipping never triggers such a halt. The proof below shows that no cap is reached on the success event. Magnitude caps may be $2G$ for $\widehat{C}$, $4V^2$ for P, and $2V$ for an inserted guard.

14.4 Updates, guards, and recursive evaluation

Rounding to a grid means rounding to the nearest grid point, with ties resolved by a fixed rule. Clipping to an interval means projection onto that interval. All operations below use the parameters and caps just specified.

A physical update. Add $\Delta_t e_{j_t}$ to every active X_v and Y_v, implemented by its row contribution. These additions implement the physical input at the root and the induced input at every descendant. An ancestor guard similarly adds its sparse contribution to both X_v and Y_v of each strict descendant. An own correction multiplies Y_v and every strict descendant's states by $1 + \alpha$. Only the appropriate

copies in each bank are modified. In particular, the parent-bank copy of Y_v receives exactly the same updates as its own-bank copy, but uses that bank's independently sampled rows.

Guard insertion. Allocate a distinct coordinate $g(v,t) = n + H(t-1) + d + 1$ to the active learner of depth d at each nonzero physical-query time. Families are disjoint across depths. No physical update uses these coordinates. Before any recursive evaluation at time t, visit all active nodes from root to leaf. Let $k = \lfloor j_v/L_1 \rfloor$ and $U = 2^{-k}U_d$. For $\widehat{C}_v > 0$, let δ be the largest power of two with

$$\delta^2 \le (\gamma^2/16)\widehat{C}_v^2 U. \tag{G}$$

Set $\delta = 0$ if this power is smaller than h_g or if $\widehat{C}_v = 0$. A nonzero guard adds $\delta e_{g(v,t)}$ to u_v and Y_v, and propagates to descendants as an incoming sparse update. It leaves X_v unchanged. Add δ^2 to its residual cache. Testing (G) requires only rational squares, not a stored irrational.

Recursive evaluation. Evaluate bottom-up. For a node of height $\ell = H - 1 - d$, obtain Q_v from its child's recursive output, or, at a leaf, by computing $\|Y_v\|^2$ and rounding to grid h_P. Set $S_v = P_v + Q_v$ and $\theta_v = (8\ell + 4)\eta$. Ask its parent bank to check X_v with candidate S_v and threshold θ_v. On acceptance return S_v. On rejection request a numerical estimate N_v of $\|X_v\|^2$. Set the sign to $+1$ for a high rejection and -1 for a low rejection. Compute

$$\alpha = \mathrm{round}_{h_\alpha}\left(\frac{\eta}{16}\,\mathrm{sign}\sqrt{P_v/Q_v}\right), \tag{U}$$

clipped to $[-1/2, 1/2]$, and use zero if either denominator or numerator is nonpositive. Apply its update, increment the count, refresh P_v, and return the grid-h_P rounding of N_v. No second evaluation of descendants is performed after this update. Their virtual inputs and caches are merely rescaled for future rounds.

At a leaf, before evaluation its outgoing vector is supported on the current physical coordinate and at most H current guard coordinates. Maintain this list explicitly for that one time step. Ancestor multipliers and the at most one own multiplier act on its entries. There are at most H such multiplications during its lifetime. Thus exact dyadic leaf arithmetic uses only polynomially many logarithmic bits; it does not retain vectors from earlier time steps.

15 Detecting zero frequency vectors

At a nonzero physical input, integer frequencies give $\|x\| \ge 1$. There is no such relative-error margin at $x = 0$, where multiplicative accuracy requires an exact zero answer. We isolate that case before using the recursive norm tests.

A polynomial fingerprint supplies an inexpensive zero test, but its queries are adaptive as well. The proof compares with an execution using the true zero decision. In that execution the queried vectors do not depend on the fingerprint seed. If no false zero occurs along that execution, the two executions agree step by step. This coupling, rather than conditioning on the actual adaptive stream as if it were independent, justifies the union bound.

Use an independent exact-zero fingerprint on the original integer stream. Choose a prime $p > \max\{2U, 8nm/\beta\}$ with at most twice this magnitude, and sample z uniformly in $\mathbb{F}_p$. Maintain $\sum_{i=1}^{n} x_i z^{i-1} \bmod p$. Existence of such a prime with this bit length follows from Bertrand's postulate [10]; it can be found by exhaustive primality checking using logarithmic workspace. If the fingerprint is zero, output zero and omit guards and recursive evaluations for that time. Births and counter updates still take place.

Lemma 15.1 (Adaptive zero test)**.** *For every admissible adversary in Definition 1.3, the independent fingerprint just specified has, with probability at least* $1 - \beta/8$*, the property*

$$\sum_{i=1}^{n} x_i^{(t)} z^{i-1} = 0 \text{ in } \mathbb{F}_p \quad \Longleftrightarrow \quad x^{(t)} = 0 \qquad \text{for every } t \in [m].$$

The algorithm may use the fingerprint decisions to select its outputs and subsequent internal operations.

Proof. Couple to an execution that makes the true zero decision and otherwise uses identical random tapes. Fix all those tapes and the adversary's coins. Its tested vectors are then independent of z. A nonzero bounded integer vector gives a nonzero polynomial over $\mathbb{F}_p$, of degree at most $n-1$, because $p > 2U$. A nonzero polynomial of degree d has at most d roots: factor out one root and induct on degree. Hence the union bound over m tests is at most $mn/p < \beta/8$. Outside this event the two executions agree by induction up to every next decision. The zero polynomial always evaluates to zero. Integrate the fixed auxiliary tapes. □

16 Correctness of the recursive estimates

There are two inductions, serving different purposes. Within a single query, induction from the leaves to the root controls estimation error. Across time, induction controls residual contraction and arithmetic amplification. Neither can be proved by assuming the other already holds for future operations.

The key local step is the gap between a child's allowed error and its parent's validation threshold. At height ℓ, the candidate error is at most roughly $8\ell\eta$, while the threshold is $(8\ell + 4)\eta$. The remaining slack implies that a rejection is caused by a true cross term, rather than merely by errors in the child or cached residual. This is what makes a rejection a valid learning step.

The arithmetic induction follows the same actual coefficients as the implementation. A previous product bound controls errors before the next test; the resulting certificate proves the next correction valid; and its fresh coordinate establishes the next product bound. This order avoids using the stability of a new correction before establishing that correction's geometric premise.

We analyze the algorithm of Section 14 with its specified parameters. At node v, write $E_v = \|X_v - Y_v\|^2$ for the residual energy in the current physical units and E_v/C_v^2 for its normalized energy. The epoch envelope bounds the latter. All vector assertions concern the shadow vectors defined using the actual selected coefficients.

The proof maintains four properties together: a decreasing normalized energy envelope, bounds on every forward multiplier product, accuracy of the stored counters and residual estimates, and accuracy of each recursive answer. The intermediate lemmas state the hypotheses needed at an individual step. Lemma 16.5 then establishes them throughout the execution by induction.

Lemma 16.1 (Ranges, support, and arithmetic)**.** *Consider an execution prefix of the algorithm in Section 14, before any declared cap failure. Assume every previous correction satisfies Lemma 13.1 and every previous epoch satisfies the envelope and interval-product bounds of Lemma 13.3. Assume every completed numerical request used to initialize or refresh a residual estimate satisfies* (N)*. For each active node at depth* d*, with the parameters defined in Section 14, the following properties hold:*

1. *Every forward product of strict-ancestor multipliers is at most* B^d*, and* $C_v \le B^d$*.*
2. *Incoming vectors have norm at most* V_d *and outgoing vectors at most* V_{d+1}*. All bank query vectors, including regularization, have norm at most* V*.*

3. *Every incoming vector is zero on that node's own reserved coordinate family.*

4. *Until the next correction, row-counter errors are at most TGh_b, residual-query errors at most $2TGh_b$, and $|\widehat{C}_v - C_v| \le TGh_C \le e_C$.*

5. *The residual cache obeys, with $E_v = \|X_v - Y_v\|^2$,*

$$|P_v - E_v| \le (\eta/2^{12})E_v + \tau_P, \qquad \tau_P = 2\omega G^2 + TG^2h_P \le \eta^2\nu^2/2^{30}. \tag{P}$$

Proof. Scalar multipliers commute. Grouping them by strict ancestor and applying its interval-product bound gives the first assertion. In normalized coordinates the residual is at most $\sqrt{U_d}$, so its physical norm is at most $2B^dV_d$. The triangle inequality then gives $\|Y_v\| \le (1 + 2B^d)V_d = V_{d+1}$. This proves the range assertion by depth induction, starting from $\|x\| \le \sqrt{n}U < W$; differences and the regularizing coordinate fit the stated V.

Support follows by depth induction. An incoming vector uses physical coordinates and guards of strict ancestors only; linear combinations and rescalings preserve this fact. A node's own outgoing vector adds only its own family. Births precede all current guards.

Every counter starts at zero. Each tracked vector receives known sparse additions and scalar multiplications; its multipliers are those of at most H learners. Their every-interval gain is at most G. Lemma 13.4 gives the counter bound, also for every hypothetical sign row. The range bound makes counter clipping inactive. The same argument for C_v, whose initial value is exact, gives the tag bound. There are fewer than T state operations: per physical time there are H births or fewer, H guards or fewer, H corrections or fewer, and each affects at most H active states; the chosen T dominates their additions, multiplications, and copies.

After a cache refresh, (N) and rounding give error at most $(\eta/2^{12})E_v + \omega + h_P$. Between refreshes, incoming updates leave E_v unchanged. Own guards add the same exact δ^2 to energy and to the unrounded cache, reducing the relative-error part. Ancestor multipliers scale both by their square; their interval products are at most G^2. Applying the scalar error identity to the additive part gives (P). The final numerical inequality follows by substituting (39); in particular $TG^2h_P \le \eta^2\nu^2/2^{40}$ and $2\omega G^2 \le \eta^2\nu^2/2^{40}$. □

Lemma 16.2 (Representable guards)**.** *Consider the guard rule (G) for a node with exact ancestor multiplier $C > 0$, stored multiplier $\widehat{C}$, and current envelope $U = 2^{-k}U_d$. Suppose $|\widehat{C} - C| \le e_C$ and $\sqrt{U} \le V$, with the precision parameters of Section 14. Every nonzero guard of physical amplitude δ then satisfies*

$$\frac{\gamma}{16}\sqrt{U} \le \frac{\delta}{C} \le \frac{\gamma}{2}\sqrt{U}.$$

Suppose additionally that the node's residual energy is at most C^2U and that its next correction satisfies Lemma 13.1 on an input of norm at least ν. Then the guard for that correction is nonzero.

Proof. The untruncated dyadic choice obeys $(\gamma/8)\widehat{C}\sqrt{U} < \delta \le (\gamma/4)\widehat{C}\sqrt{U}$. If $\delta \ge h_g$, then $\widehat{C}\sqrt{U} \ge 4h_g/\gamma$, whereas $|\widehat{C} - C|\sqrt{U} \le e_CV \le h_g/(2^{10}\gamma)$. Consequently $C/2 \le \widehat{C} \le 2C$, proving both amplitude bounds. An omitted guard injects no energy and does not invalidate an envelope.

If a genuine correction is to occur, its physical residual has norm at least $\eta\nu/16$ by Lemma 13.1. The envelope implies $C\sqrt{U} \ge \eta\nu/16$. The tag error is less than $\eta\nu/64$, so $\widehat{C}\sqrt{U} \ge \eta\nu/32$ and the untruncated dyadic amplitude exceeds $\gamma\eta\nu/256 > h_g$. It is therefore not omitted. This uses the envelope, whose proof requires only the upper bound on earlier guard injections and contraction, not the assertion that every test inserts a guard. □

Lemma 16.3 (A floor for corrected targets)**.** *Consider a round of the algorithm in Section 14 with $x^{(t)} \neq 0$. Assume that earlier corrections satisfy Lemma 13.1, that their normalized energy envelopes are valid, and that current guards satisfy the upper amplitude bound of Lemma 16.2. Every input X_v tested during this round's recursive evaluation then satisfies $\|X_v\| \geq \nu$, where $\nu = \gamma\eta 2^{-H-20}$ is the parameter of the algorithm.*

Proof. At the root $C = 1$ and a correction requires physical residual energy at least $\eta^2/256$, since the target is a nonzero integer vector. Thus an epoch containing a root correction has envelope at least $\eta^2/256$. After its halving the next envelope is at least $\eta^2/512$. Initially the envelope is at least two. It follows that every root envelope is at least $\eta^2/512$. Its tag is exactly one. Its fresh dyadic guard therefore has amplitude greater than $\gamma\eta/256$ and is never omitted.

The root validation has norm at least one. For every deeper validation, consider this round's fresh root guard coordinate. All current births precede its insertion, so it is zero in every current prefix and iterate before insertion. It enters each descendant as an incoming update. Before the descendant is evaluated, the only possible operations on that coordinate are ancestor or own correction multipliers; each is at least $1/2$, and each learner corrects at most once in the round. There are at most H such factors. Other guards use different coordinates, and physical updates use original coordinates. Thus its absolute coefficient is at least $\gamma\eta 2^{-H}/256 > \nu$. This proves the norm floor. The assertion about the root envelope uses only previous root corrections; it therefore applies inside a first-failure induction as well. □

Lemma 16.4 (Prediction and correction induction)**.** *Consider a recursive evaluation at a node v of height $\ell = H - 1 - d$ in Section 14. Assume $\|X_v\| \geq \nu$, the residual estimate obeys (P), and the node's norm-bank requests satisfy Lemma 12.2. If $\ell \geq 1$, assume its child's returned estimate of $\|Y_v\|^2$ has relative error at most $8\ell\eta$; if $\ell = 0$, assume the specified exact leaf calculation followed by grid-h_P rounding. Then its answer A satisfies*

$$|A - \|X_v\|^2| \leq 8(\ell + 1)\eta\|X_v\|^2. \tag{A}$$

Every rejection makes a correction satisfying Lemma 13.1 with the actual rounded coefficient.

Proof. Let $F = \|X_v\|^2$, $E = \|X_v - Y_v\|^2$, $Q = \|Y_v\|^2$, and $S_0 = E + Q$. The parallelogram inequality gives $S_0 \geq F/2 \geq \nu^2/2$. By induction on height, the child estimate has relative error at most $8\ell\eta$; at a leaf the only error is its grid rounding, at most h_P. Combining this with (P) gives, for $S = P_v + Q_v$,

$$|S - S_0| \leq b_\ell S_0, \qquad b_\ell = 8\ell\eta + \eta/100. \tag{S}$$

Indeed the sum of its additive errors is at most $\tau_P + h_P$, which is less than $\eta S_0/1000$, and the residual relative error is at most $\eta/4096$. In particular $S \geq \nu^2/4$.

Put $\theta = (8\ell + 4)\eta$. Our choice of η implies $\theta \leq 1/64$ and $\theta^2 \leq \eta/100$. Acceptance gives $|F - S| \leq (\theta + \eta/1000)(S + \omega) \leq (\theta + \eta/500)S$. Here ω/S is bounded by (39). For $z = \theta + \eta/500$, $z < 1/32$ and $z/(1 - z) \leq z + 2z^2 \leq \theta + \eta/10$. Thus an accepted answer has relative error less than $8(\ell + 1)\eta$.

At a rejection, the bank's indicated sign is the sign of $F - S$, and $|F - S| \geq (\theta - \eta/1000)S$. Using (S),

$$|F - S_0| \geq \big[(\theta - \eta/1000)(1 - b_\ell) - b_\ell\big]S_0 \geq \eta S_0.$$

To check the last inequality, $\theta - b_\ell \geq 3.99\eta$, $\theta b_\ell \leq \theta^2 \leq \eta/100$, and the remaining subtracted term is at most $\eta/1000$. Also $|S - S_0| < |F - S|$, so $F - S_0$ has the indicated sign. In particular Lemma 13.1's premise holds. Its last assertions give $E, Q \geq \eta^2\nu^2/256$ and $\eta/4 \leq \sqrt{E/Q} \leq 4/\eta$. By

(P), $P_v/E \in [7/8, 9/8]$; by the child guarantee and its rounding bound, $Q_v/Q \in [7/8, 9/8]$. Hence the unrounded coefficient in (U) lies between $(\eta/32)\sqrt{E/Q}$ and $(\eta/8)\sqrt{E/Q}$ with a margin at least $\eta^2/512$ from either endpoint. The rounding error $h_\alpha/2$ is smaller than this margin. Clipping at $1/2$ preserves the interval since its upper endpoint is at most $1/2$ and its lower endpoint is at most $1/8$. The rounded coefficient therefore satisfies the robust correction lemma. The returned numerical answer, including its grid rounding, has error at most $(\eta/4096)F + \omega + h_P < \eta F$, proving (A) on rejection. □

Lemma 16.5 (Nonexhaustion and joint validity)**.** *On the simultaneous shared-sample, coarse-block, and zero-test success event, all the preceding invariants hold throughout the stream and every learner makes fewer than J_* corrections. No declared cap is reached.*

Proof. Proceed in chronological order, and within a recursive call from leaves upward. At births, outgoing counters and tags have their exact initial values, and the parent bank already holds the incoming measurements. The initial normalized energy is at most $V_d^2 \le U_d/2$. All previous interval-product bounds imply current counter and tag accuracy by Lemma 16.1. The row precision in (40) gives $\xi \le 2TGh_b$ and $2\sqrt{D_*}V\xi + \xi^2 \le \eta\omega/2^{30}$. Therefore the norm-bank certificates apply to the current comparisons and numerical requests. The root guard and support give the target floor. Lemma 16.4 proves both the current answer and the validity of any correction. The geometric premise and the coefficient ratio are unchanged by dividing both residual and outgoing vector by the same $C_v > 0$. The resulting contraction therefore holds for the normalized energy used by the envelope.

Nonzero guards have the normalized amplitude bounds by Lemma 16.2. The envelope induction uses these upper bounds and the just-proved contraction, so it holds after the correction. The same lemma then shows that a correcting query did not omit its guard. The marker-coordinate argument in Lemma 13.3 proves every new interval-product bound. This in turn supplies the counter bound after the update and the accuracy needed for its residual refresh. Thus none of these assertions assumes an amplification bound for a correction before proving that correction valid. A zero physical round performs only births and updates, which preserve all the invariants and makes no corrections.

Before any correction at depth d, its normalized energy is at least

$$\frac{\eta^2\nu^2}{256C_v^2} \ge \frac{\eta^2\nu^2}{256B^{2H}}.$$

But in epoch K its envelope is at most $U_*2^{-K} \le \eta^2\nu^2/(2^{20}B^{2H})$. No correction can occur in that epoch or a later one. Thus a learner has fewer than $L_1(K+1) = J_*$ corrections. Each bank has at most two clients. Their rejections total less than $2J_*$, and their numerical requests consist of at most two birth requests and two per rejection, fewer than $4J_* + 2$. The magnitude bounds in Lemma 16.1, the cache bound, and the guard rule put all scalars strictly below their declared caps. This excludes a first cap failure and completes the induction. □

17 Adversarial robustness and space complexity

The remaining argument connects the deterministic induction to the randomized interfaces. Every bank must see only queries selected from its protected history and independent auxiliary randomness. In particular, raw row-counter values cannot be used to decide a geometric correction. Once this information-flow condition is established, the statistical success events hold for the actual adaptive execution.

The memory accounting then has three factors: the number of live banks, the number of rows per bank, and the bits retained per row. Persistence bounds the first by the tree depth. Geometric progress and the ADA interface bound the second by $\widetilde{O}(\eta^{-2})$. Guarded amplification bounds the third by a polynomial in logarithmic parameters.

Proof of Theorem 1.4. First justify applying the statistical success events. Fix one bank's shared sample. All physical updates, query identities, candidates, coefficient choices, guard choices, births, and stopping decisions are functions of permitted outputs and auxiliary randomness independent of that sample. Its row counters are rowwise deterministic functions of those quantities and the sign seed; they are not exposed to a client. Thus every bounded query used in Lemma 8.2 is a legitimate function of a hypothetical row, even though its evaluation is obtained by the maintained rounded recurrence. Internal predictor comparisons are protected by the same hierarchy as in Part I. In particular, raw sample predictions are not used for geometric correction signs. These come from protected high/low flags.

All other bank tapes and all coarse tapes can be fixed as independent auxiliary information for the shared sample's privacy proof. Conversely, before a coarse block's first use its rows have affected no decision: their counter clipping is purely rowwise and produces no observable flag. Conditional on all other randomness, its exact shadow query is fixed independently of that block. The coarse success argument is therefore valid. This establishes the statistical events for arbitrary adaptive streams, including executions that halt. There are at most $2m+1$ banks. Their allocated failure probabilities sum to less than $\beta/4$; with Lemma 15.1, their intersection has probability greater than $1-\beta$.

On that event Lemma 16.5 applies. Nonzero root queries obey $|\widehat{F}-F| \le 8H\eta F \le \varepsilon F/2$. For $0<\varepsilon<1$, $1-\varepsilon/2 \ge 1/(1+\varepsilon)$. Zero root queries return exactly zero. This proves simultaneous accuracy.

For space, put $\Lambda = \log_2(2^{100}(n+2)(m+2)(U+2)/(\varepsilon\beta))$. Then $H = O(\Lambda)$, $\log T = O(\Lambda)$, $\log B = O(\Lambda)$, and

$$\log V_d \le \log W + d\log 3 + \tfrac{1}{2}d(d-1)\log B = O(\Lambda^3).$$

It follows that $\log V, \log U_*, \log G, \log(1/\nu) = O(\Lambda^3)$, $K = O(\Lambda^3)$, and $J_* \le \eta^{-2}\,\mathrm{poly}(\Lambda)$. Every precision in (38)–(40) has $O(\Lambda^3)$ fractional bits. The bounds on V and G give $O(\Lambda^3)$ integer bits as well. Thus all geometric counters, tags, cached estimates, candidates, and leaf coordinates have $O(\Lambda^3)$-bit representations. The leaf has at most $H+1$ coordinates and at most H multiplications of dyadic coefficients; its exact numerator and denominator lengths are polynomial in Λ.

Lemma 12.2 gives at most $C\eta^{-2}\Lambda^8$ rows per bank, including all coarse blocks: its band mechanism has $a^{-1} = O(\eta^{-1}\Lambda)$, $J+R = O(\eta^{-2}\Lambda^3)$, and a fifth power of an $O(\Lambda)$ logarithm; the coarse rows contribute $O(\eta^{-2}\Lambda^4)$. Each row uses $O(s\log D_*) = O(\Lambda^2)$ seed bits and a constant number of rounded counters. The ADA scores, private thresholds, and finite samplers use the bounds proved in Lemma 8.2. There are at most $H+1$ live banks. Their total memory is at most $C\eta^{-2}\Lambda^{12}$ bits. Since $\eta^{-2} = O(\varepsilon^{-2}\Lambda^4)$, this is at most $C\varepsilon^{-2}\Lambda^{16}$ bits. The scalar and leaf workspace is smaller; the stated $C\varepsilon^{-2}\Lambda^{20}$ bound allows all temporary parameter and sampler computations as well. The prime-field zero counter uses $O(\Lambda)$ bits.

All numerical procedures are finite. Rational square comparisons implement guard selection and coefficient bisection. Dyadic rounding uses integer division. Fresh banks can be generated when their blocks start, discarding expired banks; the independent complete tapes used in the proof are a probability-space device, not stored state. This completes the implementation and space proof. □

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